

**(National Competitive Bidding)
for**

Tender no. TNPCB/TCCL/2025

**Request for Proposal (RFP) for Selection of Agency for
Execution of Interim Remediation at the Chrome contaminated
site at M/s TCCL, Ranipet, Tamil Nadu.**

Volume I to III



**TAMIL NADU POLLUTION CONTROL BOARD
76, Mount Salai, Guindy, Chennai – 600 032**

TERMS OF REFERENCE		
INSTRUCTIONS TO BIDDERS		
TENDER – TNPCB/TCCL/2025		
Request for proposal (RFP) for Selection of Agency for Execution of Interim Remediation at the Chrome Contaminated site at M/s TCCL, Ranipet, Tamil Nadu.		
SCOPE OF WORK: The work includes Capping of existing and new hazardous waste stockpiles, construction of a storm water drainage network, a rainwater collection pond, and a new boundary wall with a cut-off trench.		
A. General		
1	Tender Reference Number	TNPCB/TCCL/2025
2	Tender Type	Online
4	Name of client	Tamil Nadu Pollution Control Board
5	Method of Selection	Technical followed by Financial
6	Submission Mode	Online
7	Tender Document issue date	28.06.2025
8	Pre-bid meeting	07.07.2025 @ 3.30 PM
9	Date and Time for Bid Submission	18.07.2025 @ 11.00 AM
10	Opening of Technical Bid	18.07.2025 @ 11.30 AM
11	Payment Mode(Tender fee, EMD)	Online
12	Tender Fee	Rs. 10,000/-
13	Site Inspection Contact Address	O/o. District Environmental Engineer, Tamil Nadu pollution control Board, No. A-417, Tamil Nadu Housing Board Seekaranjapuram, Walajah Taluk, Ranipet – 632515 Email: deerpt@tnpcb.gov.in
14	Earnest Money Deposit (EMD)	Rs.10,00,000/-
15	Security Deposit	5% of agreement value
16	Consortium or Joint Venture	NOT PERMITTED
17	Period of Completion	6 months from the date of Letter of Intent
18	Tender Document	Volume I, Part I
		Instructions to bidders
		Volume I, Part II
		Documents to be supplied by the tenderer
		Volume II
		Section 1: General specifications Section 2: Technical specifications
		Volume III
		Drawings

19	Completion of commissioning	Successful all allied works and permission from TNPCB.
B. Preparation of Proposals		
20	The Tender has been issued in English Language and the Proposals shall be submitted in English Language.	
21	The proposal shall comprise of the following for each package: - Form 1: Letter of Tenderer - Form 2: Declaration of Eligibility - Form 3: Site Inspection Declaration - Form 4: General information about the tenderer - Form 5: Power of attorney of tenderer - Form 6: Financial statement of tenderer - Form 7: Organization chart & Legal capacity - Form 8: Description of approach, methodology, and work plan - Form 9: Work Schedule - Form 10: Team Composition, Task Assignment and Level of Effort - Form 11: Eligible Assignments of Bidder	
22	PRE-QUALIFICATION CRITERIA: The Tender Scrutiny Committee as a whole will evaluate the Pre- Qualification Proposals/ Forms on the basis of their Audit Turnover, Experience, Projects Executed, Projects Experience, Qualification and Experience of key personnel, applying the evaluation criteria specified in Tender Document. Eligibility Conditions for Bidders: - Bidders should have a Permanent Account Number (PAN). - GST Registration (GSTIN) valid on date of Bid submission. - The company should be a registered company (under Indian Companies Act) operating in INDIA for a minimum period of 5 years. - Average Annual Turnover of Rs. 100 crores for the last three consecutive years as on 31st March 2024. - Solvency Certificate of Rs. 8.00 Crores from nationalized bank. - Tenderer should not have been black listed/debarred by any Government organization. An affidavit to this effect should be provided by the agency on appropriate stamp paper. - The bidder should have successfully completed at least one project of secured land fill for hazardous waste with minimum capacity of 10000 m³. Or environmental related works such as STP (Sewage Treatment Plant) and ETP (Effluent Treatment Plant) worth a minimum of Rs. 50 Crores (as a single project) - Work Order and completion certificate shall be attached along with tender document. - Consortium/JV/tie-ups not allowed	

23	Tender Document should be initialized in all pages, where entries or amendments have been made.
24	Scoring System 80 % weightage for technical proposal and 20 % weightage for financial proposal a. Based on experience: i. 5 years (min): 15 marks ii. 5 – 10 years: 20 marks iii. Above 10 years: 25 marks b. Relevance of projects done earlier – 25 marks c. Team strength & experience of in – house professionals i. Up to 10 persons: 15 marks ii. 10 – 20 persons: 20 marks iii. Above 20 persons: 25 marks d. Presentation score – 25 marks
25	Payment release terms Completion of Phase 1: Capping existing HW Stockpile – 20% Completion of Phase 2: Capping new Stockpile area – 20% Completion of Phase 3: Construction of drainage work – 15% Completion of Phase 4: Construction of collection pond – 15% Completion of Phase 5: Construction of boundary wall, Access Pathway and cut – off trench – 15% Submission of completion report – 15%
26	Progress on execution of work to be reported every 15 days.

**TENDER DOCUMENT
(National Competitive Bidding)**

For
Tender no. TNPCB/TCCL/2025

**Request for Proposal (RFP) for Execution of Interim
Remediation at the Chrome Contaminated site at M/s TCCL,
Ranipet, Tamil Nadu.**

**Volume-I
Instructions to Bidders**



**TAMIL NADU POLLUTION CONTROL BOARD
76, Mount Salai, Guindy, Chennai – 600 032**

Acronyms

S.No.	Abbreviation	Expansion
1.	BOQ	Bill of Quantity
2.	CPCB	Central Pollution Control Board
3.	CS	Combined Score
4.	CV	Curriculum Vitae
5.	DEE	District Environmental Engineer
6.	DTP	Desktop Publishing
7.	EMD	Earnest Money Deposit
8.	EPR	Extended Producers Responsibility
9.	FS	Financial Score
10.	FW	Financial Weightage
11.	FX	Financial Quote of other proposal
12.	GoI	Government of India
13.	GST	Goods & Service Tax
14.	HCF	Health Care Facilities
15.	IEC	Information, Education & Communication
16.	JV	Joint Venture
17.	LFQ	Lowest Financial Quote
18.	LOA	Letter Of Award
19.	MoEF & CC	Ministry of Environment Forest and Climate Change
20.	MoU	Memorandum of Understanding
21.	NHWTS	National Hazardous Waste Tracking System
22.	OCMMS	Online Consent Management and Monitoring System
23.	PDD	Proposal Due Date
24.	PDF	Portable Document Format
25.	PMU	Project Management Unit
26.	PPT	Power Point Presentation
27.	PIBOs	Producers, Importers, Brand Owners
28.	PWPs	Plastic Waste Processors
29.	QCBS	Quality Cum Cost Based Selection
30.	RFP	Request For Proposal
31.	STP	Sewage Treatment Plant
32.	SD	Security Deposit
33.	SUP	Single Use Plastic
34.	TN	Tamil Nadu
35.	TNPCB	Tamil Nadu Pollution Control Board
36.	TOR	Terms Of Reference
37.	FML	Flexible Membrane Liner

DISCLAIMER

The information contained in this Request for Proposal document ("RFP") or subsequently provided to Bidder (s), whether verbally or in documentary or any other form by or on behalf of the Member Secretary, TNPCB, or any of their employees or advisors, is provided to Bidder (S) on the terms and conditions set out in this RFP and such other terms and conditions subject to which such information is provided. This RFP is not an agreement and is neither an offer nor invitation by the MEMBER SECRETARY, TNPCB, to the prospective Bidders or any other person. The purpose of this RFP is to provide interested parties with information that may be useful to them in preparing their technical proposals and financial offers pursuant to this RFP. This RFP includes statements, which reflect various assumptions and assessments arrived at by the MEMBER SECRETARY, TNPCB, in relation to the Assignment. Such assumptions, assessments and statements do not purport to contain all the information that each Bidder may require. This RFP may not be appropriate for all persons, and it is not possible for the MEMBER SECRETARY, TNPCB, its employees or advisors to consider the investment objectives, financial situation and particular needs of each party who reads or uses this RFP. The assumptions, assessments, statements, and information contained in this RFP may not be complete, accurate, adequate or correct. Each Bidder should, therefore, conduct its own investigations and analysis and should check the accuracy, adequacy, correctness, reliability and completeness of the assumptions, assessments, statements, and information contained in this RFP and obtains independent advice from appropriate sources. Information provided in this RFP to the Bidder (s) is on a wide range of matters, some of which depends upon interpretation of law. The information given is not an exhaustive account of statutory requirements and should not be regarded as a complete or authoritative statement of law. THE MEMBER SECRETARY, TNPCB, accepts no responsibility for the accuracy or otherwise for any interpretation or opinion on law expressed herein.

THE MEMBER SECRETARY, TNPCB, its employees and advisors make no representation or warranty and shall have no liability to any person, including any Bidder or Bidders under any law, statute, rules or regulations or tort, principles of restitution or unjust enrichment or otherwise for any loss, damages, cost or expense which may arise from or be incurred or suffered on account of anything contained in this RFP or otherwise, including the accuracy, adequacy, correctness, completeness or reliability of the RFP and any assessment, assumption, statement or information contained therein or deemed to form part of this RFP or arising in any way in this Bid State. THE MEMBER SECRETARY, TNPCB also accepts no liability of any nature whether resulting from negligence or otherwise howsoever caused arising from reliance of any Bidder upon the statements contained in this RFP. THE MEMBER SECRETARY, TNPCB, may in its absolute discretion, but without being under any obligation to do so, update, amend or supplement the information, assessment or assumptions contained in this RFP. The issue of this RFP does not imply that THE MEMBER SECRETARY, TNPCB, is bound to select a Bidder or to appoint the Selected Bidder and THE MEMBER SECRETARY, TNPCB, reserves the right to reject all or any of the Bidders or Proposals without assigning any reason whatsoever. The Bidder shall bear all its costs associated with or relating to the preparation and submission of its Proposal including but not limited to preparation, copying, postage, delivery fees, expenses associated with any demonstrations or presentations which may be required by THE MEMBER SECRETARY, TNPCB, or any other costs incurred in connection with or relating to its Proposal. All such costs and expenses will remain with the Bidder.

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INTRODUCTION

1.1 Background

Tamil Nadu Chromates and Chemicals Limited (TCCL) is located at Ranipet, SIPCOT Industrial Complex, Vellore, Tamil Nadu, India. During its operational period 1975 – 1995, chromium bearing process waste of 2.27 Lakh Tons was dumped in the open backyard in an area of about 2 hectares. The leachate generation is causing chrome contamination of land, ground water in that area.

TNPCB has decided to implement the interim remedial measure of capping the site as suggested in the DPR submitted by M/s ERM India Pvt Ltd to stop further contamination since current & previous management is not traceable.

The Hon'ble NGT (SZ) in OA. No. 186 of 2020 has suo moto taken the matter of M/s. TCCL, Ranipet and passed its final order dated: 02.09.2022 wherein directed TNPCB to implement the interim remedial measure within a period of one year.

1.2. Request for proposals

TNPCB invites proposals from interested and eligible firms for Execution of Interim Remediation at the Chrome contaminated site at M/s TCCL, Ranipet, Tamil Nadu.

1.3. Due diligence by bidders

- 1.3.1. Bidders are encouraged to inform themselves fully about the assignment and the local conditions before submitting the Proposal by paying a visit or sending written queries to TNPCB and attending a Pre-bid meeting on the date and time specified in this RFP.
- 1.3.2. Submission of bid shall be deemed to have been done after careful study and examination of the RFP document with full understanding of its implications. The response to this RFP should be full and complete in all respects. Failure to furnish all information required by the RFP documents or submission of a proposal not substantially responsive to the RFP documents in every respect will be at the bidder's risk and may result in rejection of the bidder's Proposal.

1.4. RFP documents

RFP document can be downloaded from the website of www.tnpcb.gov.in and <https://tntenders.gov.in/> after paying the tender fee.

1.5. Validity of the proposal

The Proposal shall be valid for a period of not less than 180 days from the Proposal Due Date (the "PDD").

1.6. Schedule of selection process and other details

S No	Information related to bid process	Details
1	Publication of Request for Proposal	28.06.2025 @ 10.00 AM
2	Tender fee	Rs.10,000/-
3	Earnest Money Deposit (EMD)	Rs.10,00,000/-
4	Last date for submission for queries for clarification to TNPCB	07.07.2025
5	Bid validity period	180 Days
6	Submission deadline or Proposal Due Date(PDD)	18.07.2025 @ 11.00 AM
7	Contact person and email id	The Member Secretary, TNPCB, e-mail ID :acee1chn@tnpcb.gov.in
8	Pre-bid meeting - Date, time, and venue	07.07.2025 @ 3.30 PM at TNPCB, Corporate Office, Guindy
9	Opening of Technical Proposal – date, time	18.07.2025 @ 11.30 AM
10	Opening of Financial Proposal– date, time (Tentative)	25.07.2025
11	Letter of Intent (LoI)	7 th day after opening Financial bid
12	Payment of Security Deposit (5%)	5 days after issuing LoI
13	Letter of Award (LoA)	5 days after receiving Security Deposit (SD)
14	Signing of agreement	10 days after date of LoA

1.7. Schedule of works for the project:

Work	Period
(i) Pre- bid meeting (ii) Opening and assessment of the technical bid and financial bid	30 Days (after the release of online RFP)
(i) Selection of suitable agency (ii) Drafting of work order and contract agreement	15 Days
Detailed construction and engineering	20 Days from date of award of contract
Purchasing supplies	10 Days
Demolition, Levelling and preparing landfills	30 Days
(i) Capping and draining (ii) Collection pond and boundary wall construction (iii) Finish the work	60 Days
Draft report submission with presentation	Within 5 days of completion of interim remediation work
(i) Verifying and certifying completion of work (ii) Report on project completion	10 Days

1.8. Communications and address

All communications, including proposal documents should be addressed to:

**The Member Secretary, TNPCB,
76, Mount Salai, Guindy, Chennai – 600 032.
Ph. No: 044 – 22353145**

e-mail ID :acee1chn@tnpcb.gov.in

All communications should contain the following information:

Tender no. TNPCB/TCCL/2025

Request for Proposal (RFP) for Execution of Interim Remediation at the Chrome Contaminated site at M/s TCCL, Ranipet, Tamil Nadu.”

**Member Secretary
Tamil Nadu Pollution Control Board**

Instruction to Bidders (General)

General

1.9. Scope of proposal

- 1.9.1. The objectives, scope of services, deliverables and other requirements relating to this Contract are specified in this RFP. In case a bidding firm possesses the requisite experience and capabilities required for undertaking the Contract, it may participate in the selection process individually as the **“Sole Firm”** in response to this invitation. The term **“Bidder”** means the Sole Firm. The manner in which the Proposal is required to be submitted, evaluated and accepted is explained in this RFP.
- 1.9.2. Bidders are advised that the selection of Contractor shall be on the basis of an evaluation by the TNPCB through the Selection Process specified in this RFP. Bidders shall be deemed to have understood and agreed that no explanation or justification for any aspect of the Selection Process will be given and that TNPCB's decisions are without any right of appeal whatsoever.
- 1.9.3. The Bidder shall submit its Proposal in the form and manner specified in this RFP. The Technical proposal shall be submitted in the format specified in *Volume-I Part-II* and the Financial Proposal shall be submitted in the BOQ format provided in the e-Tender website. Upon selection, the Bidder shall be required to enter into an agreement with TNPCB in the format specified in this RFP.

1.10. Eligibility conditions for contractors

- 1.10.1. Proposals of only those Bidders who satisfy the Conditions of Eligibility, stated in 1.10.2 will be considered for evaluation by the TNPCB.
- 1.10.2. Basic Eligibility criteria
- a) The Bidder shall be a private company incorporated in India under the (Indian) Companies Act 1956/2013. The Bidder shall be required to submit a true copy of its Certificate of Incorporation.
 - b) The Bidder must have a valid Goods and Service Tax (GST) registration.
 - c) The Bidder must have at least one office in Chennai which has been operational for the last five years or more.
 - d) Joint ventures/ Consortium bids will not be accepted.
- 1.10.3. The other eligibility criteria are as given below
- a) **Technical capacity:**
The agency should have demonstrated experience of at least five projects as a Contractor Agency with a minimum duration of 1 year with Central Government, State Government, Multilateral financing institutions, donor agencies, ULBs, UDAs etc. within last 10 years.

b) **Financial capacity:** The Bidder must have an average annual turnover of **Rs. 100 crores (Rs. One Hundred crores only)** per annum, from contracting assignments, for the last three consecutive years.

c) The bidder shall offer in the proposal and make available all the key resources, as per Clause 1.10.4

1.10.4. The Bidder shall enclose with its Proposal, certificate(s) from its Statutory Auditors stating its total revenues from professional fees during each of the 3 (three) financial years preceding the PDD. If the Bidder does not have a statutory auditor, it shall provide the requisite certificate(s) from the firm of Chartered Accountants that ordinarily audits the annual accounts of the Bidder.

1.10.5. The Bidder should submit a Power of Attorney as per the format provided in Volume I – Part-2.

1.10.6. The bidder should not have been blacklisted/ debarred by any Government organization.

1.10.7. The bidder must have valid PAN and GST,

1.10.8. The bidder must have filed income tax for the last 5 years.

1.11. Conflict of interest

1.11.1. Bidders shall not have a conflict of interest that may affect the Selection Process (the “**Conflict of Interest**”). Any Bidder found to have a Conflict of Interest shall be disqualified.

1.11.2. TNPCB requires that the Contractor provides professional, objective, and timely work at all times hold the TNPCB’s interest’s paramount, avoid conflicts with other assignments or its own interests, and act without any consideration for future work.

1.12. Number of proposals

A Bidder applying individually or as an associate shall not be entitled to submit another application either individually or as a member of any other consortium, as the case may be.

1.13. Acknowledgement by the Bidder

1.13.1. It shall be deemed that by submitting the proposal, the Bidder has,

- a) Made a complete and careful examination of the RFP
- b) Received all relevant information requested from TNPCB
- c) Accepted the risk of inadequacy, error or mistake in the information provided in the RFP or furnished by or on behalf of TNPCB or relating to any of the matters
- d) Agreed to be bound by the undertaking provided by it under and in terms hereof.

1.14. Right to reject any proposal

1.14.1. Notwithstanding anything contained in this RFP, TNPCB reserves the right to accept or reject any Proposal and to annul the Selection Process and reject all Proposals, at any time without any liability or any obligation for such acceptance, rejection or annulment, and

without assigning any reasons therefore.

- 1.14.2. Without prejudice to the generality of D)2.5.2, TNPCB reserves the right to reject any Proposal if,
- a) at any time, a material misrepresentation is made or discovered, or
 - b) the Bidder does not provide, within the time specified by TNPCB, the supplemental information sought by TNPCB for evaluation of the Proposal
 - c) Misrepresentation/ improper response by the Bidder may lead to the disqualification. If such disqualification / rejection occurs after the Proposals have been opened and the highest ranking Bidder gets disqualified / rejected, then TNPCB reserves the right to consider the next best Bidder, or take any other measure as may be deemed fit in the sole discretion of TNPCB, including annulment of the Selection Process
- 1.14.3. The entire Proposal shall be strictly as per the format specified in the Request for Proposal. Any deviation from the specified format will lead to disqualification of the Bidder.

1.15. Clarifications to queries

- 1.15.1. Bidders requiring any clarification on the RFP may send their queries to TNPCB in writing by e-mail (given in 1.8) so as to reach before the date mentioned in the Schedule of Selection Process at 1.6 of this RFP.
- 1.15.2. TNPCB shall endeavor to respond to the queries within the period specified therein but no later than 7 (seven) days prior to the PDD. TNPCB will post the reply to all such queries on the official website and copies thereof will also be circulated to all Bidders who have raised queries on the RFP document without identifying the source of queries.
- 1.15.3. TNPCB reserves the right not to respond to any query or provide any clarifications, in its sole discretion, and nothing in this clause 1.15 shall be construed as obliging TNPCB to respond or provide clarifications.

1.16. Amendment to RFP

- 1.16.1. At any time prior to the deadline for submission of Proposal, TNPCB may, for any reason, whether at its own initiative or in response to clarifications requested by a Bidder, modify the RFP document by the issuance of Addendum/ Amendment and posting it on the official website.
- 1.16.2. In order to give the Bidders a reasonable time for taking an amendment into account, or for any other reason, TNPCB may, in its sole discretion, extend the PDD.

SUBMISSION OF BID DOCUMENTS

1.17. Language

- 1.17.1. The Proposal with all accompanying documents (the “**Documents**”) and all communications in relation to or concerning the Selection Process shall be in English language and strictly on the forms provided in this RFP.
- 1.17.2. No supporting document or printed literature shall be submitted with the Proposal unless specifically asked for.

1.18. Format of Proposal

- 1.18.1. The Bidder shall provide all the information sought under this RFP. TNPCB would evaluate only those Proposals that are received in the specified forms and complete in all respects. The proposals shall be submitted as soft copy only in the TN e-tender's website.
- 1.18.2. The Bidder shall prepare original set of the Technical Proposal in soft copy and Financial Proposal in BOQ format provided (together with originals/ copies of Documents required to be submitted through TN e-tender's website along therewith pursuant to this RFP).
- 1.18.3. The Proposal shall be typed or written in indelible ink and signed in all pages/documents by the authorized signatory of the Bidder. All the alterations, omissions, additions, or any other amendments made to the Proposal shall be initialed by the person(s) signing the Proposal. The Proposals must be signed by the Authorised signatory (the "**Authorised Signatory**") as detailed below:
- a) by the proprietor, in case of a proprietary firm; or
 - b) by a partner, in case of a partnership firm and/or a limited liability partnership; or
 - c) by a duly Authorised person holding the Power of Attorney, in case of a Limited Company or a corporation; or

A copy of the Power of Attorney certified by a notary public shall accompany the Proposal (if required).

- 1.18.4.** Bidders should note the PDD, as specified in Clause 1.6, for submission of Proposals. Except as specifically provided in this RFP, no supplementary material will be entertained by TNPCB, and that evaluation will be carried out only on the basis of Documents submitted by the closing time of PDD. Bidders will ordinarily not be asked to provide additional material information or documents subsequent to the date of submission, and unsolicited material if submitted will be summarily rejected.

1.18.5. Technical proposal

- 1.18.6. Bidders shall upload the soft copy of technical proposal through TN e-tender's website in the formats at Volume I- Part-2 (the "**Technical Proposal**") on or before PDD.
- 1.18.7. While submitting the Technical Proposal, the Bidder shall, in particular, ensure that:
- a) Power of attorney, if applicable, is executed as per Applicable Laws.
 - b) All contents of the Proposal should be clearly numbered indexed and arranged in a sequence and shall be bound firmly.
- 1.18.8. The Technical Proposal shall not include any financial information relating to the Financial Proposal.
- 1.18.9. TNPCB reserves the right to verify all statements, information and documents, submitted to TNPCB in response to the RFP. The lack of such verification by TNPCB shall not relieve the Bidder of its obligations or liabilities hereunder nor will it affect any rights of TNPCB thereunder.
- 1.18.10. In case it is found during the evaluation or at any time before signing of the Agreement or after its execution and during the period of subsistence thereof, that one or more of the eligibility conditions have not been met by the Bidder or the Bidder has made material

misrepresentation or has given any materially incorrect or false information, the Bidder shall be disqualified forthwith if not yet appointed as the Contractor either by issue of the LOA or entering into of the Agreement, and if the Selected Bidder has already been issued the LOA or has entered into the Agreement, as the case may be, the same shall, notwithstanding anything to the contrary contained therein or in this RFP, be liable to be terminated, by a communication in writing by TNPCB without TNPCB being liable in any manner whatsoever to the Selected Bidder or Contractors, as the case may be.

1.19. Financial proposal

- 1.19.1. The Financial Proposal shall be submitted in the BOQ formats provided in the TN e-tenders website, clearly indicating the total cost of the Contract in both figures and words, in Indian Rupees. In the event of any difference between figures and words, the amount indicated in words shall prevail.
- 1.19.2. While submitting the Financial Proposal, the Bidder shall ensure the following;
 - a) All the costs associated with the assignment shall be included in the Financial Proposal.
 - b) The total amount indicated in the Financial Proposal shall be without any condition attached or subject to any assumption and shall be final and binding. In case any assumption or condition is indicated in the Financial Proposal, it shall be considered non-responsive and liable to be rejected.
 - c) The Financial Proposal shall take into account all expenses excluding tax liabilities. All applicable taxes shall be paid in addition to the financial quote and calculated as per applicable laws at the time of payment. All payments to contractors shall be subject to deduction of taxes at source as per Applicable Laws.

1.20. Submission of proposal

- 1.20.1. The Proposals as mentioned in this RFP shall be submitted in two covers system in the formats/schedules given in the TN e-tenders website.
- 1.20.2. The first cover marked "Technical Proposal" shall be submitted with necessary attachments.
- 1.20.3. The "Financial Proposal" shall be submitted in BOQ format provided in the E-tender website.

1.21. Withdrawal of proposals

- 1.21.1. The Bidder may withdraw its Proposal after it has been submitted by sending a written Withdrawal Notice, duly signed by an authorised Signatory, and including a copy of the authorization document. The Withdrawal Notice must be submitted at least one day prior to the PDD.
- 1.21.2. Proposals that are withdrawn in accordance with Clause 1.21.1 shall not be considered for evaluation.
- 1.21.3. No Proposal shall be withdrawn in the interval between the deadline for submission of Proposals and the expiration of the period of Proposal validity specified in the RFP or any extension thereof, except in the case of a request by the Client to extend the Proposal validity.

1.22. Earnest Money Deposit (EMD)

- 1.22.1. The Bidder shall pay as part of its Proposal, Earnest Money Deposit of Rs. 10 Lakhs/- (Rupees ten lakhs only) in the online payment mode available in the TN e- tender's website, returnable not later than 90 (Ninety) days from PDD except in case of the two highest ranked Bidders.
- 1.22.2. In the event that the first ranked Bidder commences the assignment, the second ranked Bidder, who has been kept in reserve, shall be returned its Earnest Money Deposit forthwith, but in no case not later than 90 (Ninety) days from PDD. The Selected Bidder's Earnest Money Deposit shall be returned, upon the Bidder signing the Agreement and completing the Deliverables assigned to it for the first 2 (two) months of the Contractor in accordance with the provisions thereof.
- 1.22.3. Any Bid not accompanied by the Earnest Money Deposit shall be rejected by TNPCB as non- responsive.
- 1.22.4. TNPCB shall not be liable to pay any interest on the Earnest Money Deposit and the same shall be interest free.
- 1.22.5. The Bidder, by submitting its Proposal pursuant to this RFP, shall be deemed to have acknowledged that without prejudice to TNPCB's any other right or remedy hereunder or in law or otherwise, the Earnest Money Deposit shall be forfeited and appropriated by TNPCB as the mutually agreed pre-estimated compensation and damage payable to TNPCB for, *inter alia*, the time, cost and effort of TNPCB in regard to the RFP including the consideration and evaluation of the Proposal under the following conditions:
 - a) If a Bidder submits a non-responsive Proposal.
 - b) If a Bidder engages in any of the Prohibited Practices.
 - c) If a Bidder withdraws its Proposal during the period of its validity.
 - d) In the case of the Selected Bidder, if the Bidder fails to reconfirm its commitments during negotiations.
 - e) In the case of a Selected Bidder, if the Bidder fails to sign the Agreement or commence the assignment.
 - f) If the Bidder is found to have a Conflict of Interest.

EVALUATION PROCESS

1.23. Evaluation of proposals

- 1.23.1. TNPCB shall open the Proposals online on the PDD as specified in Clause 1.6 The Technical Proposals shall be opened first.
- 1.23.2. Prior to evaluation of Proposals, TNPCB will determine whether each Proposal is responsive to the requirements of the RFP. A Proposal shall be considered responsive only if:
 - a) The Technical Proposal is uploaded in the form specified at Volume-I-Part -2.
 - b) It is received within the PDD including any extension thereof.

- c) It is accompanied by the Power of Attorney or Authorized Signatory.
- d) It contains all the information (complete in all respects) as requested in the RFP.

1.23.3. Scrutinizing process. After the financial evaluation, the final ranking of the Proposals shall be carried out.

1.23.4. Bidders are advised that Selection shall be entirely at the discretion of the TNPCB. Bidders shall be deemed to have understood and agreed that TNPCB shall not be required to provide any explanation or justification in respect of any aspect of the Selection Process or Selection.

1.24.5 Any information contained in the Proposal shall not in any way be construed as binding on TNPCB, its agents, successors or assigns, but shall be binding against the Bidder if the Contractor is subsequently awarded to it.

1.24. Confidentiality

Information relating to the examination, clarification, evaluation, and recommendation for the selection of Bidders shall not be disclosed to any person who is not officially concerned with the process or is not a retained professional adviser advising TNPCB in relation to matters arising out of or concerning the Selection Process. TNPCB shall treat all information, submitted as part of the Proposal, in confidence and shall require all those who have access to such material to treat the same in confidence. TNPCB may not divulge any such information unless it is directed to do so by any statutory entity that has the power under law to require its disclosure or to enforce or assert any right or privilege of the statutory entity and/or TNPCB or as may be required by law or in connection with any legal process.

Appointment of Third Party Agency

1.25. Negotiations

The Selected Bidder may, if necessary, be invited for negotiations. The negotiations shall generally not be for reducing the price of the Proposal but will be for re-confirming the obligations of the Contractor under this RFP. Issues such as deployment of Key Personnel, understanding of the RFP, methodology and quality of the work plan shall be discussed during negotiations.

1.26. Indemnity

1.26.1. The Contractor shall, subject to the provisions of the Agreement, indemnify TNPCB, for an amount not exceeding the value of the Agreement, for any direct loss or damage that is caused due to any deficiency in Services.

1.27. Issuance of Letter of Intent and Award of Contract

- a) **Letter of Intent:** After selection, a letter of Intent (LoI) will be issued in duplicate, by TNPCB to the Selected Bidder and the Selected Bidder shall, within 5 (five) working days of the receipt of the LOI, sign and return the duplicate copy of the LOI along with 5% of the Security Deposit as mentioned in TOR. In the event the duplicate copy of the LOI duly signed by the Selected Bidder and Security Deposit is not received within the stipulated date, TNPCB may, unless it consents to extension of time for submission

thereof, cancel the LOI and the next highest-ranking Bidder may be considered.

- b) Upon receipt of the duplicate copy of the duly signed LOI and Security Deposit Letter of Award (the “**LOA**”) shall be issued, in duplicate, by TNPCB to the Selected Bidder and the Selected Bidder shall, within 2 (two) days of the receipt of the LOA, sign and return the duplicate copy of the LOA in acknowledgement thereof. In the event the duplicate copy of the LOA duly signed by the Selected Bidder is not received within the stipulated date, TNPCB may, unless it consents to extension of time for submission thereof, cancel the LOA and the next highest-ranking Bidder may be considered.

1.28. Execution of Agreement

After acknowledgement of the LOA as aforesaid by the Selected Bidder, it shall execute the Agreement within the period prescribed in this RFP.

1.29. Commencement of the Assignment

The Contractor shall commence the Services at the Project site within 15 (fifteen) days of the date of the Agreement, or such other date as may be mutually agreed. If the Contractor fails to either sign the Agreement as specified in Clause 1.28 or commence the assignment as specified herein, TNPCB may invite the second ranked Bidder for negotiations. In such an event, the Earnest Money Deposit of the first ranked Bidder shall be forfeited and appropriated in accordance with the provisions of Clause 1.22

1.30. Proprietary data

Subject to the provisions of Clause D)3.8, all documents and other information provided by the TNPCB or submitted by a Bidder to the TNPCB shall remain or become the property of the TNPCB. Bidders and the Contractors, as the case may be, are to treat all information as strictly confidential. The TNPCB will not return any Proposal or any information related thereto. All information collected, analyzed, processed or in whatever manner provided by the Contractor to the TNPCB in relation to the Contractor shall be the property of TNPCB.

1.31. Short listing of Bidders

TNPCB shall pre-qualify not less than 2 (two) Bidders and shortlist for financial evaluation. However, if the number of such pre-qualified Bidders is less than 2 (two), TNPCB may, in its sole discretion, pre-qualify the Bidder(s) with lower Technical Score (TS).

1.32. Evaluation of Financial Proposal

- i) For financial evaluation, the total Lump-sum cost of the Contractor assignment indicated in the Financial Proposal, excluding all applicable taxes, shall be considered.

Fraudulent and Corrupt Practices

- 1.33. The Bidders and their respective officers, employees, agents and advisers shall observe the highest standard of ethics during the Selection Process. Notwithstanding anything to the contrary contained in this RFP, TNPCB shall reject a Proposal without being liable in any manner whatsoever to the Bidder, if it determines that the Bidder has, directly or

indirectly or through an agent, engaged in corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice (collectively the “**Prohibited Practices**”) in the Selection Process. In such an event, TNPCB shall, without prejudice to its any other rights or remedies, forfeit and appropriate the Bid Security, if available, as mutually agreed genuine pre-estimated compensation and damages payable to TNPCB for, *inter alia*, time, cost and effort of TNPCB, in regard to the RFP, including consideration and evaluation of such Bidder’s Proposal.

- 1.34. Without prejudice to the rights of TNPCB under Clause D)3.2.6 hereinabove and the rights and remedies which TNPCB may have under the LOA or the Agreement, if a Bidder or Contractors, as the case may be, is found by TNPCB to have directly or indirectly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice during the Selection Process, or after the issue of the LOA or the execution of the Agreement, such Bidder or Contractor shall not be eligible to participate in any tender or RFP issued by TNPCB during a period of two years from the date such Bidder or Contractors, as the case may be, is found to have directly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice, as the case may be.
- 1.35. For the purposes of this Clause, the following terms shall have the meaning hereinafter respectively assigned to them
- a) “**corrupt practice**” means (i) the offering, giving, receiving, or soliciting, directly or indirectly, of anything of value to influence the action of any person connected with the Selection Process (for avoidance of doubt, offering of employment to or employing or engaging in any manner whatsoever, directly or indirectly, any official of the TNPCB who is or has been associated in any manner, directly or indirectly with the Selection Process or the LOA or has dealt with matters concerning the Agreement or arising there from, before or after the execution thereof, at any time prior to the expiry of one year from the date such official resigns or retires from or otherwise ceases to be in the service of the TNPCB, shall be deemed to constitute influencing the actions of a person connected with the Selection Process; or (ii) save as provided herein, engaging in any manner whatsoever, whether during the Selection Process or after the issue of the LOA or after the execution of the Agreement, as the case may be, any person in respect of any matter relating to the Project or the LOA or the Agreement, who at any time has been or is a legal, financial or technical contractors/ adviser of TNPCB in relation to any matter concerning the Project;
 - b) “**fraudulent practice**” means a misrepresentation or omission of facts or disclosure of incomplete facts, in order to influence the Selection Process;
 - c) “**coercive practice**” means impairing or harming or threatening to impair or harm, directly or indirectly, any persons or property to influence any person’s participation or action in the Selection Process;
 - d) “**undesirable practice**” means (i) establishing contact with any person connected with or employed or engaged by the TNPCB with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Selection Process; or (ii)

having a Conflict of Interest; and

- e) **“restrictive practice”** means forming a cartel or arriving at any understanding or arrangement among Bidders with the objective of restricting or manipulating a full and fair competition in the Selection Process.

Miscellaneous

- 1.36.** The Selection Process shall be governed by, and construed in accordance with, the laws of India and the Courts in the State of Tamil Nadu in which the TNPCB has its headquarters shall have exclusive jurisdiction over all disputes arising under, pursuant to and/or in connection with the Selection Process.
- 1.37.** TNPCB, in its sole discretion and without incurring any obligation or liability, reserves the right, at any time, to
 - a) suspend and/or cancel the Selection Process and/or amend and/or supplement the Selection Process or modify the dates or other terms and conditions relating thereto;
 - b) consult with any Bidder in order to receive clarification or further information;
 - c) retain any information and/or evidence submitted to the TNPCB by, on behalf of and/or in relation to any Bidder; and/or
 - d) Independently verify, disqualify, reject and/or accept any and all submissions or other information and/or evidence submitted by or on behalf of any Bidder.
- 1.38.** It shall be deemed that by submitting the Proposal, the Bidder agrees and releases the TNPCB, its employees, agents and advisers, irrevocably, unconditionally, fully and finally from any and all liability for claims, losses, damages, costs, expenses or liabilities in any way related to or arising from the exercise of any rights and/or performance of any obligations hereunder, pursuant hereto and/or in connection herewith and waives any and all rights and/or claims it may have in this respect, whether actual or contingent, whether present or future.
- 1.39.** All documents and other information supplied by the TNPCB or submitted by a Bidder shall remain or become, as the case may be, the property of the TNPCB. The TNPCB will not return any submissions made hereunder. Bidders are required to treat all such documents and information as strictly confidential.
- 1.40.** The TNPCB reserves the right to make inquiries with any of the clients listed by the Bidders in their previous experience record.

Schedule 1: Terms of Reference

1. Background

M/s. Tamil Nadu Chromates and Chemicals Limited (TCCL) is an identified chromium contaminated site at Ranipet. During its operational period 1975 – 1995, chromium bearing process waste of 2.27 Lakh Tons was dumped in the open backyard in an area of about 2 hectares. The leachate generation is causing chrome contamination of land, ground water in that area.

TNPCB has decided to implement the interim remedial measure of capping the site as suggested in the DPR submitted by M/s ERM India Pvt Ltd to stop further contamination since current & previous management is not traceable.

The Hon'ble NGT (SZ) in OA. No. 186 of 2020 has suo moto taken the matter of M/s. TCCL, Ranipet and passed its final order dated: 02.09.2022 wherein directed TNPCB to implement the interim remedial measure within a period of one year.

2. Scope of services

The work involves Capping of existing and new hazardous waste stockpiles, construction of a storm water drainage network, a rainwater collection pond, and a new boundary wall with a cut-off trench.

3. The detailed scope under each component is as follows:

(Detailed Scope of the Project explained in Volume II Section -2)

Phase 1 - Capping the Existing Hazardous Waste Stockpile

Phase 2 - Capping New Stockpile Area

Phase 3 - Construction of Drainage Network

Phase 4 - Construction of a Collection Pond for Rainwater Collection

Phase 5 - Construction of New Boundary Wall, Access Pathway and Cut-off Trench

4. Reporting

The Contractor will work under the head of the Department assigned in TNPCB under the direct supervision of the Member Secretary, TNPCB, or a person appointed by them.

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Schedule 2: AGREEMENT

This AGREEMENT (hereinafter called the “**Agreement**”) is made on the...day of the month of20..., between, on the one hand, the [The Member Secretary, TNPCB, Chennai]

(hereinafter called the “**Authority**” which expression shall include their respective successors and permitted assigns, unless the context otherwise requires) and, on the other hand,(Hereinafter called the “**Contractor**” which expression shall include their respective successors and permitted assigns). WHEREAS

- A) The Authority vide its Request for Proposal for Execution of interim remediation at the chrome contaminated site at M/s TCCL (hereinafter called the “**Project**”) to the..... (hereinafter called the “**Contractor**”)
- B) the Contractor submitted its proposals for the aforesaid work, whereby the Contractor represented to the Authority that it had the required professional skills, and in the said proposals the Contractor also agreed to provide the Services to the Authority on the terms and conditions as set forth in the RFP and this Agreement; and
- C) the Authority, on acceptance of the aforesaid proposals of the Contractors, awarded the Contractor to the Contractor vide its Letter of Award dated (the “**LOA**”); and
- D) in pursuance of the LOA, the parties have agreed to enter into this Agreement

NOW, THEREFORE, the parties hereby agree as follows:

1. GENERAL

1.1. Definitions and interpretations

1.1.1. The words and expressions beginning with capital letters and defined in this Agreement shall, unless the context otherwise requires, have the meaning hereinafter respectively assigned to them

- A) “**Additional Costs**” shall have the meaning set forth in Clause 6.1.1;
- B) “**Agreement**” means this Agreement, together with all the Annexes;
- C) “**Agreement Value**” shall have the meaning set forth in Clause D)6.1.1;
- D) “**Applicable Laws**” means the laws and any other instruments having the force of law in India as they may be issued and in force from time to time;
- E) “**Confidential Information**” shall have the meaning set forth in Clause D)3.3;
- F) “**Conflict of Interest**” shall have the meaning set forth in Clause D)3.2 read with the provisions of RFP;
- G) “**Dispute**” shall have the meaning set forth in Clause D)8.2.1;
- H) “**Effective Date**” means the date on which this Agreement comes into force and

effect pursuant to Clause 2.1

- I) **“Government”** means the Government of Tamil Nadu;
- J) **“INR, Re. or Rs.”** means Indian Rupees;
- K) **“Member”**, in case the Contractor consists of a joint venture or consortium of more than one entity, means any of these entities, and **“Members”** means all of these entities;
- L) **“Party”** means the Authority or the Contractors, as the case may be, and **“Parties”** means both of them;
- M) **“Personnel”** means persons hired by the Contractor or by any Sub- Contractors employees and assigned to the performance of the Services or any part thereof;
- N) **“Resident Personnel”** means such persons who at the time of being so hired had their domicile inside India;
- O) **“RFP”** means the Request for Proposal document in response to which the Contractors’ proposal for providing Services was accepted;
- P) **“Services”** means the work to be performed by the Contractor pursuant to this Agreement, as described in the Terms of Reference hereto;
- Q) **“Sub-Contractors”** means any entity to which the Contractor subcontracts any part of the Services; and

All terms and words not defined herein shall, unless the context otherwise requires, have the meaning assigned to them in the RFP.

- 1.1.2. The following documents along with all addenda issued thereto shall be deemed to form and be read and construed as integral parts of this Agreement and in case of any contradiction between or among them the priority in which a document would prevail over another would be as laid down below beginning from the highest priority to the lowest priority:

- a. Agreement;
- b. Annexes of Agreement;
- c. RFP; and
- d. Letter of Award

1.2. Relation between the Parties

Nothing contained herein shall be construed as establishing a relation of master and servant or of agent and principal as between the Authority and the Contractors. The Contractor shall, subject to this Agreement, have complete charge of Personnel performing the Services and shall be fully responsible for the Services performed by them or on their behalf hereunder.

1.3. Rights and Obligations

The mutual rights and obligations of the Authority and the Contractor shall be as set forth in the Agreement, in particular:

- a. the Contractor shall carry out the Services in accordance with the provisions of the Agreement; and
- b. the Authority shall make payments to the Contractor in accordance with the provisions of the Agreement.

1.4. Governing law and jurisdiction

This Agreement shall be construed and interpreted in accordance with and governed by the laws of India, and the courts in the State of Tamil Nadu.

1.5. Language

All notices required to be given by one Party to the other Party and all other communications, documentation and proceedings which are in any way relevant to this Agreement shall be in writing and in English language.

1.6. Table of contents and headings

The table of contents, headings or sub-headings in this Agreement is for convenience of reference only and shall not be used in, and shall not affect, the construction or interpretation of this Agreement.

1.7. Location

The Services shall be performed at the site of the Project in Ranipet, in accordance with the provisions of RFP and at such locations as are incidental thereto, including the offices of the Contractors.

1.8. Authorized Representatives

1.8.1. Any action required or permitted to be taken, and any document required or permitted to be executed, under this Agreement by the Authority or the Contractors, as the case may be taken or executed by the officials specified in this Clause D) 1.8.2

1.8.2. The Authority may, from time to time, designate one of its officials as the Authority Representative. Unless otherwise notified, the Authority Representative shall be:

Name :

Designation :

Mobile :

Email :

- 1.8.3. The Contractor may designate one of its employees as Representative. Unless otherwise notified, the Contractors' Representative shall be

Name :

Designation :

Mobile :

Email :

1.9. Taxes and duties

Unless otherwise specified in the Agreement, the Contractor shall pay all such taxes, duties, fees and other impositions as may be levied under the Applicable Laws and the Authority shall perform such duties in regard to the deduction of such taxes as may be lawfully imposed on it. Service Tax shall be payable over and above the Agreement Value in accordance with the governing tax laws of India.

2. COMMENCEMENT, COMPLETION, AND TERMINATION OF AGREEMENT

2.1. Effectiveness of Agreement

This Agreement shall come into force and effect on the date of this Agreement (the "Effective Date").

2.2. Commencement of Services

The Contractor shall commence the Services within a period of 15 (fifteen) days from the Effective Date, unless otherwise agreed by the Parties.

2.3. Termination of Agreement for failure to commence Services

If the Contractor does not commence the Services within the period specified in Clause D)2.2 above, the Authority may, by not less than 2 (two) weeks' notice to the Contractors, declare this Agreement to be null and void, and in the event of such a declaration, the Earnest Money Deposit of the Contractor shall stand forfeited

2.4. Expiry of Agreement

Unless terminated earlier pursuant to Clauses D)2.3 or D)2.9 hereof, this Agreement shall, unless extended by the Parties by mutual consent, expire upon the earlier of (i) expiry of a period of 90 (ninety) days after the delivery of the final deliverable to the Authority. Upon Termination, the Authority shall make payments of all amounts due to the Contractor here under.

2.5. Entire Agreement

- 2.5.1. This Agreement and the Annexes together constitute a complete and exclusive statement of the terms of the agreement between the Parties on the subject hereof, and

no amendment or modification hereto shall be valid and effective unless such modification or amendment is agreed to in writing by the Parties and duly executed by persons especially empowered in this behalf by the respective Parties. All prior written or oral understandings, offers or other communications of every kind pertaining to this Agreement are abrogated and withdrawn; provided, however, that the obligations of the Contractor arising out of the provisions of the RFP shall continue to subsist and shall be deemed to form part of this Agreement.

2.5.2. Without prejudice to the generality of the provisions of Clause D)2.5.1, on matters not covered by this Agreement, the provisions of RFP shall apply.

2.6. Modifications of Agreement

Modification of the terms and conditions of this Agreement, including any modification of the scope of the Services, may only be made by written agreement between the Parties. However, each Party shall give due consideration to any proposals for modification made by the other Party.

2.7. Force Majeure

2.7.1. Definition

- a. For the purposes of this Agreement, “**Force Majeure**” means an event which is beyond the reasonable control of a Party, and which makes a Party’s performance of its obligations hereunder impossible or so impractical as reasonably to be considered impossible in the circumstances, and includes, but is not limited to, war, riots, civil disorder, earthquake, fire, explosion, storm, flood or other adverse weather conditions, strikes, lockouts or other industrial action (except where such strikes, lockouts or other industrial action are within the power of the Party invoking Force Majeure to prevent), confiscation or any other action by government agencies.
- b. Force Majeure shall not include (i) any event which is caused by the negligence or intentional action of a Party or such Party’s Sub-Contractor or agents or employees, nor (ii) any event which a diligent Party could reasonably have been expected to both (A) take into account at the time of the conclusion of this Agreement, and (B) avoid or overcome in the carrying out of its obligations hereunder.
- c. Force Majeure shall not include insufficiency of funds or failure to make any payment required hereunder.

2.7.2. No Breach of Agreement

The failure of a Party to fulfil any of its obligations hereunder shall not be considered to be a breach of, or default under, this Agreement insofar as such inability arises from an event of Force Majeure, provided that the Party affected by such an event has taken all reasonable precautions, due care and reasonable alternative measures, all with the objective of carrying out the terms and conditions of this Agreement.

2.7.3. Measures to be taken

- a. A Party affected by an event of Force Majeure shall take all reasonable measures to remove such Party's inability to fulfil its obligations hereunder with a minimum of delay.
- b. A Party affected by an event of Force Majeure shall notify the other Party of such event as soon as possible, and in any event not later than 14 (fourteen) days following the occurrence of such event, providing evidence of the nature and cause of such event, and shall similarly give notice of the restoration of normal conditions as soon as possible.
- c. The Parties shall take all reasonable measures to minimize the consequences of any event of Force Majeure.

2.7.4. Extension of time

Any period within which a Party shall, pursuant to this Agreement, complete any action or task, shall be extended for a period equal to the time during which such Party was unable to perform such action as a result of Force Majeure.

2.7.5. Payments

During the period of its inability to perform the Services as a result of an event of Force Majeure, the Contractor shall be entitled to be reimbursed for Additional Costs reasonably and necessarily incurred by it during such period for the purposes of the Services and in reactivating the Services after the end of such period.

2.7.6. Consultation

Not later than 30 (thirty) days after the Contractor has, as the result of an event of Force Majeure, become unable to perform a material portion of the Services, the Parties shall consult with each other with a view to agreeing on appropriate measures to be taken in the circumstances.

2.8. Suspension of Agreement

The Authority may, by written notice of suspension to the Contractors, suspend all payments to the Contractor here under if the Contractor shall be in breach of this Agreement or shall fail to perform any of its obligations under this Agreement, including the carrying out of the Services; provided that such notice of suspension (i) shall specify the nature of the breach or failure, and (ii) shall provide an opportunity to the Contractor to remedy such breach or failure within a period not exceeding 30 (thirty) days after receipt by the Contractor of such notice of suspension.

2.9. Termination of Agreement

2.9.1. By the Authority

The Authority may, by not less than 30 (thirty) days' written notice of termination to the Contractors, such notice to be given after the occurrence of any of the events

specified in this Clause D)2.9.1, terminate this Agreement if:

- a. the Contractor fails to remedy any breach hereof or any failure in the performance of its obligations hereunder, as specified in a notice of suspension pursuant to Clause D)2.8 hereinabove, within 30 (thirty) days of receipt of such notice of suspension or within such further period as the Authority may have subsequently granted in writing;
- b. the Contractor becomes insolvent or bankrupt or enters into any agreement with its creditors for relief of debt or take advantage of any law for the benefit of debtors or goes into liquidation or receivership whether compulsory or voluntary;
- c. the Contractor fails to comply with any final decision reached as a result of arbitration proceedings pursuant to Clause D)8.4 hereof;
- d. the Contractor submits to the Authority a statement which has a material effect on the rights, obligations or interests of the Authority and which the Contractor knows to be false;
- e. any document, information, data or statement submitted by the Contractor in its Proposals, based on which the Contractor was considered eligible or successful, is found to be false, incorrect or misleading;
- f. as the result of Force Majeure, the Contractor is unable to perform a material portion of the Services for a period of not less than 60 (sixty) days; or
- g. the Authority, in its sole discretion and for any reason whatsoever, decides to terminate this Agreement.

2.9.2. Cessation of rights and obligations

Upon termination of this Agreement pursuant to Clauses 2.3 or 2.9 hereof, or upon expiration of this Agreement pursuant to Clause 2.4 hereof, all rights and obligations of the Parties hereunder shall cease, except (i) such rights and obligations as may have accrued on the date of termination or expiration, or which expressly survive such Termination; (ii) the obligation of confidentiality set forth in Clause 3.3 hereof; (iii) the Contractors' obligation to permit inspection, copying and auditing of such of its accounts and records set forth in Clause 3.6, as relate to the Contractors' Services provided under this Agreement; and (iv) any right or remedy which a Party may have under this Agreement or the Applicable Law.

2.9.3. Cessation of Services

Upon termination of this Agreement by notice of either Party to the other pursuant to Clauses D)2.9.1 or D)2.9.2 hereof, the Contractor shall, immediately upon dispatch or receipt of such notice, take all necessary steps to bring the Services to a close in a prompt and orderly manner and shall make every reasonable effort to keep expenditures for this purpose to a minimum. With respect to documents prepared by the Contractor and equipment and materials furnished by the Authority, the Contractor shall proceed as provided respectively by Clauses D)3.9 or D)3.10 hereof.

2.9.4. Payment upon Termination

Upon termination of this Agreement pursuant to Clauses 2.9.1 or 2.9.2 hereof, the Authority shall make the following payments to the Contractor (after offsetting against these payments any amount that may be due from the Contractor to the Authority):

- a. remuneration pursuant to Clause 6 hereof for Services satisfactorily performed prior to the date of termination.
- b. reimbursable expenditures pursuant to Clause 6 hereof for expenditures actually incurred prior to the date of termination; and
- c. except in the case of termination pursuant to sub-clauses (a) through (e) of Clause 2.9.1 hereof, reimbursement of any reasonable cost incidental to the prompt and orderly termination of the Agreement including the cost of the return travel of the Contractors' personnel.

2.9.5. Disputes about Events of Termination

If either Party disputes whether an event specified in Clause 2.9.1 or in Clause 2.9.2 hereof has occurred, such Party may, within 30 (thirty) days after receipt of notice of termination from the other Party, refer the matter to arbitration pursuant to Clause 8.4 hereof, and this Agreement shall not be terminated on account of such event except in accordance with the terms of any resulting arbitral award.

3. OBLIGATIONS OF THE CONTRACTOR

3.1. General

3.1.1. Standards of Performance

The Contractor shall perform the Services and carry out its obligations hereunder with all due diligence, efficiency and economy, in accordance with generally accepted professional techniques and practices, and shall observe sound management practices, and employ appropriate advanced technology and safe and effective equipment, machinery, materials and methods. The Contractor shall always act, in respect of any matter relating to this Agreement or to the Services, as a faithful adviser to the Authority, and shall at all times support and safeguard the Authority's legitimate interests in any dealings with Sub-Contractors or Third Parties.

3.1.2. Terms of Reference

The scope of services to be performed by the Contractor is specified in the Terms of Reference (the "**TOR**") at 1.1 The Contractor shall provide the Deliverables specified therein in conformity with the time schedule stated therein.

3.1.3. Applicable Laws

The Contractor shall perform the Services in accordance with the Applicable Laws and shall take all practicable steps to ensure that any Sub-Contractors, as well as the Personnel and agents of the Contractor and any Sub-Contractors, comply with the Applicable Laws.

3.2. Conflict of Interest

3.2.1. The Contractor shall not have a Conflict of Interest and any breach hereof shall constitute a breach of the Agreement.

3.2.2. Contractor and Affiliates not to be otherwise interested in the Project

The Contractor agrees that, during the term of this Agreement and after its termination, the Contractor or any Associate thereof and any entity affiliated with the Contractors, shall be disqualified from providing goods, works, services, loans or equity for any project resulting from or closely related to the Services and any breach of this obligation shall amount to a Conflict of Interest; provided that the restriction herein shall not apply after a period of five years from the completion of this assignment or to consulting assignments granted by banks/ lenders at any time; provided further that this restriction shall not apply to Contractor/ advisory services provided to the Authority in continuation of this Contractor or to any subsequent Contractor/ advisory services provided to the Authority in accordance with the rules of the Authority. For the avoidance of doubt, an entity affiliated with the Contractor shall include a partner in the Contractors' firm or a person who holds more than 5% (five per cent) of the subscribed and paid up share capital of the Contractors, as the case may be, and any Associate thereof.

3.2.3. Prohibition of conflicting activities

Neither the Contractor nor the Personnel of either of them shall engage, either directly or indirectly, in any of the following activities:

Any project or assignment that the contractor has been undertaking prior to signing this contract will not be considered as conflict and will be allowed to carry on such project/ assignment.

- a. during the term of this Agreement, any business or professional activities which would conflict with the activities assigned to them under this Agreement;
- b. after the termination of this Agreement, such other activities as may be specified in the Agreement; or
- c. at any time, such other activities as have been specified in the RFP as Conflict of Interest.

3.2.4. Contractor not to benefit from commissions discounts, etc.

The remuneration of the Contractor pursuant to Clause 6 hereof shall constitute the

Contractor's sole remuneration in connection with this Agreement or the Services and the Contractor shall not accept for its own benefit any trade commission, discount or similar payment in connection with activities pursuant to this Agreement or to the Services or in the discharge of its obligations hereunder, and the Contractor shall use its best efforts to ensure that any Sub-Contractors, as well as the Personnel and agents of either of them, similarly shall not receive any such additional remuneration.

- 3.2.5. The Contractor and its Personnel shall observe the highest standards of ethics and shall not have engaged in and shall not hereafter engage in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice (collectively the "**Prohibited Practices**"). Notwithstanding anything to the contrary contained in this Agreement, the Authority shall be entitled to terminate this Agreement forthwith by a communication in writing to the Contractors, without being liable in any manner whatsoever to the Contractors, if it determines that the Contractor has, directly or indirectly or through an agent, engaged in any Prohibited Practices in the Selection Process or before or after entering into of this Agreement. In such an event, the Authority shall forfeit and appropriate the performance security, if any, as mutually agreed genuine pre-estimated compensation and damages payable to the Authority towards, inter alia, the time, cost and effort of the Authority, without prejudice to the Authority's any other rights or remedy hereunder or in law.
- 3.2.6. Without prejudice to the rights of the Authority under Clause 3.2.5 above and the other rights and remedies which the Authority may have under this Agreement, if the Contractors found by the Authority to have directly or indirectly or through an agent, engaged or indulged in any Prohibited Practices, during the Selection Process or before or after the execution of this Agreement, the Contractor shall not be eligible to participate in any tender or RFP issued during a period of 2 (two) years from the date the Contractors found by the Authority to have directly or indirectly or through an agent, engaged or indulged in any Prohibited Practices.
- 3.2.7. For the purposes of Clauses 3.2.5 and 3.2.6, the following terms shall have the meaning hereinafter respectively assigned to them:
- a. "**corrupt practice**" means (i) the offering, giving, receiving or soliciting, directly or indirectly, of anything of value to influence the actions of any person connected with the Selection Process (for removal of doubt, offering of employment or employing or engaging in any manner whatsoever, directly or indirectly, any official of the Authority who is or has been associated in any manner, directly or indirectly with Selection Process or LOA or dealing with matters concerning the Agreement before or after the execution thereof, at any time prior to the expiry of one year from the date such official resigns or retires from or otherwise ceases to be in the service of the Authority, shall be deemed to constitute influencing the actions of a person connected with the Selection Process); or (ii) engaging in any manner whatsoever, whether during the Selection Process or after the issue of LOA or after the execution of the Agreement, as the case may be, any person in respect of any matter relating to the Project or the LOA or the Agreement, who at any time has been or is a legal,

financial or technical adviser the Authority in relation to any matter concerning the Project;

- b. **“fraudulent practice”** means a misrepresentation or omission of facts or suppression of facts or disclosure of incomplete facts, in order to influence the Selection Process;
- c. **“coercive practice”** means impairing or harming, or threatening to impair or harm, directly or indirectly, any person or property to influence any person’s participation or action in the Selection Process or the exercise of its rights or performance of its obligations by the Authority under this Agreement;
- d. **“undesirable practice”** means (i) establishing contact with any person connected with or employed or engaged by the Authority with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Selection Process; or (ii) having a Conflict of Interest; and
- e. **“Restrictive practice”** means forming a cartel or arriving at any understanding or arrangement among Bidders with the objective of restricting or manipulating a full and fair competition in the Selection Process.

3.3. Confidentiality

The Contractor agency and the Personnel of either of them shall not, either during the term or within two years after the expiration or termination of this Agreement disclose any proprietary information, including information relating to reports, data, drawings, design software or other material, whether written or oral, in electronic or magnetic format, and the contents thereof; and any reports, digests or summaries created or derived from any of the foregoing that is provided by the Authority to the Contractors, its Sub-Contractors and the Personnel; any information provided by or relating to the Authority, its technology, technical processes, business affairs or finances or any information relating to the Authority’s employees, officers or other professionals or suppliers, customers, or contractors of the Authority; and any other information which the contractor is under an obligation to keep confidential in relation to the Project, the Services or this Agreement (**“Confidential Information”**), without the prior written consent of the Authority.

Notwithstanding the aforesaid, the Contractors, its Sub-Contractors and the Personnel of either of them may disclose Confidential Information to the extent that such Confidential Information:

- ii) was in the public domain prior to its delivery to the Contractors, its Sub-Contractors and the Personnel of either of them or becomes a part of the public knowledge from a source other than the Contractors, its Sub-Contractors and the Personnel of either of them;
- iii) was obtained from a third party with no known duty to maintain its confidentiality;
- iv) is required to be disclosed by Applicable Laws or judicial or administrative or

arbitral process or by any governmental instrumentalities, provided that for any such disclosure, the Contractors, its Sub-Contractors and the Personnel of either of them shall give the Authority, prompt written notice, and use reasonable efforts to ensure that such disclosure is accorded confidential treatment; and

- v) is provided to the professional advisers, agents, auditors or representatives of the Contractor or its Sub-Contractors or Personnel of either of them, as is reasonable under the circumstances; provided, however, that the Contractor or its Sub-Contractors or Personnel of either of them, as the case may be, shall require their professional advisers, agents, auditors or its representatives, to undertake in writing to keep such Confidential Information, confidential and shall use its best efforts to ensure compliance with such undertaking.

3.4. Liability of the Contractors

- 3.4.1. The liability of the contractor as defined in this RFP will be limited to the Agreement Value, as specified in this RFP

3.5. Insurance to be taken by the Contractors

- 3.5.1. The following are the insurance to be taken by the contractors

- i) The Contractor shall, for the duration of this Agreement, take out and maintain, own cost, but on terms and conditions approved by the Authority, insurance against the risks, and for the coverages, as specified in the Agreement and in accordance with good industry practice.
- ii) Within 15 (fifteen) days of receiving any insurance policy certificate in respect of insurances required to be obtained and maintained under this clause, the Contractor shall furnish to the Authority, copies of such policy certificates, copies of the insurance policies and evidence that the insurance premium have been paid in respect of such insurance. No insurance shall be cancelled, modified or allowed to expire or lapse during the term of this Agreement.
- iii) If the Contractor fails to effect and keep in force the aforesaid insurances for which it is responsible pursuant hereto, the Authority shall, apart from having other recourse available under this Agreement, have the option, without prejudice to the obligations of the Contractors, to take out the aforesaid insurance, to keep in force any such insurances, and pay such premium and recover the costs thereof from the Contractors, and the Contractor shall be liable to pay such amounts on demand by the Authority.
- iv) Except in case of Third Party liabilities, the insurance policies so procured shall mention the Authority as the beneficiary of the Contractor and the Contractor shall procure an undertaking from the insurance company to this effect; provided that in the event the Contractor has a general insurance policy that covers the risks specified in this Agreement and the amount of insurance cover is equivalent to 3 (three) times the cover required hereunder, such insurance policy may not mention the Authority as the sole beneficiary of the Contractor or require an

undertaking to that effect.

3.5.2. The Parties agree that the risks and coverages shall include but not be limited to the following:

- i) Third Party liability insurance as required under Applicable Laws, with a minimum coverage equivalent to Agreement Value;
- ii) Employer's liability and workers' compensation insurance in respect of the Personnel of the Contractor and of any Sub-Contractors, in accordance with Applicable Laws; and
- iii) Professional liability insurance for an amount no less than the Agreement Value.

The indemnity limit in terms of "Any One Accident" (AOA) and "Aggregate limit on the policy period" (AOP) should not be less than the amount stated in Clause 6.1.1 of the Agreement. In case of consortium, the policy should be in the name of Lead Member and not in the name of individual Members of the consortium.

3.6. Contractors' actions requiring the Authority's prior approval

The Contractor shall obtain the Authority's prior approval in writing before taking any of the following actions:

- i) Appointing such members of the Professional Personnel as are not listed in Annexure II: Deployment of Key Personnel (*Form 8-Section-d of Volume I part II of the RFP*)
- ii) any other action that is specified in this Agreement

3.7. Reporting obligations

The Contractor shall submit to the Authority the reports and documents specified in the Agreement, in the form, in the numbers and within the time periods set forth therein.

3.8. Documents prepared by the Contractor to be property of the Authority

- 3.8.1. All documents prepared by the contractor (referred to as "**Contractor Documents**") in performing the Services shall become and remain the property of the Authority, and all intellectual property rights in such Contractor Documents shall vest with the Authority. Any Contractor Document, of which the ownership or the intellectual property rights do not vest with the Authority under law, shall automatically stand assigned to the Authority as and when such Contractor Document is created.
- 3.8.2. The Contractor shall, not later than termination or expiration of this Agreement, deliver all Contractor Documents to the Authority, together with a detailed inventory thereof. The Contractor may retain a copy of such Contractor Documents.
- 3.8.3. The Contractor shall hold the Authority harmless and indemnified for any losses, claims, damages, expenses (including all legal expenses), awards, penalties or injuries (collectively referred to as '**Claims**') which may arise from or due to any unauthorized use of such Contractor Documents.

3.8.4. TNPCB shall hold the Contractor indemnified for any Claims which may arise from or due to unintended use of such Contractor Documents.

3.9. Equipment and materials furnished by the Authority

Equipment and materials made available to the Contractor by the Authority shall be the property of the Authority and shall be marked accordingly. Upon termination or expiration of this Agreement, the Contractor shall furnish forthwith to the Authority, an inventory of such equipment and materials and shall dispose of such equipment and materials in accordance with the instructions of the Authority.

3.10. Providing access to Project Office and Personnel

The Contractor shall ensure that the Authority, and officials of the Authority having authority from the Authority, are provided unrestricted access to the Project Office and to all Personnel during office hours. The Authority's official, who has been authorized by the Authority in this behalf, shall have the right to inspect the Services in progress, interact with Personnel of the Contractor and verify the records relating to the Services for his satisfaction.

4. CONTRACTORS'S PERSONNEL

4.1. General

The Contractor shall employ and provide such qualified and experienced Personnel as may be required to carry out the Services

4.2. Deployment of Personnel

4.2.1. The designations, names and the estimated periods of engagement in carrying out the Services by each of the contractor's Personnel are described in Annexure II: Deployment of Key Personnel(*Form 8-Section-d of Volume I part II of the RFP*) of this Agreement.

4.2.2. If additional work is required beyond the scope of the Services specified in the Terms of Reference, the estimated periods of engagement of Personnel, set forth in the Annexes of the Agreement may be increased by agreement in writing between the Authority and the Contractors, provided that any such increase shall not, except as otherwise agreed, cause payments under this Agreement to exceed the Agreement Value set forth in Clause 6.1.1

4.3. Approval of Personnel

If the Contractor hereafter proposes to engage any person as Professional Personnel, it shall submit to the Authority its proposal along with a CV of such person in the form provided at Annexure II: Deployment of Key Personnel (*Form 8-Section-d of Volume I part II of the RFP*). The Authority may approve or reject such proposal within 14 (fourteen) days of receipt thereof. In case the proposal is rejected, the Contractor may propose an alternative person for the Authority's consideration. In the event the Authority does not reject a proposal within 14 (fourteen) days of the date of receipt thereof under this Clause 4.3, it shall be deemed to have been approved by the

Authority.

4.4. Substitution of key personnel

The Authority expects all the Key Personnel specified in the Proposal to be available during implementation of the Agreement. The Authority will not consider any substitution of Key Personnel except under compelling circumstances. Such substitution shall be subject to an equally or better qualified and experienced personnel being provided to the satisfaction of the Authority.

5. OBLIGATIONS OF THE AUTHORITY

5.1. Assistance in clearances, etc.,

Unless otherwise specified in the Agreement, the Authority shall make best efforts to ensure that the Government shall:

- i) provide the Contractors and Personnel with work permits and such other documents as may be necessary to enable the Contractors or Personnel to perform the Services;
- ii) Issue to officials, agents and representatives of the Government all such instructions as may be necessary or appropriate for the prompt and effective implementation of the Services.

5.2. Access to land and property

The Authority warrants that the Contractor shall have, free of charge, unimpeded access to the site of the project in respect of which access is required for the performance of Services; provided that if such access shall not be made available to the contractor as and when so required, the Parties shall agree on the time extension, as may be appropriate, for the performance of Services.

5.3. Payment

In consideration of the Services performed by the Contractor under this Agreement, the Authority shall make to the Contractor such payments and in such manner as is provided in Clause 6 of this Agreement

6. PAYMENT TO THE CONTRACTORS

6.1. Cost estimates and Agreement value

- 6.1.1. An abstract of the cost of the Services payable to the Contractor as mentioned in the LoA (Letter of Award).
- 6.1.2. Additional payments shall NOT be made to the Contractor in order to cover any additional expenditures not envisaged in the cost estimates referred to in Clause 6.1.1 above the Agreement Value.

6.1.3. Payment release terms

Completion of Phase 1: Capping existing HW Stockpile – 20%

Completion of Phase 2: Capping new Stockpile area – 20%

Completion of Phase 3: Construction of drainage work – 15%

Completion of Phase 4: Construction of collection pond – 15%

Completion of Phase 5: Construction of boundary wall, Access Pathway and cut – off trench – 15%

Submission of completion report – 15%

6.2. Currency of Payments

All payments shall be made in Indian Rupees.

6.3. Mode of billing and payment

6.3.1. The Contractor shall be paid for its services as per the Payment Schedule at Annexure III: Payment Schedule of this Agreement.

6.3.2. The Authority shall cause the payment due to the Contractor to be made within 30 (thirty) days after the receipt by the Authority of duly completed bills with necessary particulars (the “Due Date”).

6.3.3. Any amount which the Authority has paid or caused to be paid in excess of the amounts actually payable in accordance with the provisions of this Agreement shall be reimbursed by the Contractor to the Authority within 30 (thirty) days after receipt by the Contractor of notice thereof. Any delay by the Contractor in reimbursement by the due date shall attract simple interest @ 10% (ten per cent) per annum

6.3.4. All payments under this Agreement shall be made to the account of the contractor as maybe notified to the Authority by the Contractors.

7. FAIRNESS AND GOOD FAITH

7.1. Good Faith

The Parties undertake to act in good faith with respect to each other’s rights under this Agreement and to adopt all reasonable measures to ensure the realization of the objectives of this Agreement.

7.2. Operations of the Agreement

The Parties recognize that it is impractical in this Agreement to provide for every contingency which may arise during the life of the Agreement, and the Parties hereby agree that it is their intention that this Agreement shall operate fairly as between them, and without detriment to the interest of either of them, and that, if during the term of

this Agreement either Party believes that this Agreement is operating unfairly, the Parties will use their best efforts to agree on such action as may be necessary to remove the cause or causes of such unfairness, but failure to agree on any action pursuant to this Clause shall not give rise to a dispute subject to arbitration in accordance with Clause 8.4 hereof.

8. SETTLEMENT OF DISPUTES

8.1. Amicable settlement

The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement or the interpretation thereof.

8.2. Dispute resolution

8.2.1. Any dispute, difference or controversy of whatever nature howsoever arising under or out of or in relation to this Agreement (including its interpretation) between the Parties, and so notified in writing by either Party to the other Party (the “**Dispute**”) shall, in the first instance, be attempted to be resolved amicably in accordance with the conciliation procedure set forth in Clause 8.3

8.2.2. The Parties agree to use their best efforts for resolving all Disputes arising under or in respect of this Agreement promptly, equitably and in good faith, and further agree to provide each other with reasonable access during normal business hours to all non-privileged records, information and data pertaining to any Dispute.

8.3. Conciliation

In the event of any Dispute between the Parties, either Party may call upon Chairman, Tamil Nadu Pollution Control Board and the Chairman of the Board of Directors of the Contractor or a substitute thereof for amicable settlement, and upon such reference, the said persons shall meet no later than 10 (ten) days from the date of reference to discuss and attempt to amicably resolve the Dispute. If such meeting does not take place within the 10 (ten) day period or the Dispute is not amicably settled within 15 (fifteen) days of the meeting or the Dispute is not resolved as evidenced by the signing of written terms of settlement within 30 (thirty) days of the notice in writing or such longer period as may be mutually agreed by the Parties, either Party may refer the Dispute to arbitration in accordance with the provisions of Clause D)8.4

8.4. Arbitration

8.4.1. Any Dispute which is not resolved amicably by conciliation, as provided in Clause 8.3, shall be finally decided by reference to arbitration by an Arbitral Tribunal appointed in accordance with Clause 8.4.2. Such arbitration shall be held in accordance with the Rules of Arbitration of the International Centre for Alternative Dispute Resolution, New Delhi (the “**Rules**”), or such other rules as may be mutually agreed by the Parties, and shall be subject to the provisions of the Arbitration and Conciliation Act, 1996. The place of such arbitration shall be the capital of the State where the Authority has its headquarters and the language of arbitration proceedings shall be English.

- 8.4.2. There shall be an Arbitral Tribunal of three arbitrators, of whom each Party shall select one, and the third arbitrator shall be appointed by the two arbitrators so selected, and in the event of disagreement between the two arbitrators, the appointment shall be made in accordance with the Rules.
- 8.4.3. The arbitrators shall make a reasoned award (the “**Award**”). Any Award made in any arbitration held pursuant to this Clause 8 shall be final and binding on the Parties as from the date it is made, and the Contractor and the Authority agree and undertake to carry out such Award without delay.
- 8.4.4. The Contractor and the Authority agree that an Award may be enforced against the Contractor and/or the Authority, as the case may be.
- 8.4.5. This Agreement and the rights and obligations of the Parties shall remain in full force and effect, pending the Award in any arbitration proceedings hereunder

IN WITNESS WHEREOF, the Parties hereto have caused this Agreement to be signed in their respective names as of the day and year first above written.

SIGNED, SEALED AND DELIVERED

SIGNED, SEALED AND DELIVERED

For and on behalf of

for and on behalf of

Contractors:

Authority:

Signature:

Signature:

Name:

Name:

Designation:

Designation:

Address:

Address:

In the presence of:

1)

2)

Annexure I: Terms of Reference

Background

M/s. Tamil Nadu Chromates and Chemicals Limited (TCCL) is an identified chromium contaminated site at Ranipet. During its operational period 1975 – 1995, chromium bearing process waste of 2.27 Lakh Tons was dumped in the open backyard in an area of about 2 hectares. The leachate generation is causing chrome contamination of land, ground water in that area.

TNPCB has decided to implement the interim remedial measure of capping the site as suggested in the DPR submitted by M/s ERM India Pvt Ltd to stop further contamination since current & previous management is not traceable.

The Hon'ble NGT (SZ) in OA. No. 186 of 2020 has suo moto taken the matter of M/s. TCCL, Ranipet and passed its final order dated: 02.09.2022 wherein directed TNPCB to implement the interim remedial measure within a period of one year.

Scope of services

The work involves Capping of existing and new hazardous waste stockpiles, construction of a storm water drainage network, a rainwater collection pond, and a new boundary wall with a cut-off trench.

The detailed scope under each component is as follows:

Phase 1 - Capping the Existing Hazardous Waste Stockpile

Phase 2 - Capping New Stockpile Area

Phase 3 - Construction of Drainage Network

Phase 4 - Construction of A Collection Pond for Rainwater Collection

Phase 5 - Construction of New Boundary Wall, Access Pathway and Cut-off Trench
(Detailed Scope of the Project explained in Volume II Section -2)

Reporting

The Contractor will work under the head of the Department assigned in TNPCB under the direct supervision of the Member Secretary, TNPCB, or a person appointed by them.

Annexure II: Deployment of Key Personnel

(Reproduce as per Form-8 –Section d of RFP)

d. Team Composition, Task Assignment and Level of Effort (CV to be included)

S.No	Name of Staff & firm associated with	Area of expertise	Designation	Assigned tasks

Annexure III: Payment Schedule

(as per 6.1.2)

Payment release terms

Completion of Phase 1: Capping existing HW Stockpile – 20%

Completion of Phase 2: Capping new Stockpile area – 20%

Completion of Phase 3: Construction of drainage work – 15%

Completion of Phase 4: Construction of collection pond – 15%

Completion of Phase 5: Construction of boundary wall, Access Pathway and cut – off trench – 15%

Submission of completion report – 15%

PART II - DOCUMENTS TO BE SUPPLIED BY THE TENDERER

VOLUME 1 - CONDITIONS OF TENDER

Part II: Documents to be supplied by the Tenderer:

The first part - Technical Proposal should contain the scanned copy of the following documents:

Pre-qualification documents: The bidder is to submit work order clearly mentioning the value of work, nature of work along with completion certificate, turnover, incorporation certificate, GST registration certificate, PAN.

Technical Proposal

- a) Form 1: Letter of Tenderer
- b) Form 2: Declaration of Eligibility
- c) Form 3: Site Inspection Declaration
- d) Form 4: General information about the tenderer
- e) Form 5: Power of attorney of tenderer
- f) Form 6: Financial statement of tenderer
- g) Form 7: Organisation chart & Legal capacity
- h) Form 8: Description of approach, methodology, and work plan
- i) Form 9: Work Schedule
- j) Form 10: Team Composition, Task Assignment and Level of Effort
- k) Form 11: Eligible Assignments of Bidder

The Second Part - Financial Proposal should contain the contents as below:
Financial Proposal

- a. Form 12: Letter of Proposal
- b. Form 13-Financial Proposal Submission Form (To be submitted in BOQ format provided in e- tenders website only)

Form-1 LETTER OF TENDERER

Name of contract: **Execution of interim remediation at the chrome contaminated site at M/s TCCL, Ranipet, Tamil Nadu.**

**To: The Member Secretary, TNPCB,
76, Mount Salai, Guindy, Chennai – 600 032.
Ph. No: 044 - 22353145**

We have examined the Conditions of Contract; Specifications, Drawings, Bill of Quantities, the other Schedules, the attached Appendix and Addenda Nos. for the execution of the above-named Works. We offer to design, execute and complete the Works and remedy any defects therein in conformity with this tender, which includes these documents, for the sum given in the price bid or such other sum as may be determined in accordance with the Conditions of Contract.

We agree to abide by this tender until _____ (*date*) and it shall remain binding upon us and may be accepted at any time before that date. We acknowledge that the Appendix forms part of this Letter of tender.

If this offer is accepted, we will provide the specified Security Deposit, commence the works as soon as is reasonably practicable after the commencement date, and complete the Works in accordance with the above-named documents within the time specified for completion.

Unless and until a formal Agreement is prepared and executed this Letter of tender, together with your written acceptance thereof, shall constitute a binding contract between us.

We understand that you are not bound to accept the lowest or any tender you may receive.

Signature _____ in the capacity of _____

Duly authorised to sign tenders for and on behalf of _____

Address _____

Date: _____

Form -2 DECLARATION OF ELIGIBILITY

Name of contract: Execution of interim remediation at the chrome contaminated site at M/s TCCL, Ranipet, Tamil Nadu.

Name and address of Employer The Member Secretary, TNPCB,
76, Mount Salai, Guindy, Chennai – 600 032.
Ph. No: 044 - 22353145

_____(Together with successors and assigns, all as defined in the Contract as the Employer)

Name and address of Tenderer: _____

Whereas the tenderer declares in accordance of Instructions to tenderers; Conditions of tender; that all eligibility criteria set forth in tender document and Instructions to Tenderers; Conditions of Tender are met, namely that:

(i) None of the following applies to us, that:

- a) We are bankrupt;
- b) Payments to us have been suspended in accordance with the judgement of a court or a judgement declaring bankruptcy and resulting, in accordance with our national laws, in total or partial loss of the right to administer and dispose of our property;
- c) Legal proceedings have been instituted against us involving an order suspending payments and which may result, in accordance with our national laws, in declaration of bankruptcy or in any other situation entailing the total or partial loss of the right to administer and dispose of our property;
- d) We are guilty of serious misrepresentation with regard to information required for participation in an invitation to tender;
- e) We are in breach of contract on another contract with the Employer and/or the State of Tamil Nadu;
- f) We have any connection with a firm or entity which has provided consulting services during the preparatory stages of the works or of the project of which the works form a part, or which has been hired (or is intended to be hired) as the Employer's Representative for the Contract.

(ii) The above criteria also apply to our Sub-Contractors and Suppliers.

(iii) In particular we meet and conform to the following criteria:

1. We have Permanent Account Number (PAN) and GST registration (GSTIN) valid on date of bid submission.
2. We have minimum average annual turnover of INR..... in the last 3 financial years as on 31.03.2025
3. We have successfully completed Number of project of secured land fill for hazardous waste with minimum capacity of 10000 m³.
4. We have completed Number of projects in environmental related works such as _____ worth a minimum of Rs 50 Crores.
5. We have executed number of works in construction of secured land fill / scientific closure of secured land fill each of value in the last 7 years.
6. We are not a part of any Consortium or Joint-Venture.
7. We have not been black listed/debarred by any Government organization.
8. Our company is a registered company (under Indian Companies Act) operating in INDIA for a period of more than 5 years.
9. We are filing income tax returns for the following years (last 5 years)

Signature(s) for and on behalf of the Tenderer _____

Date _____

Form -3 SITE INSPECTION DECLARATION

Name of contract: **Execution of interim remediation at the chrome contaminated site at M/s TCCL, Ranipet, Tamil Nadu.**

Name and address of Employer **THE MEMBER SECRETARY, TNPCB
76, Mount Salai, Guindy, Chennai – 600 032.
Ph. No: 044 - 22353145**

_____ (Together with successors and assigns, all as defined in the Contract as the Employer)

Name and address of Tenderer: _____

We/I as tenderer declare that; we have inspected the Site on _____ (date)
Represented by Messrs

_____ (Names and addresses representatives)

We have familiarised ourselves, so far as practicable, with the local conditions in general, and the situation with regard to the supply of local materials and electricity, gas, water and other services, and we fully understand the nature and scope of the Works.

We have also acquainted ourselves with the conditions in the State of Tamil Nadu, and the access and transport routes and facilities to the Sites

Signature(s) for and on behalf of the Tenderer _____

Date _____

Form – 4: GENERAL INFORMATION ABOUT THE TENDERER

- i - Name of Company
.....
- ii - Registered Address
.....
.....
Tel.
E-mail
Mobile no. of contact person.....
- iii - Name of Principals/Directors & Associates
(Nationality to be stated in the case of personnel not a citizen of India)
.....
.....
.....
.....
- iv - Type of Company
(Individual, Partnership, Corporation, etc.)
.....
- v - Company's Original Nationality
- vi - Number of Years in Business as Contractor
- vii - Registration Particulars
.....
(Please attach copy of the Registration Certificate)
- | viii - Participation in the Company | Share (%) |
|-------------------------------------|-----------|
| | |
| | |
| | |
- ix - Name(s) and address(es) of associated Company(ies) to be involved in the project and whether
parent/subsidiary/other:
.....
.....
.....

Initials of signatory of Tenderer: _____

1.	Name of the Bidder	
2.	Legal status (Proprietorship/ partnership/etc.)	
3.	Country of incorporation	
4.	Registered address of Bidder	
5.	Year of incorporation	
6.	Year of commencement of business	
7.	Principal place of Business	
8.	Name of the authorized signatory	
9.	Designation of authorized signatory	
10.	Address of authorized signatory	
11.	Phone no:	
12.	Email address	
13.	Details of other firms in the consortia (Name, legal status, country of incorporation, registered address, principal place of business)	
14.	In case of non-Indian firm, does the firm have business presence in India	Yes/ No
15.	Has the Bidder or any member of the consortium been blacklisted by any Government department/Public Sector Undertaking in the last five years	
16.	Has the bidder filed income tax for the past 5 years.	
	(signature, name, designation of the authorised signatory) For and on behalf of 	

Form- 5: POWER OF ATTORNEY OF TENDERER

Know all men by these presents, we (Name of the firm and address of the registered office) do hereby irrevocably constitute, nominate, appoint and authorise Mr/ Ms (name), son/daughter/wife of and presently residing at, who is presently employed with us/ retained by us and holding the position of, as our true and lawful attorney (hereinafter referred to as the "Authorised Signatory") to do in our name and on our behalf, all such acts, deeds and things as are necessary or required in connection with or incidental to submission of our proposal for qualification and submission of our proposal for engaging Independent Engineer but not limited to signing and submission of all applications, bids and other documents and writings, participate in Pre-proposal and other conferences and providing information/ responses to the /TNPCCB representing us in all matters before the /TNPCCB, signing and execution of all documents including the Contract and undertakings consequent to acceptance of our bid, and generally dealing with the /TNPCCB in all matters in connection with or relating to or arising out of our bid for the said Project and/ or upon award thereof to us and/or till the entering into of the Contract.

AND we hereby agree to ratify and confirm and do hereby ratify and confirm all acts, deeds and things done or caused to be done by our said Authorised representative pursuant to and in exercise of the powers conferred by this Power of Attorney and that all acts, deeds and things done by our said Authorised representative in exercise of the powers hereby conferred shall and shall always be deemed to have been done by us.

IN WITNESS WHERE OF WE....., THE ABOVE NAMED PRINCIPAL HAVE EXECUTED THIS POWER OF ATTORNEY ON THIS DAY OF 20.....

For.....
(Signature, name, designation and address)

Witnesses:

1.

2.

Accepted

...

Signature)

(Name, Title and Address of the Attorney)

(Notarised)

Form- 6 : FINANCIAL STATEMENT OF TENDERER

i - Basic Capital:
 Authorised:
 Issued:

ii - Annual value of works undertaken for each of the last five years (as on 31 March)

	2020-2021	2021-2022	2022-2023	2023-2024	2024-2025	TOTAL
Home						
Abroad						
Total						

iii - Approximate value of Work in hand (home + abroad)

iv - Please attach copies of the Company's previous three years certified statements of account (with translation in English) from which the following basic data shall be abstracted:

Financial Year	2022-2023	2023-2024	2024-2025
1. Total Assets			
2. Total Liabilities			
<i>Net Value (1-2)</i>			
4. Current Assets			
5. Current Liabilities			
<i>Circulating Capital (4-5)</i>			
6. Profit (before taxation)			
7. Loss			

v - Name and address of Bankers:

.....

vi - Please enclose a reference/certificate about the financial situation of the Company and access to credit facilities (maximum amount of credit facility to be stated):
 (Equivalent of Indian Rupees).

vii - Please provide financial projections for the forthcoming 2 years.

Initials of signatory of Tenderer: _____

S No	Financial Year	Annual Revenue from Advisory and Consultancy services (Rs)
1		
2		
3		
Average		

Certificate from the Statutory Auditor*

This is to certify that.....
 (Name of the Bidder)
 has received the payments shown above against the respective years on account of professional fees from advisory and consulting services.

Signature, name, designation of the authorised signatory

Date: _____ Name and seal of the firm _____

* In case the Bidder does not have a statutory auditor, it shall provide the certificate from its chartered accountant that ordinarily audits the annual accounts of the Bidder.

Form 7: Organization chart & Statement of legal capacity

(To be forwarded on the letterhead of the Bidder)

Reference Date:

To

.....

.....

.....

Dear Sir,

Sub:

I/We have agreed that.....(*insert individual's name*) will act as our Authorised Representative/ on our behalf and has been duly authorized to submit our Proposal. Further, the authorised signatory is vested with requisite powers to furnish such proposal and all other documents, information or communication and authenticate the same.

Yours faithfully,

(Signature, name, designation of the authorised signatory)

For and on behalf of

.....

Form 8: Description of approach, methodology, and work plan

A description of the approach, methodology and work plan for performing the assignment, including a detailed description of the proposed methodology and staffing.

Suggested structure:

- a. **Technical Approach and Methodology:** Explain understanding of the objectives of the assignment as outlined in the Terms of Reference (TORs), the technical approach, and the methodology for implementing the tasks to deliver the expected output(s), and the degree of detail of such output. Please do not repeat/copy the TORs in here.
- b. **Work Plan:** Outline the plan for the implementation of the main activities/tasks of the assignment, their content and duration, phasing and interrelations and tentative delivery dates of the monthly progress reports. The proposed work plan should be consistent with the technical approach and methodology, showing understanding of the TOR and ability to translate them into a feasible working plan. A list of the final documents (including reports) to be delivered as final output(s) should be included here.
- c. **Organization and Staffing:** Describe the structure and composition of your team, including the list of the Key Personnel, Sector Experts, Support Team and relevant technical and administrative support staff.
- d. **Team Composition, Task Assignment and Level of Effort (CV to be included)**

S.No	Name of Staff & firm associated with	Area of expertise	Designation	Assigned tasks

Form 9: Work Schedule

S No	Activity	Months												
		1	2	3	4	5	6	7	8	9	10	11	12	n
1														
2														
3														
4														
5														
n														

- 1 **Indicate all main activities of the assignment, including delivery of reports during study period and construction period, and other benchmarks such as PMC approvals. For phased assignments indicate activities, delivery of reports, and benchmarks separately for each phase.**
- 2 **Duration of activities shall be indicated in the form of a bar chart.**

Form :10 Abstract of relevant assignments of the Bidder

S No	Name of Project	Name of the Client
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

The Bidder should provide details of only those assignments that have been undertaken by it under its own name.

Form 11: Eligible Assignments of Bidder

S. No	Project Name	Client's Name and address	Total Contract Value in INR Crores	Project Start date	Date of completion	Project Duration	Main features of the Project (Description of the Scope of work)	Client Contact Information (Name, Phone No. & Email ID)
1								
2								
3								
4								
5								

Note:

1. A copy of contract/ engagement letter/LOA/ Completion certificate from client shall be submitted as documentary evidence for the experience on project delivered for Government, local bodies, multi-lateral agencies, etc.

FORM 12: Financial proposal submission form

[Location, Date]

To: [Name and address of Authority]

Dear Sirs,

We, the undersigned, offer to provide the contract services for [Insert title of assignment] in accordance with your Request for Proposal dated [Insert Date] and our Technical Proposal. Our attached Financial Proposal is given.

Our Financial Proposal shall be binding upon us subject to the modifications resulting from Contract negotiations, up to expiration of the validity period of the Proposal, i.e. before the date indicated in the RFP Document.

We hereby certify that we have taken steps to ensure that no person acting for us or on our behalf will engage in bribery.

We undertake that, in competing for (and, if the award is made to us, in executing) the above contract, we will strictly observe the laws against fraud and corruption in force in India namely "Prevention of Corruption Act, 1988."

We understand you are not bound to accept any Proposal you receive.

We remain, Yours

sincerely,

Authorized Signature [In full and initials]: _____

Name and Title of Signatory: _____

Name of Firm: _____

Address: _____

*Amounts must coincide with the ones indicated under Total Cost of Financial proposal. The Bidder shall provide the financial bid only through online mode.

TENDER DOCUMENT
(National Competitive Bidding)
For
Tender no. TNPCB/TCCL/2025

**Request for Proposal (RFP) for Execution of Interim
Remediation at the Chrome Contaminated site at M/s TCCL,
Ranipet, Tamil Nadu.**

Volume-II, Section I
Civil Works – General Specifications



TAMIL NADU POLLUTION CONTROL BOARD
76, Mount Salai, Guindy, Chennai – 600 032

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CIVIL WORKS - GENERAL SPECIFICATIONS

1.1 GEO-TECHNICAL SURVEY

A detailed geo-technical survey of the site has to be carried out before finalising the structural design for construction.

1.2 GENERAL GUIDELINES AND DESIGN CRITERIA

- All RCC designs of the foundations and constructions shall be conform to IS 456, IS 800 and IS 3370.
- Minimum concrete grade used in foundation is of M25 grade and for water retaining structure is of design mix M30.
- Minimum cement content for foundation should be of 300 kg/m³ and for other RCC works should be provided with 320 kg/m³ of cement.
- Reinforcement used in RCC structure should have minimum characteristic strength of 500N/mm². The TMT bar should be coated with fusion bonded epoxy. Preferred powder band: Valspar, Jotun, 3M. The density of steel should be as standards.
- Nominal Clear Cover all RCC structure are provided with a clear cover of 75-mm.
- Levelling concrete for foundations of load bearing brick masonry walls shall be of 150 mm thick PCC M20 1:1.5:3 with 40 mm down size graded stone aggregate.
- All tanks resting on ground shall be provided with minimum 100 mm thick PCC 1:1.5:3 with 40 mm size aggregates levelling concrete below the base slab.
- Levelling concrete below RCC footings shall be of PCC 1:1.5:3 using 40 mm down size aggregates.
- Load bearing brick masonry walls shall be provided with damp proof course and Admixture of CM 1:4, 25 mm thick.
- Water retaining structures are added with water proofing admixtures while concreting.
- Cube sampling has to done at site as per IS 456.
- Test for materials specified in the Specifications should be carried out at site and report for all construction materials have to be submitted to TNPCB.
- All the works shall be carried out after approval from the Engineer in charge of TNPCB.
- All water retaining structures should pass leak test.
- Wherever necessary construction joints are to be provided as per the specifications.
- Preferred MoC of all Shuttering works shall be Plywood.
- For concrete more than 10 m³, ready mix concrete of specified grade has to be adopted.

Preferred Make:

Epoxy coating for TMT bars	Valspar, Jotun, 3M.
For Cement	Ultratech,/Zuari/Coromandel/Sankar/Ramco
TMT bar	Tata tison/isteel/SAIL/jindal steel

1.3 DEMOLISHING

1. Demolition of All Existing Buildings on Site:

Demolition should be systematically planned and executed in reverse order of construction to minimize damage and nuisance to adjacent structures and the public. Necessary measures, such as propping, shoring, and underpinning, must be implemented to ensure the safety of adjoining properties before commencing demolition.

2. Debris Size and Uniformity:

Materials prone to damage from height drops should be carefully removed first. Dismantled items must be manually lowered to the ground and not thrown, ensuring proper stacking as directed by the Engineer-in-Charge. This approach aids in achieving debris that is no more than half an inch in size and fairly uniform.

3. Scraping of Sluffed Waste Up to 2 m Depth:

While the CPWD specifications do not explicitly detail the scraping of sluffed waste up to a 2-meter depth, such activities should be conducted under the supervision of the Engineer-in-Charge, adhering to standard safety and operational protocols.

4. Controlling Fugitive Emissions During Demolition:

To minimize dust emissions, water may be used during the removal of plaster from brickwork. Additionally, screens should be erected where necessary to prevent injuries and contain debris. All demolition work must be performed under the guidance of the Engineer-in-Charge, utilizing appropriate safety gear such as helmets, goggles, and safety belts.

For comprehensive details and any updates, please refer to the latest CPWD Works Manual and Specifications.

CIVIL WORKS

1.3.1 Tolerances

Tolerance is a specified permissible variation from lines, grade or dimensions given in approved drawings. No tolerance specified for horizontal or vertical building lines or footings shall be constructed to permit encroachment beyond the legal boundaries. Unless otherwise specified, the following tolerances will be permitted:

Tolerances for R.C.C. Buildings

- (i) Variation from the plump -
In the lines and surfaces of columns, piers, walls and in arises: 5mm per 2.5m or 25mm whichever is less.
For exposed corner columns and other conspicuous lines -

- | | |
|---------------------------|-------|
| In any bay or 5m maximum: | 5 mm |
| In 10m or more: | 10 mm |
- (ii) Variation from the level or from the grades indicated on approved drawings –
 In slab soffits, ceilings, beam soffit, and in arises -

In 2.5m	5 mm
In any bay or 5m maximum:	10 mm
In 10m or more	15 mm

 For exposed lintels, sills, parapets, horizontal grooves and other conspicuous lines -

In any bay or 5m maximum:	5 mm
In 10 m or more:	10 mm

 (iii) Variation of the linear building lines from established position in plan and related position of columns, wall and partitions -

In any bay or 5m maximum:	10 mm
In 10m or more:	20 mm

 (iv) Variation in the sizes and location of sleeves, openings in walls and floors - 5mm except in the case of and for anchor bolts.

 (v) Variation in cross-sectional dimensions of columns and beams and in the thickness of slabs and walls -

Minus	5 mm
Plus	10 mm

 (vi) Footings -
 Variation in dimension in plan:

Minus	5 mm
Plus	50 mm

 Misplacement or eccentricity - 2% of footing width in the direction of misplacement but not more than 50mm. Reduction in thickness - minus (-) 5% of specified thickness subject to a maximum of 50mm
 (v) Variation in steps -
 In a flight of stairs -

Rise	3 mm
Tread	5 mm

 In consecutive steps -

Rise	1.5 mm
Tread	3.0 mm

Tolerances in other Concrete Structures

- (i) All Structures –

Variation of the Constructed linear outline from established position in plan –

In 5m	10 mm
In 10m or more ¹	5 mm

Variations of dimensions to individual structural features from established positions-

In 20m more	25 mm
In buried construction	50 mm

Variation from plumb, from specified batter or from curved surfaces of all structures

In 2.5 m	10 mm
In 5m	15 mm
In 10m or more	25 mm

In buried construction - twice the above values

Variation from level or grade indicated on approved drawings in slab, beams, soffits, horizontal grooves and visible arises -

In 2.5 m	5 mm
In 7.5 m or more	10 mm

In buried construction - twice the above values

Variation in cross-sectional dimensions of columns, beams, buttresses, piers and similar members -

Minus	5 mm
Plus	10 mm

Variation in the thickness of slabs, walls and sections and similar members -

Minus	5 mm
Plus	10 mm

(ii) Footings for columns, piers, walls buttresses and similar members -

Variation of dimensions in plan -

Minus	10 mm
Plus	50 mm

Misplacement or eccentricity - 2% of footing width in the direction of misplacement but not more than 50 mm.

Reduction in thickness - 5% of specified thickness subject to a maximum of 50 mm

(iii) Tolerances in other types of structures shall generally conform to those given in Clause 3.3 of Recommended Practice for concrete Form work (American Concrete Institute - ACI 347).

(iv) Tolerances in fixing anchor bolts shall be as follows -

- Anchor bolts without sleeves - + 5 mm
- Anchor bolts with sleeves - + 5mm up to 28mm dia. bolts, + 3mm for 32mm dia. bolts.
- Embedded parts- + 5mm in all directions.

1.4 EARTHWORK

(Excavation in trenches for pipes, cables etc. and refilling)

This shall comprise excavation not exceeding 1.5 mts in width and to any depth trenches for

pipes. Cables etc. and returning the excavated material to fill the trenches after pipes, cables etc. are laid and their joints tested and passed and disposal of surplus excavated material up to 50 m lead.

1.4.1 Setting out and making profiles

Prior to starting of earth works the Contractor shall set out the necessary alignment and grade boards for proper control of the works. All earth works shall have to be made to the required grades and sizes as per the drawings, specifications and directions of the Engineer. The line and level of excavations shall be as shown in the Drawings or as may be directed by the Engineer. The Contractor shall maintain the setting-out undisturbed during the course of earth work and subsequent construction, and arrangements shall be provided for reference bench marks / reduced levels so that the levels can be checked and measured as and when necessary. The Contractor shall take and record the levels in a manner agreed by or specified by the Engineer. Two working days' notice is to be given to the Engineer so that he can be present during taking and recording of the levels.

1.4.2 Width of trench

- (a) Up to one metre depth the authorized width of trench for excavation shall be arrived at by adding 25 cm to the external diameter of pipe (not socket/ collar) cable, conduit etc. Where a pipe is laid on concrete bed/ cushioning layer, the authorized width shall be the external diameter of pipe (not socket/ collar) plus 25 cm or the width of concrete bed/ cushioning layer whichever is more.
- (b) For depths exceeding one metre, an allowance of 5 cm per metre of depth for each side of the trench shall be added to the authorized width (that is external diameter of pipe plus 25 cm) for excavation. This allowance shall apply to the entire depth of the trench. In firm soils the sides of the trenches shall be kept vertical upto depth of 2 metres from the bottom. For depths greater than 2 metres, the excavation profiles shall be widened by allowing steps of 50 cm on either side after every two metres from bottom.
- (c) Where more than one pipe, cable, conduit etc, are laid, the diameter shall be reckoned as the horizontal distance from outside to outside of the outermost pipes, cable, conduit etc.
- (d) Where the soil is soft, loose or slushy, width of trench shall be suitably increased or side sloped or the soil shored up as directed by the Engineer-in-Charge. It shall be the responsibility of the contractor to take complete instructions in writing from the Engineer-in-Charge regarding increase in the width of trench. Sloping or shoring to be done for excavation in soft, loose or slushy soils.

1.4.3 Excavation in all kinds of soils

1. All excavation operations manually or by mechanical means shall include excavation and 'getting out' the excavated materials. In case of excavation for trenches, basements, water tanks etc. 'getting out' shall include throwing the excavated materials at a distance of at least one metre or half the depth of excavation, whichever is more, clear off the edge of excavation. In all other cases 'getting out' shall include depositing the excavated materials as specified. The subsequent disposal of the excavated material shall be either stated as a separate item or included with the items of excavation stating lead.
2. During the excavation the natural drainage of the area shall be maintained. Excavation shall be done from top to bottom. Undermining or undercutting shall not be done.
3. In firm soils, the sides of the trenches shall be kept vertical upto a depth of 2 metres from the bottom. For greater depths, the excavation profiles shall be widened by allowing steps of 50 cms on either side after every 2 metres from the bottom. Alternatively, the excavation can be done so as to give slope of 1:4 (1 horizontal: 4 vertical). Where the soil is soft, loose or slushy, the width of steps shall be suitably increased or sides sloped or the soil shored up as directed by the Engineer-in- Charge. It shall be the responsibility of the contractor to take complete instructions in writing from the Engineer-in-Charge regarding the stepping, sloping or shoring to be done for excavation deeper than 2 metres.
4. The excavation shall be done true to levels, slope, shape and pattern indicated by the Engineer-in- Charge. Only the excavation shown on the drawings with additional allowances for centering and shuttering or as required by the Engineer-in-Charge shall be measured and recorded for payment.
5. In case of excavation for foundation in trenches or over areas, the bed of excavation shall be to the correct level or slope and consolidated by watering and ramming. If the excavation for foundation is done to a depth greater than that shown in the drawings or as required by the Engineer-in-Charge, the excess depth shall be made good by the contractor at his own cost with the concrete of the mix used for levelling/ bed concrete for foundations. Soft/defective spots at the bed of the foundations shall be dug out and filled with concrete (to be paid separately) as directed by the Engineer-in-Charge.

1.4.4 Earth work

1. While carrying out the excavation for drain work care shall be taken to cut the side and bottom to the required shape, slope and gradient. The surface shall then be properly dressed. If the excavation is done to a depth greater than that shown on the drawing or as required by the Engineer-in-Charge, the excess depth shall be made good by the contractor at his own cost with stiff clay puddle at places where the drains are required to be pitched and with ordinary earth, properly watered and rammed, where the drains are not required to be pitched. In case the drain is required to be pitched, the back filling with clay puddle, if required, shall be done simultaneously as the pitching work

- proceeds. The brick pitched storm water drains should be avoided as far as possible in filled-up areas and loose soils.
2. In all other cases where the excavation is taken deeper by the contractor, it shall be brought to the required level by the contractor at his own cost by filling in with earth duly watered, consolidated and rammed.
 3. In case the excavation is done wider than that shown on the drawings or as required by the Engineer-in-Charge, additional filling wherever required on the account shall be done by the contractor at his own cost.
 4. The excavation shall be done manually or by mechanical means as directed by Engineer-in-charge considering feasibility, urgency of work, availability of labour /mechanical equipment's and other factors involved. Contractor shall ensure every safety measures for the workers. Neither any deduction will be made nor will any extra payment be made on this account.

1.4.5 Site clearance

All area of the Site marked in the Specification Drawings for clearance or from which material is to be excavated or upon which filling is to be deposited shall be cleared to the extent required by the Engineer of all buildings, walls, gates, fence and other structures and obstructions and of all bushes, hedges, trees, stumps, roots and other vegetation except for trees marked for preservation. Material so cleared shall so far as suitable be preserved and stacked for further use but shall otherwise be disposed off the Site as directed by the Engineer.

1.4.6 Excess excavation to be made good

The Contractor shall, if directed, remove from the Site all material resulting from excess excavation and shall make good the same with such kind of fill material or in such class of concrete as may be reasonably required by the Engineer having regard to the circumstances.

1.4.7 Stripping top soil

Where ordered by the Engineer, topsoil shall be stripped to such depths and over such areas as he may direct, as a separate operation prior to any further excavation which may be required.

1.4.8 Supporting excavations

1. The Contractor shall well and effectually support the sides and ends of all excavations to prevent any fall or run from any portion of the ground outside the excavation and to prevent settlement or damage to structures adjacent to the excavation. Any excavation necessary to provide space for such support or other working space shall be carried out. If, for any reason, any portion of the bottoms, sides or ends of any excavations shall give way, the Contractor shall at his own expense take all necessary remedial measures including the excavation and removal of all the ground thereby distributed.

2. Where the Contractor elects and is permitted by the Engineer to perform excavations with sloping faces (other than sloping excavations shown on the Drawings or required as permanent features of the Works) and without shoring, the excavated faces shall be to stable slopes and heights.

1.4.9 Trimming excavations

1. When excavating to specified or required levels for the foundation of any structure or to specified or required limits for the face of any structure required to abut undisturbed ground, the Contractor shall not excavate the last 12.10 mm until immediately before commencing the constructional work, except where the Engineer shall permit otherwise. Should the Contractor have excavated to within 150 mm above these specified levels or to within 150 mm of these specified limits before he is ready or able to commence the constructional work he shall, where required by the Engineer, excavate further so as to remove not less than 150 mm of material immediately before commencing the constructional work.
2. Before commencement of any constructional work all shattered and loose materials shall be removed from the excavations by hand so as to ensure that the work rests on a solid and perfectly clean foundation or abuts against solid ground.

1.4.10 Inspection by the engineer

1. When the specified levels or limits of excavation are reached the Engineer will inspect the ground exposed, and if he considers that any part of the ground is by its nature unsuitable he may direct the Contractor to excavate further. Such further excavation shall be refilled to the specified levels or limits with concrete, selected excavated material or selected imported material as directed by the Engineer without any extra payment.
2. Should the material forming the bottom of any excavation, while acceptable to the Engineer at the time of his inspection, subsequently become unacceptable to him due to exposure to weather conditions or due to flooding or have puddled, soft or loose during the progress of the works, the Contractor shall remove such damaged, softened or loosened material and excavate further by hand.

1.4.11 disposing excavated material

Excavated material including rock which is not required for or is unsuitable for re-use in the works shall be formed as a new stockpile. Such material shall be transported and deposited at places designated by the Engineer. Material so deposited shall be shaped up or spread and levelled as directed by the Engineer. Any necessary work to provide access to the Engineer's tips or other preliminary work in connection therewith shall be carried out by the Contractor to the approval of the Engineer. The quantity of earth work to be transported depends on the amount of earth work involved based on the design for various units to be submitted by the contractor and hence quantity cannot be indicated.

1. All excavated material at any cause, should not be transported to any location by the contractor for his personal wealth, but disposal of surplus excavated materials including rock should be carted & deposited & levelled by the contractor at the location specified by the Engineer in charge of TNPCB or
2. The Contractor shall ensure that no excavated material which is suitable for and is required for re-use in the Works is transported unless so ordered by the Engineer.
3. The term "excavation" shall be deemed to include for disposing excavated material within the Site in any of the following ways :
 - (i) Back-filling to excavations and completed structures using suitable excavated material and including placing in temporary spoil tips and any double handling required all as specified hereafter,
 - or
 - (ii) Transporting and placing approved excavated material in permanent spoil tips, as directed by the Engineer of TNPCB or ,including the shaping and drainage of such tips all as specified hereafter,

1.4.12 Back-filling, general site grading & sand filling

1. Filling in trenches shall be commenced soon after the joints of pipes, cables; conduits etc. have been tested and passed. The space around the pipes, cables conduits etc. shall be cleared of all debris, brick bats etc. Where the trenches are excavated in hard/ soft soil, the filling shall be done with earth on the side and top of pipes in layers not exceeding 20 cm in depth.
2. Each layer shall be watered, rammed and consolidated. All clods and lumps of earth exceeding 8 cm in any direction shall be broken or removed before the excavated earth is used for filling. In case of excavation trenches in ordinary/ hard rock, the filling upto a depth of 30cm above the crown of pipe, cable, conduits etc. shall be done with fine material like earth, moorum or pulverized/ decomposed rock according to the availability at site.
3. The remaining filling shall be done with boulders of size not exceeding 15cm mixed with fine material like decomposed rock, moorum or earth as available to fill up the voids, watered, rammed and consolidated in layers not exceeding 30cm. Excavated material containing deleterious material, salt peter earth etc. shall not be used for filling. Ramming shall be done with iron rammers where feasible and with blunt ends of crow bars where rammers cannot be used. Special care shall be taken to ensure that no damage is caused to the pipes, Cables, Conduits etc. laid in the trenches.

a. Fill material

- (i) All fill material whether such material is brought from outside borrow areas or

- excavation within the site, will be subject to Engineer's approval. Not with standing any approval given to the fill material or borrow areas from which fill material is proposed to be brought, the Engineer reserves the right to reject such material which in his opinion either does not meet the specification requirements or unsuitable for the purpose for which it is intended.
- (ii) It shall be the Contractor's responsibilities to locate suitable borrow areas for borrowing fill material. Such areas will be inspected by Engineer and approved before Contractor makes arrangements to borrow the fill material. The top soil which may contain vegetation, rubbish, slush, etc. shall not be used. If requested by the Engineer, the Contractor shall arrange to have trial pits of specified dimensions and numbers dug at locations specified, for the Engineer to examine the nature and type of material likely to be obtained from the borrow areas.
 - (iii) The borrowed soil shall be generally granular, and non-cohesive. It shall consist of sand, silty sand, morrum, ordinary soil, gravel and shingle. Dredged material, free from clayey deposit, may be accepted. Fill material shall also be free from sulphates, salts, organic, foreign and other harmful or objectionable materials. Any material rejected by the Engineer shall be removed from the site immediately.
 - (iv) Roads, of a temporary nature, required to be constructed for access and for movement of men, materials, equipment, transport vehicles, vehicles carrying fill material, etc. to or over borrow areas and/or to or over areas on which fill has to be deposited shall be constructed by the Contractor. Such access roads shall be maintained in good condition during all seasons to ensure completion of the work according to the time schedule.

b. Backfilling

- (i) Excavated material used as backfilling to excavations or completed structures shall be free from rubbish, vegetation, clods and lumps and shall be approved by the Engineer. The approved materials shall be placed in layers, not exceeding 225 mm in depth before compaction and shall be compacted to a dry density not less than ninety-five percent of the maximum dry density obtained by the test in Part VII of I.S. 2720 or to such higher density as is specified hereinafter. During compaction the backfill shall have a uniform moisture content equal to or a little above the optimum moisture content recorded in the I.S. Compaction Test. Where necessary the Contractor shall adjust the moisture content of the backfill either by drying out or by adding water. After such drying out or adding of water the backfill shall be thoroughly mixed until the moisture content is uniform.
- (ii) Soft material shall not be used as backfilling around structures in rock. The Contractor shall backfill such excess excavation with concrete, rubble, stone or rock fill as directed by the Engineer. Filling other than concrete shall be placed in layers not exceeding 225 mm in thickness, shall be thoroughly compacted and have adequate fines content to fill the voids.
- (iii) Should the material being placed as backfilling, while acceptable at the time of selection,

- become unacceptable to the Engineer due to exposure to weather conditions or due to flooding or have become puddle, soft or segregated during the progress of the Works, the Contractor shall remove such damaged, softened or segregated material and replaced it with fresh approved material.
- (iv) The Contractor shall when placing the backfilling make due allowance for any settlement that may occur before the end of the Defects Liability Period, remove any excess material or make up any deficiency by backfilling to the specified levels. As a rule, material to be backfilled shall be stacked temporarily within the basic lead of 300 meters unless otherwise directed by the Engineer.
 - (v) Compaction shall be carried out to achieve at least 95% of standard Proctor Dry Density at an optimum moisture content determined in accordance with the relevant I.S. Specification. It shall be ensured however that the minimum compacted dry density is not less than 1.6 T/cum. As the work progresses field density tests shall be conducted on each layer at the rate of one test for every 1000 square meters to check whether the desired compaction has actually been achieved.

c. General site Grading

- (i) Site grading shall be carried out as directed by the Engineer. Excavation shall be carried out as specified in the specification. Filling and compaction shall be carried out as specified under (ii) of this Clause unless otherwise indicated below.
- (ii) The approved material shall be placed in layers not exceeding 225 mm in depth before compaction and shall be compacted to a dry density not less than 95 percent of the maximum dry density obtained by the test in Part VII of IS 2720.
- (iii) To ensure that the fill has been compacted as specified, field and laboratory tests shall be carried out by the Contractor.

- (iv) Field compaction test shall be carried out at different stages of filling and also after the fill to the entire height has been completed. This shall hold good for embankments as well.
- (v) The Contractor shall protect the earth fill from being washed away by rain or damaged in any other way. Should any slip occur, the Contractor shall remove the affected materials and make good the slip.
- (vi) The fill shall be carried out to such dimensions and levels as directed by the Engineer, after the stipulated compaction. The fill will be considered as incomplete if the desired compaction has not been obtained.
- (vii) If specifically permitted by Engineer, compaction can be obtained by allowing loaded trucks conveying fill or other material to ply over the fill area. Even if such a method is permitted, it will be for the Contractor to demonstrate that the desired/specified compaction has been obtained. In order that the fill may be reasonably uniform throughout, the material should be dumped in place in approximately uniform layers. Traffic over the fill shall then be so routed to compact the area uniformly throughout.
- (viii) If so specified, the rock as obtained from the excavation may be used for filling and levelling to indicate grades without further breaking. In such event, filling shall be done in layers not exceeding 50 cms approximately. After rock filling to the approximate level indicated above has been carried out, the void in the rock fill shall be filled with finer materials such as earth, broken stone, etc. and the area flooded so that the finer materials fill up voids. Care shall be taken to ensure that the finer fill material does not get washed out. Over the layer so filled, a 100 mm thick mixed layer of broken material and earth shall be laid and consolidation carried out by a 12 tonne roller. Not less than twelve passes of the roller shall be accepted before subsequent similar operations are taken up.

d. Sand filling below Plinth and other places:

Backfilling shall be carried out with sand at places as directed by the Engineer. The sand used shall be clean, medium grained and free from impurities. The filled-in sand shall be kept flooded with water for 24 hours to ensure maximum consolidation. Any temporary work required to contain sand under flooded conditions shall be to the Contractor's account. The surface of the consolidated sand shall be dressed to the required level or slope. Construction of floors or other structures on sand fill shall not be started until the Engineer has inspected and approved the fill.

1.4.13 Measurements

Trenches for pipes, cables, conduits etc. shall be measured in running metre correct to the nearest cm in stages of 1.5 m depth and described separately as under:

- (a) Pipes, cables, conduits, etc. not exceeding 80 mm dia.
- (b) Pipes, cables, conduits etc. exceeding 80 mm dia but not exceeding 300mm dia.
- (c) Pipes, cables, conduits etc. exceeding 300 mm dia.

Where two or more categories of each work are involved due to different classification of soil within the same stage of trench depth or where the soil is soft loose or slushy requiring increase in the width of trench or sloping sides or shoring, trenches for pipes, cables, conduits, etc. shall be measured in cubic metres as specified in 2.10 (CPWD Manual). Extra excavation, if any, on account of collar/ socket of pipes shall neither be measured nor paid for separately.

1.4.14 Partial back-filling of pipeline trench for testing

1. After laying and jointing of pipes partial backfilling of the trench shall be done leaving the joints exposed. Testing of pipes shall be carried out expeditiously and after successful testing the trench shall be backfilled and road surface reinstated.
2. The Contractor shall strictly comply with Clause 6.1.20 of CPWD specifications Volume 1 to avoid any accident to road users due to exposed trench. Where the pipes are laid across the road complete backfilling shall be done immediately after laying and jointing of pipes pending the pressure test.

1.4.15 Trench excavation in roads

All trench excavation and other work carried out within the limits of any existing road or highway shall be completed as rapidly as possible and, in the case of roads capable of carrying two or more lanes of traffic, not more than one half of the width of the carriageway shall be obstructed at any time. In single lane roads, the Contractor shall programme his work in such a manner that the minimum inconvenience is caused to those persons who have reasonable grounds for using the road and, in the case of private or, restricted roads, which have authority to use them. Road drains and channels shall be kept free from obstructions at all times.

The Engineer shall normally direct that trench excavation along roads be located in footpaths or verges adjacent to the road rather than in the carriageway itself. Trench excavation shall wherever practicable be carried out in such a way that every part of the excavation is at least 0.9 m clear of existing edges of the carriageway. In any event the Contractor shall take special precautions, which shall include the continuous support of the sides of the excavation, from the time that excavation begins until the refilling of the trench, to ensure protection of the adjacent road and its foundation.

Where trench excavation or any other part of the Works obstructs any footpath or right of way, the Contractor shall provide, at his own cost, a temporary footpath around the obstruction to the satisfaction of the Engineer. Where applicable, this temporary footpath shall include stout bridges of wooden planks or other approved construction across any open trenches.

Where excavated material has temporarily been deposited on a grass margin or verge, the margin shall on completion of refill be restored entirely to its original condition and left free

from loose stones.

1.4.16 Trench excavation in surfaces other than roads

Trench excavation in surfaces other than roads shall include all surfaces except those asphalt surfaces that require road reinstatement. These surfaces include but are not limited to fields, paddy fields, pastureland and the like, footpaths, verges, non-asphalted roads, lanes, alleys, and all private lands. Trench excavation shall, if the Engineer so requires, have temporary fencing erected around that length. Temporary fencing shall not be removed without the Engineer's permission, which will not normally be given until the trench excavation has been refilled and reinstated. The Contractor shall have particular regard to the safety of livestock, and shall ensure that all open excavation, access routes and steep or loose slopes arising from the Contractor's operations are adequately fenced and protected. After the erection of temporary fencing where required, the Contractor shall remove topsoil to such depth and over such area as may be necessary to provide sufficient material to ensure adequate surface reinstatement of the working areas occupied by the Contractor for construction of the pipeline.

1.4.17 Refilling of trench excavations

- a) Trench Excavations shall be refilled using suitable materials selected from excavations carried out at site or suitable borrow areas as directed by the Engineer.
- b) Soft Material (free from stones greater than 75 mm in size for pipes without bitumen sheathing and 20 mm in size for pipes with bitumen sheathing) shall be deposited in 150 mm layers and thoroughly rammed with rammers working alternately on either side of the pipe (particular care being taken to avoid damage to the pipe and any sheathing) until the trench has been refilled up to the crown of the pipe, and thereafter until the soft filling has been carried up at least 300 mm above the top of the pipe.
- c) The remainder of the refilling may consist of coarse material (including broken rock from excavation in Hard Rock) free from boulders and clods of earth larger than 150 mm in size provided that the compacted backfill is, in the opinion of the Engineer sufficiently dense to prevent material from the superimposed layers being washed into the voids in such backfill. This coarse material shall be spread in layers of not greater depth than 225 mm and be thoroughly rammed by an approved mechanical rammer. The coarse filling is to be carried up to the level at which (in roads and footpaths) surface reinstatement is to commence or (elsewhere) to such level as with the surface reinstatement of the whole of the topsoil will leave the finished work sufficiently "proud" to allow for future settlement to the original ground level.
- d) Hard material such as broken rock and original road metaling shall normally be used only for the surface reinstatement of roads as specified but where it is suitable and available in sufficient quantity it may be used in place of or as well as the aforesaid coarse material.

- e) Where necessary, the Contractor shall adjust the moisture content of the refill material either by drying out or by adding water to assist the compaction of the material.
- f) Should the material being placed as refilling, while acceptable at the time when approved, become unacceptable to the Engineer due to exposure to weather conditions or due to flooding or have become puddled, soft or segregated during the progress of the Works, the Contractor shall at his own expense remove such damaged, softened or segregated material and replace it with fresh approved material. Where directed by the Engineer trench excavations shall be refilled with concrete.

1.4.18 Transporting equipment

This implies horizontal movement primarily but it can involve some vertical movement too. Transporting Equipment's specified in CPWD Specifications Volume 1 can be deployed for the required works.

1.4.19 Shoring and boxing

1. Shoring and boxing work should be done in cases where loose soil is encountered, masonry structures are close to the excavations, depth of cutting exceeds 1.5 metres, and/or utilities require safeguarding, etc., to avoid side collapse, and should be done simultaneously with the earthwork excavation. Design of the shoring or boxing shall be determined by the site conditions and shall be subject to the approval of the Engineer. The process of boxing and shoring shall be continued until the excavation reaches the required depth of cutting. All boxing and shoring materials such as planks, runners and struts shall be withdrawn from the excavations prior to completion of backfilling.
2. Close timbering shall be done by completely covering the sites of the trenches and pits generally with short, upright members called 'polling boards'. The boards shall generally be placed in position vertically side by side without any gap on each side of the Excavation and shall be secured by horizontal walling of strong wood at maximum 1.2 mtrs., spacing and suitably strutted. If the soil is very soft and loose, the boards shall be placed horizontally against each side of the excavation and supported by vertical walling, which in turn shall be suitably strutted. The lowest boards supporting the sides shall be taken into the ground and no portion of the vertical side of the trench or pit shall remain exposed, so as to render the earth liable to slip out.
3. The shoring material shall not be sizes less than those specified below unless steel sheet piling is used or unless otherwise approved by the Engineer in writing:
 - i. Planks - 5 cm x 25 cm
 - ii. Waling pieces - 10 cm x 20 cm
 - iii. Struts - 15 cm x 20 cm
4. Timber shoring shall be 'close' or 'open' type, depending on the nature of soil and the depth of pit or trench. The type of timbering shall be as approved by Engineer. It shall be the responsibility of the Contractor to take all necessary steps to prevent the sides of

excavations, trenches, pits, etc., from collapsing.

5. Timber shoring may be required to keep the sides of excavations vertical to ensure safety of adjoining structures or to limit the slope of excavations, or due to space restrictions or for other reasons. Such shoring shall be carried out, except in an emergency, only under instructions from the Engineer.
6. The withdrawal of the timber shall be done very carefully to prevent the collapse of the pit or trench. It shall be started at one end and proceeded systematically to the other end. Concrete or masonry shall not be damaged during the removal of the timber. No claim shall be entertained for any timber which cannot be retrieved.
7. In the case of open timbering, the entire surface of the side of trench or pit is not required to be covered. The vertical boards of minimum 25 cm x 5 cm sections shall be spaced sufficiently apart to leave unsupported strips of maximum 50 cm average width. The detailed arrangement, sizes of the timber and the spacing shall be subject to the approval of the Engineer. In all other respects, the specification for close timbering shall apply to open timbering.
8. In case of large pits and open excavations, where shoring is required for securing safety of adjoining structures or for any other reasons and where the planking across sides of excavations/pits cannot be strutted against, suitable inclined struts supported on the excavated bed shall be provided. Load from such struts shall be suitably distributed on the bed to ensure no yielding of the strut.

1.4.20 Dewatering

1. All excavations shall be kept free of water. Grading in the vicinity of excavations shall be controlled to prevent surface water running into excavated areas. The contractor shall remove by pumping or other means approved by Engineer any water inclusive of rain water and sub-soil water accumulated in excavation and keep all excavations dewatered until the foundation work is completed and backfilled. Sumps made for dewatering must be kept clear of the excavations/trenches required for further work.
2. Method of pumping shall be approved by Engineer; but in any case, the pumping arrangement shall be such that there shall be no movement of sub-soil or blowing in due to differential head of water during pumping. Pumping arrangements shall be adequate to ensure no delays in construction.
3. When there is a continuous inflow of water and the quantum of water to be handled is large in the opinion of the Engineer, the well point system single stage, or multi stage, shall be adopted. The contractor shall submit to the Engineer his scheme of the well point system including the stages, the spacing, number and diameter of well points, headers etc., and the number, capacity and location of pumps for approval.
4. All excavations shall have sufficient clearance for the purpose of making sumps for dewatering and boxing arrangement. Arrangement for sumps shall be made in such a way that the water level remains below the bed level, and bailing out of water from the sumps should be continuous. Where necessary in trenches, sumps may be made away

from the trenches and connected by laying pipes.

5. During construction of the works existing watercourses or flows of drains may have to be diverted through pipes or open channels so that no water can seep into the excavations and interfere with the works, or cause collapse of the walls. The diversion channels should be large enough to take the load of rains, and should be leak proof. The diversion arrangement should be continuous from section to section and the slope should be such that it maintains a gravity flow.
6. In case the soil on which the diversion pipes are to be laid is not suitable for the purpose the diversion pipes should be laid on proper staging. All diversion lines shall be maintained till completion of the works as a whole. The Contractor will, in no case, be allowed to put up bunds that may cause stagnation of water anywhere within the excavation.
7. Where gravity drainage is not possible or practical, the Contractor shall provide adequate pumping facilities to maintain the excavation in a safe and workmanlike condition so that water does not interfere with the progress and quality of the works. Unless specifically provided for by a separate item in the contract, any costs of pumping shall be deemed to be included in the contract rates, and no extra payment will be made on this account.

1.5 CONCRETE AND ALLIED WORKS

1.5.1 General

- a) The quality of materials and method and control of manufacture and transportation of all concrete work irrespective of mix, whether reinforced or otherwise, shall conform to the applicable portions of this specification.
- b) The Engineer shall have the right to inspect the sources of material/s, the layout and operation of procurement and storage of materials, the concrete batching and mixing equipment, and the quality control system. Such an inspection shall be arranged and Engineer's approval obtained, prior to starting of concrete work.

1.5.2 Materials for Standard Concrete

- a) The ingredients to be used in the manufacture of concrete shall consist solely of Portland cement, clean sand, natural coarse aggregate, clean water, and admixtures, if specifically called for on conditions at site warrant its use.
- b) Cement
 - (i) Unless otherwise specified in the Specification or called for by the Engineer, cement shall be ordinary Portland cement in 50 kg. bags. The use of bulk cement will be permitted only with the approval of the Engineer. Changing of brands or type of cement within the same structure should be avoided as far as possible.

(ii) The Contractor will have to make his own arrangements for procurement and storage of adequate quantity of cement. Cement in bulk may be stored in bins or silos which will provide complete protection from dampness, contamination and minimize caking and false set. Cement bags shall be stored in a dry enclosed shed (storage under tarpaulins will not be permitted), well away from the outer walls and insulated from the floor to avoid contact with moisture from the ground and so arranged as to provide ready access. Damaged or reclaimed or partly set cement will not be permitted to be used and shall be removed from the Site. The storage bins and storage arrangements shall be such that there is no dead storage. Not more than 12 bags shall be stacked in any tier. The storage arrangement shall be approved by the Engineer. Consignments of cement shall be stored as received and shall be consumed in the order of their delivery.

(iii) Cement held in storage for a period of ninety (90) days or longer shall be tested. Should at any time the Engineer have reasons to consider that any cement is defective, then irrespective of its origin date of manufacture and /or manufacturer's test certificate, such cement shall be tested immediately at the Contractor's cost at an approved laboratory and until the results of such tests are found satisfactory, it shall not be used in any work. The Contractor shall not be entitled to any claim of any nature on this account.

c) Coarse Aggregate

General: Aggregate most of which is retained on 4.75mm IS Sieve and contains only as much fine material as is permitted in IS 383 for various sizes and grading is known as coarse aggregate. Coarse aggregate shall be specified as stone aggregate and it shall be obtained from approved /authorized sources.

Stone Aggregate: It shall consist of naturally occurring (uncrushed, crushed or broken) stones. It shall be hard, strong, dense, durable and clean. It shall be free from veins, adherent coating, and injurious amounts of disintegrated pieces, alkali, vegetable matter and other deleterious substances. It shall be roughly cubical in shape. Flaky and elongated pieces shall be avoided. It shall conform to IS 383 unless otherwise specified.

Deleterious Material: Coarse aggregate shall not contain any deleterious material, such as pyrites, coal, lignite, mica, shale or similar laminated material, clay, alkali, soft fragments, seashells and organic impurities in such quantity as to affect the strength or durability of the concrete. Coarse aggregate to be used for reinforced cement concrete. Coarse aggregate to be used for reinforced cement concrete shall not contain any material liable to attack the steel reinforcement. Aggregates which are chemically reactive with alkalis of cement shall not be used. The maximum quantity of deleterious material shall not be more than five percent of the weight of coarse aggregate when determined in accordance with IS 2386.

Size and Grading of aggregates

The maximum size of coarse aggregate shall be 40mm for M-10 concrete and 20 mm for M-

- 15 to M-40 concrete, or as directed by the Engineer or specified otherwise. The maximum size of coarse aggregate shall be the maximum size specified above, but in no case greater than $\frac{1}{4}$ of the minimum thickness of the member, provided that the concrete can be placed without difficulty so as to surround all reinforcement thoroughly and fill the corners of the form. For plain concrete the maximum size of aggregate shall be of 40 mm. For heavily reinforced concrete members, the nominal maximum size of the aggregate shall be 5 mm less than the minimum clear distance between the reinforcing main bars or 5 mm less than the minimum cover to reinforcement whichever is smaller. The amount of fine particles occurring in the free state or as loose adherent shall not exceed 1% when determined by laboratory sedimentation tests as per I.S. 2386. After 24 hours immersion in water, a previously dried sample shall not have gained more than 10% of its oven dry weight in air, as determined by I.S. 2386.
- (i) Stone aggregate and gravel: It shall be either graded or single sized as specified. Nominal size and grading shall be as per CPWD Specifications (Volume-1)
 - a) Nominal sizes of graded stone aggregate or gravel shall be 40, 20, 16, or 12.5 mm as specified in Table 4.1 CPWD Specifications (Volume 1).
 - b) Nominal sizes of single sized stone aggregate or gravel shall be 63, 40, 20, 16, 12.5 or 10 mm as specified in Table 4.2 CPWD Specifications (Volume 1)
 - c) When stone aggregate or gravel brought to site is single sized (ungraded), it shall be mixed with single sized aggregate of different sizes in the proportion to be determined by field tests to obtain graded aggregate of specified nominal size.
 - (ii) For the required nominal size, the proportion of other sizes in mixed aggregate as determined by method prescribed in Appendix 'A' of Chapter 4 (CPWD Specifications volume 1) shall be in accordance with Table 4.1.
 - (iii) Recommended proportions by volume for mixing of different sizes of single size (ungraded) aggregate to obtain the required nominal size of graded aggregate are given in Table 4.3 of CPWD Specifications volume 1.

1.5.3 Sampling and Testing

1. Samples of the aggregates for mix design and determination of suitability shall be taken under the supervision of Engineer and delivered to the laboratory, well in advance of the scheduled placing of concrete. Records of tests which have been made on proposed aggregates and on concrete made from this source of aggregates shall be furnished to Engineer in advance or the work for use in determining aggregate suitability. The costs of all such tests, sampling, etc., shall be borne by Contractor.
2. **Storage of aggregates:** All coarse and fine aggregates shall be stacked separately in stock piles in the material yard near the work site in bins properly constructed to avoid inter mixing of different aggregates. Contamination with foreign material and earth during storage and while heaping the materials shall be avoided. The aggregate must be of specified quality not only at the time of receiving at Site but more so at the time of loading into mixer. Rakers shall be used for lifting the coarse aggregates from bins or

stock piles. Coarse aggregate shall be piled in layers not exceeding 1.20 meters in height to prevent coning or segregation. Each layer shall cover the entire area of the stock pile before succeeding layers are started. Aggregates that have become segregated shall be rejected. Rejected material after remixing may be accepted, if subsequent tests demonstrate conformance with required gradation.

3. **Specific Gravity:** Aggregates having a specific gravity below 2.6 (saturated surface dry basis) shall not be used without special permission of the Engineer.

1.5.4 Fine aggregate

Fine aggregate shall consist of natural or crushed sand conforming to IS 383. The sand shall be clean, sharp, hard, strong and durable and shall be free from dust, vegetable substances, adherent coating, clay, alkali, organic matter, mica, salt, or other deleterious substances, which can be injurious to the setting qualities/strength /durability of concrete.

1. **Machine-made Sand:** Machine-made sand will be acceptable, provided the constituent rock-gravel composition shall be sound, hard, dense, non-organic, uncoated and durable against weathering.
2. **Screening and Washing:** Sand shall be prepared for use by such screening or washing, or both, as necessary, to remove all objectionable foreign matter while separating the sand grains to the required size fractions.
3. **Fineness Modulus:** The sand shall have a fineness modulus of not less than 2.2 or more than 4.2. The fineness modulus is determined by adding the cumulative percentages retained on the following I.S. sieve sizes (4.75 mm, 2.36 mm, 1.18 mm, 600-micron, 300 micron and 150 micron) and dividing the sum by 100.
4. **Foreign material limitations:** the foreign material limitations should not be more than 5 percent by weight for uncrushed fine aggregate and about 3 percent by weight for crushed fine aggregate
5. **Gradation:** Unless otherwise directed or approved by the Engineer, the grading of sand shall be within the limits indicated in Table 4 of (IS 383-1970). Where the grading falls outside the limits of any particular grading zone of sieves, other than 600 micron I.S sieve, by total amount not exceeding 5 percent, it shall be regarded as falling within that grading zone. This tolerance shall not be applied to percentage passing the 600-micron I.S. sieve or to percentage passing any other sieve size on the coarser limit of Grading Zone I or the finer limit of Grading Zone IV. Fine aggregates conforming to Grading Zone IV shall not be used. Mix designs and preliminary tests shall show its suitability for producing concrete of specified strength and workability.

IS SIEVE DESIGNATION	Percentage passing for			
	Grading Zone-I	Grading Zone- II	Grading Zone-III	Grading Zone - IV
10 mm	100	100	100	100
4.75 mm	90 – 100	90 - 100	90 - 100	95 - 100
2.36 mm	60 – 95	75 - 100	85 - 100	95 - 100
1.18 mm	30 – 70	55 – 90	75 – 100	90 – 100
600 micron	15 – 34	35 – 59	60 – 79	80 – 100
300 micron	5 – 20	8 – 30	12 – 40	15 – 50
150 micron	0 – 10	0 – 10	0 – 10	0 - 15

1.5.5 Water

- a) Water used for washing, mixing and curing shall be free from injurious amounts of deleterious materials. Potable water is generally satisfactory for mixing and curing concrete.
- b) In case of doubt, the suitability of water for making concrete shall be ascertained by the compressive strength and initial setting time test specified in IS 456. The sample of water taken for testing shall be typical of the water proposed to be used for concreting, due account being paid to seasonal variation. The sample shall not receive any treatment before testing other than that envisaged in the regular supply of water proposed for use in concrete. The sample shall be stored in a clean container previously rinsed out with similar water.
- c) Average 28 days compressive strength of at least three 15 cm concrete cubes prepared with water proposed to be used shall not be less than 90% of the average strength of three similar concrete cubes prepared with distilled water. The cubes shall be prepared, cured and tested in accordance with the requirements of IS 516.
- d) The initial setting time of test block made with the appropriate test cement and the water proposed to be used shall not be less than 30 minutes and shall not differ by more than +30 minutes from the initial setting time of control test block prepared with the appropriate test cement and distilled water. The test blocks shall be prepared and tested in accordance with the requirements of IS 4031.
- e) Where water can be shown to contain an excess of acid, alkali, sugar or salt, Engineer will refuse to permit its use. As a guide, the following concentrations represent the maximum permissible values:
 - i. To neutralize 200 ml. sample of water, using phenolphthalein as indicator, it should not require more than 2 ml. of 0.1 Normal NaOH. The details of test shall be as given in IS 3025.

- ii. To neutralize 200 ml. sample of water, using methyl orange as an indicator, it should not require more than 10ml. of 0.1 Normal HCl. The details of test shall be as given in IS 3025.
- f) Percentage of solids, when tested in accordance with the method indicated below shall not exceed indicated in Clause 10, 11, 12, 20, 24 of IS 3025.
- g) The pH value of water shall not generally be less than 6.

1.5.6 Brick aggregate

Stacking of Aggregates: Aggregate shall be stacked on a hard, dry and level patch of ground. When stack piling, the aggregate shall not form pyramids resulting in segregation of different sized materials. It shall be stacked separately according to nominal size of coarse aggregates. Stacking shall be done in regular stacks, of height not exceeding 100 cm.

Testing of Aggregates: Coarse aggregate shall be tested for the followings (as per IS 2386)

- (a) Determination of particle size and shape (Appendix 'A' of Chapter 4 in CPWD Specifications Volume 1)
- (b) Estimation of organic impurities (as per IS 2386 - Part II)
- (c) Surface moisture (Appendix 'A' of Chapter 4 in CPWD Specifications Volume 1)
- (d) Determination of 10% fine value (Appendix 'A' of Chapter 4 in CPWD Specifications Volume 1)

Measurement of Aggregates:

1. The aggregates shall be measured in stacks and paid for after making a deduction of 7.5% of the gross measurements of stacks in respect of aggregates of nominal size 40 mm and above.
2. No deduction from the gross measurements of the stacks is to be made in respect of aggregate of nominal size below 40 mm.

1.5.7 Admixtures

When required, admixtures of approved quality shall be mixed with concrete, as specified. The admixtures shall conform to IS 9103 and as specified in Chapter 5 - R.C.C in CPWD Specifications Volume 1.

Dosage of these admixtures may vary according to manufacturer's specification.

1. Two or more admixtures may not be compatible in the same solution. It is therefore mandatory that when two admixtures manufactured by the same manufacturers is being used simultaneously, the manufacturer shall certify their compatibility. In case the two or more admixtures are produced by different manufacturers, then, before their use in concrete, test shall be performed by the manufacturer to establish their compatibility, all such test reports shall be furnished to the Engineer-in-Charge for his approval before

- their use in concrete. Some admixture may be in the form of powder, particle or high concentration liquids which may require mixing with water prior to dosing. Under these conditions water in solution shall be considered as part of total water content in the batch in order to maintain the water-cement ratio.
2. Admixture manufacturer's recommendation shall be carefully followed so as to ensure complete solution of the product or to prepare a standard solution of uniform strength for easier use.
 3. Certain admixtures may contain significant amounts of finely divided insoluble materials or active ingredients which may or may not be readily soluble. It is essential for such admixtures that precautions be taken to ensure that these constituents be kept in a state of uniform suspension before actual batching. When relatively small amounts of powdered admixtures are to be used directly, these shall be pre-blended with cement.
 4. Admixtures are sold under various trade names and may be in the form of liquids or powders. The proprietary name and the net quantity of content shall be clearly indicated in each package or container of admixtures. The admixtures shall be uniform within each batch and uniform between all batches.
 5. No admixtures shall be accepted for use in concrete unless these are tested in accordance with IS 9103 and the test results are approved by the Engineer-in-Charge.

1.6 CEMENT CONCRETE

1.6.1 Grades of Cement Concrete

1. The concrete shall be in grade designated in Table 2 of IS 456:2000
2. The characteristic strength is defined as the strength of material below which not more than 5 percent of the test results are expected to fall.
3. Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal, Maximum Size will be as specified in Table 5 of IS 456: 2000

Notes:

1. Cement content prescribed in this Table is irrespective of the grades of cement. The additions such as fly or ground granulated blast furnace slag may be considered in the concrete composition with respect to the cement content and water-cement ratio, if the suitability is established and as long as the maximum amounts considered do not exceed the limit of pozzolona and slag specified in IS 1489 (Part 1) and IS 455 respectively.
2. Minimum grade for plain concrete under mild exposure condition is not specified.
3. The above minimum cement content and maximum water cement ratio apply only to 20 mm nominal maximum size aggregate. For other sizes of aggregate, these should be changed as per Table 6 of IS 456: 2000.
4. The minimum grade of concrete for plain and reinforced concrete shall be as per Table 5 of IS 456: 2000. Concrete of grades lower than those given in Table 6 of IS 456: 2000 may be used for lean concrete, foundation for masonry walls or temporary reinforced concrete construction.

1.6.2 Workability of concrete

1. The concrete mix proportion chosen should be such that the concrete is of adequate workability for the placing conditions of the concrete and can properly be compacted with the means available. Suggested ranges of workability of concrete measured in accordance with IS 1199, should be in the range specified in chapter 7 – IS 456: 2000.
2. In the 'very low' category of workability where strict control is necessary, for example, pavement quality concrete, measurement of workability by determination of compacting factor will be more appropriate than slump (see IS 1199) and a value of compacting factor of 0.75 to 0.80 is suggested.
3. In the 'very high' category of workability, measurement of workability by determination of flow will be appropriate (see IS 9103).

1.6.3 Concrete mix proportioning

The determination of the proportion of cement, aggregate and water to attain the required strength shall be made as follows:

- (a) By designing the concrete mix: such concrete shall be called 'Design mix concrete', for details reference may be made to chapter 9 of IS 456: 2000 or RCC Chapter of CPWD Specifications Volume 1.
 - (b) By adopting nominal concrete mix: such concrete shall be called 'Nominal mix concrete'.
1. Design mix concrete is preferred to nominal mix. If design mix concrete cannot be used for any reason on the work for grades of M20 or lower, nominal mixes may be used with the permission of Engineer-in-Charge, which, however, is likely to involve higher cement content.
 2. Nominal Mix Concrete may be used for concrete of M20 or lower. The proportions of materials for nominal mix concrete shall be in accordance with Table 9 of IS456:2000.
 3. The cement content of the mix specified in Table 9 of IS 456:2000 for any nominal mix shall be proportionately increased if the quantity of water in the mix has to be increased to overcome the difficulty of placement and compaction, so that the water cement ratio as specified is not exceeded.

Note:

1. The proportion of the fine to coarse aggregate should be adjusted from upper limit progressively the grading of fine aggregate becomes finer and the maximum size of coarse aggregate becomes larger. Graded coarse aggregate shall be used.
2. Quantity of water required from durability point of view may be less than the value given above.

1.6.4 Batching

1. To avoid confusion and error in batching, consideration should be given to using the smallest practical number of different concretes mixed on any site or in any one plant. In batching concrete, the quantity of both cement and aggregate shall be determined by

- mass; admixture, if solid, by mass: liquid admixture may however be measured in volume or mass: water shall be weighed or measured by volume in a calibrated tank (see also IS 4925).
2. Ready-mixed concrete supplied by ready-mixed concrete plant shall be preferred. For large and medium project sites the concrete shall be sourced from ready-mixed concrete plants or from on site or off-site batching and mixing plants (see IS 4926).
 3. Except where it can be shown to the satisfaction of the Engineer-in-Charge that supply of properly graded aggregate of uniform quality can be maintained over a period of work, the grading aggregate should be controlled by obtaining the coarse aggregate in different sizes and blending them in the right proportions when required, the different sizes being stocked in separate stock-piles. The material should be stock-piled for several hours preferably a day before use. The grading of coarse and fine aggregate should be checked as frequently as possible, the frequency for a given job being determined by the Engineer-in-Charge to ensure that the specified grading is maintained.
 4. The accuracy of the measuring equipment shall be within ± 2 percent of the quantity of cement being measured and within ± 3 percent of the quantity of aggregate, admixtures and water being measured.
 5. Proportion/Type and grading of aggregates shall be made by trial in such a way so as to obtain densest possible concrete. All ingredients of the concrete should be used by mass only.
 6. Volume batching may be allowed only where weigh-batching is not practicable and provided accurate used in concrete have earlier been established. Allowance for bulking shall be made in accordance with IS 2386 (Part 3). The mass volume relationship should be checked as frequently as necessary, the frequency for the given job being determined by Engineer-in-Charge to ensure that the specified grading is maintained.
 7. It is important to maintain the water cement ratio constant at its correct value. To this end, determination of moisture contents in both fine and coarse aggregates shall be made as frequently as possible, the frequency for a given job being determined by the Engineer-in-Charge according to weather conditions. The amount of the added water shall be adjusted to compensate for any observed variations in the moisture contents. For the determination of moisture content in the aggregates, IS 2386 (Part 3) may be referred to. To allow for the variation in mass for aggregate due to variations in their moisture content, suitable adjustments in the masses of aggregates shall be made. In the absence of exact data, only in the case of nominal mixes, the amount of surface water may be estimated from the values given in Table 10 of IS 456: 2000.
- No substitutions in materials used on the work or alteration in the established proportions, except as permitted in above mentioned conditions shall be made without additional tests to show that the quality and strength of concrete are satisfactory.

1.6.5 Mixing of concrete

1. Mixing of Concrete will be carried out as specified in Clause 10.3 of IS 456:2000.
2. Concrete shall be mixed in mechanical batch type concrete mixers conforming to IS 1791 having two blades and fitted with power loader (lifting hopper type). Half bag mixers and mixers without lifting hoppers shall not be used for mixing concrete. In exceptional circumstances, such as mechanical breakdown of mixer, work in remote areas or power breakdown and when the quantity of concrete work is very small, hand mixing may be done with the specific prior permission of the Engineer-in-Charge in writing subject to adding 10% extra cement. When hand mixing is permitted, it shall be carried out on a water tight platform and care shall be taken to ensure that mixing is continued until the concrete is uniform in colour and consistency.
3. Before mixing the brick, aggregate shall be well soaked with water for a minimum period of two hours and stone aggregate or gravel shall be washed with water to remove, dirt, dust and other foreign materials. For guidance, the mixing time may be 1½ to 2 minutes, for hydrophobic cement it may be taken as 2½ to 3 minutes.

Power Loader: Mixer will be fitted with a power loader complying with the following requirements.

- (a) The hopper shall be of adequate capacity to receive and discharge the maximum nominal batch of unmixed materials without spillage under normal operating conditions on a level site.

Note: In such a case the volume of the maximum nominal batch of mixed material is 50% greater than the nominal mixed batch capacity.

- (b) The minimum inside width of the feeding edge of the hopper shall be as specified below in Table 4.9 of CPWD Specifications Volume 1.
- (c) The hopper shall be of adequate capacity to receive and discharge the maximum nominal batch of unmixed materials without spillage under normal operating conditions on a level site.

Note: In such a case the volume of the maximum nominal batch of mixed material is 50% greater than the nominal mixed batch capacity.

- (d) The minimum inside width of the feeding edge of the hopper shall be as specified below in Table 4.9 of CPWD Specifications Volume 1.
- (e) The design of the loader shall be such that it allows the loading hopper to be elevated to such a height that the center line of the chute plate of the hopper when in discharge position, is at an angle of not less than 50° to the horizontal. A mechanical device to aid discharge of the contents as quickly as possible from the hopper to the drum may also be provided. Even when a mechanical device is provided, it is recommended that the

- angle of center line of the chute plate of the hopper when in discharge position, should be as large as practicable, preferably not less than 40° to horizontal.
- (f) When the means of raising and lowering the loading hopper includes flexible wire ropes winding on to a drum or drums, the method of fastening the wire to rope to the drums shall be such as to avoid, as far as possible any tendency to cut the strands of the ropes and the fastening should preferably be positioned clear of the barrel of the drum for example, outside the drums flange. When the loading hopper is lowered to its normal loading position, these should be at least one and a half drums of rope on the drum.
 - (g) Clutch brake and hydraulic control lever shall be designed so as to prevent displacement by liberation or by accidental contact with any person.
 - (h) The clutch and brake control arrangements shall also be so designed that the operator can control the falling speed of the loader.
 - (i) Safety device shall be provided to secure the hopper in raised position when not in use.

1.6.6 Mixing efficiency of mixer:

1. The mixer shall be tested under normal working conditions in accordance with the method specified in IS 4643 with a view to check its ability to mix the ingredients to obtain concrete having uniformity within the prescribed limits. The uniformity of mixed concrete shall be evaluated by finding the percentage variation in quantity (mass in water) of cement, fine aggregate and coarse aggregate in a freshly mixed batch of concrete.
2. The percentage variation between the quantities of cement, fine aggregate and coarse aggregates (as found by weighing in water) in the two halves of a batch and average of the two halves of the batch shall not be more than the following limits: Cement (8%), Fine aggregate (6%), Coarse aggregate (5%)

1.6.7 Machine mixing:

1. The mixer drum shall be flushed clean with water. Measured quantity of coarse aggregate shall be placed first in the hopper. This shall be followed with measured quantity of fine aggregate and then cement. In case fine aggregate is damp, half the required quantity of coarse aggregate shall be placed in the hopper, followed by fine aggregate and cement. Finally the balance quantity of coarse aggregate shall be fed in the hopper, & then the dry materials are slipped into the drum by raising the hopper. The dry material shall be mixed for at least four turns of the drum. While the drum is rotating, water shall be added gradually to achieve the water cement ratio as specified or as required by the Engineer-in-Charge. After adding water, the mixing shall be continued until concrete of uniform colour, uniformly distributed material and consistency is obtained. Mixing shall be done for at least two minutes after adding water. If there is segregation after unloading from the mixer, the concrete should be remixed.
2. The drum shall be emptied before recharging. When the mixer is closed down for the day or at any time exceeding 20 minutes, the drum shall be flushed cleaned with water.

1.6.8 Hand mixing:

When hand mixing has been specifically permitted in exceptional circumstances by the Engineer-in-Charge in writing, subject to adding 10% extra cement, it shall be carried out on a smooth, clean and water tight platform of suitable size. Measured quantity of sand shall be spread evenly on the platform and the cement shall be dumped on the sand and distributed evenly. Sand and cement shall be mixed intimately with spade until mixture is of even colour throughout. Measured quantity of coarse aggregate shall be spread on top of cement sand mixture and mixing done by showlling and turning till the coarse aggregate gets evenly distributed the cement sand mixture. Three quarters of the total quantity of water required shall be added in a hollow made in the middle of the mixed pile and the material is turned towards the middle of pile with spade. The whole mixture is turned slowly over and again and the remaining quantity of water is added gradually. The mixing shall be continued until concrete of uniform colour and consistency is obtained. The mixing platform shall be washed and cleaned at the end of the day.

1.6.9 Transportation and handling:

Concrete shall be transported from the mixer to the place of laying as rapidly as possible by methods which will prevent the segregation or loss of any of the ingredients and maintaining the required workability. During hot or cold weather, concrete shall be transported in deep containers, other suitable methods to reduce the loss of water by evaporation in hot weather and heat loss in cold weather may also be adopted.

1.6.10 PLACING OF CONCRETE

The concrete shall be deposited as nearly as practicable in its final position to avoid handling. It shall be laid gently (not thrown) and shall be thoroughly vibrated and compacted before setting commences and should not be subsequently disturbed. Method of placing shall be such as to preclude segregation. Care shall be taken to avoid displacement of reinforcement or movement of form work and damage due to rains. As a general guidance, the maximum free fall of concrete may be taken as 1.5 metre.

Compaction of Concrete

1. Concrete shall be thoroughly compacted and fully worked around embedded fixtures and into corners of the form work. Compaction shall be done by mechanical vibrator of appropriate type till a dense concrete is obtained. The mechanical vibrators shall conform to IS 2505, IS 2506, IS 2514 and IS 4656. To prevent segregation, over vibration shall be avoided.
2. Compaction shall be completed before the initial setting starts. For the items where mechanical vibrators are not to be used, the contractor shall take permission of the Engineer-in-Charge in writing before the start of the work. After compaction the top surface shall be finished even and smooth with wooden trowel before the concrete begins to set.

1.6.11 Construction joints

1. Concreting shall be carried out continuously upto construction joints. The position and arrangement of construction joints shall be as shown in the structural drawings or as directed by the Engineer-in Charge. Number of such joints shall be kept minimum. Joints shall be kept as straight as possible. Construction joints should comply with IS 11817.
2. When the work has to be resumed on a surface which has hardened, such surface shall be roughened. It shall then be swept clean and thoroughly wetted. For vertical joints, neat cement slurry, of workable consistency by using 2 kgs of cement per sqm shall be applied on the surface before it is dry. For horizontal joints, the surface shall be covered with a layer of mortar about 10-15 mm thick composed of cement and sand in the same ratio as the cement and sand in concrete mix. This layer of cement slurry of mortar shall be freshly mixed and applied immediately before placing of the concrete.
3. Where the concrete has not fully hardened, all laitance shall be removed by scrubbing the wet surface with wire or bristle brushes, care being taken to avoid dislodgement of particles of coarse aggregate. The surface shall be thoroughly wetted and all free water removed. The surface shall then be coated with neat cement slurry @ 2 kgs of cement per sqm. On this surface, a layer of concrete not exceeding 150 mm in thickness shall first be placed and shall be well rammed against old work particular attention being paid to corners and close spots; work, thereafter, shall proceed in the normal way.

1.6.12 Concreting under special conditions

Work in Extreme Weather Conditions: During hot and cold weather, the concreting shall be done as per the procedure set out in IS 7861 (Part-I)-1975 and IS 7861 (Part II)-1981 respectively. Concreting shall not be done when the temperature falls below 4.5°C. In cold weather, the concrete placed shall be protected against frost. During hot weather, it shall be ensured that the temperature of wet concrete does not exceed 38°C.

Under Water Concreting: Concrete shall not be deposited under water if it is practicable to dewater the area and place concrete in the regular manner. When it is necessary to deposit concrete under water, the methods, equipment, materials and proportions of the mix to be used shall be submitted to and approved by the Engineer-in-Charge before the work is started. Under-water concrete should have a slump recommended in 4.2.2 of CPWD Specifications Volume 1. The water-cement ratio shall not exceed 0.6 and may need to be smaller, depending on the grade of concrete or the type of chemical attack. For aggregates of 40 mm maximum particle size, the cement content shall be atleast 350 kg/m³ of concrete.

1.6.13 Curing

1. Curing is the process of preventing loss of moisture from the concrete. The following methods shall be employed for effecting curing.
2. **Moist Curing:** Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering with a layer of sacking, canvas, Hessian or similar materials and kept constantly wet for at least 7 days from the date of placing concrete in

case of ordinary Portland cement and at least 10 days where mineral admixtures or blended cements are used. The period of curing shall not be less than 10 days for concrete exposed to dry and hot weather conditions. In the case of concrete where mineral admixtures or blended cements are used, it is recommended that above minimum periods may be extended to 14 days.

3. Freshly laid concrete shall be protected from rain by suitable covering.
4. Over the foundation concrete, the masonry work may be started after 48 hours of its compaction but the curing of exposed surfaces of cement concrete shall be continued along with the masonry work for at least 7 days. And where cement concrete is used as base concrete for flooring, the flooring may be commenced before the curing period of base concrete is over but the curing of base concrete shall be continued along with top layer of flooring for a minimum period of 7 days.

1.6.14 Testing of concrete

Testing of concrete shall be done as described in chapter of R.C.C of CPWD Specifications Volume 1.

1.6.15 Measurements

Dimensions of length, breadth and thickness shall be measured correct to nearest cm. except for the thickness of slab and partition which shall be measured to nearest 5mm. Areas shall be worked out to nearest 0.01sq. m and the cubic contents of consolidated concrete shall be worked out to nearest 0.01cum. Any work done in excess over the specified dimension or sections shown in the drawing shall be ignored.

1.7 REINFORCED CEMENT CONCRETE

1. Reinforced Cement Concrete in the Works shall be "Design Mix Concrete" as defined in IS 456/2000, for M20 and M30 grades. Nominal mix concrete may be used for concrete of lower grades i.e. lower than M20 grade concrete.
2. The specified characteristic compressive strength of 150 mm size cube at 28 days attained for M20 grade concrete shall be 20 N/sqmm and for M25 grade concrete shall be 25 N/sq mm respectively. The contractor is required to carry out the mix design and the mix so designed need to be got approved by the concerned Engineer, TNPCB /, within the limitations of parameters and other stipulations laid down in IS-456/2000.
3. The mix shall be designed to produce the grade of concrete having the required workability and a characteristic strength at 28 days not less than the appropriate values mentioned in Table-2 of IS-456/2000 i.e. 20 N/sqmm for M20 grade concrete and 25N/sqmm for M25 grade concrete respectively. The target mean strength of the concrete mix should be equal to the characteristic strength plus 1.65 times the standard deviation.
4. IS-3370 (Part-I), 1965, prescribes that the minimum quantity of cement used for the reinforced concrete work in the case of water retaining structures shall not be less than 330 Kg/cum. IS-456/2000 prescribes the minimum cement quantity to be used for M20

grade reinforced cement concrete as 300 Kg/cum and M30 grade reinforced cement concrete as 320 Kg/cum respectively.

1.7.1 Mix Design

a) General

1. This is to investigate the grading of aggregates, water-cement ratio, workability and the quality of cement required to give preliminary and works cubes of the minimum strength specified. The proportions of the mix shall be determined by weight. Adjustment of aggregate proportions due to moisture present in the aggregate shall be made. Determination of mix proportions shall be carried out according to 'Recommended guidelines for Concrete Mix Design' conforming to IS 10262.
2. Whenever there is a change either in required strength of concrete, or water-cement ratio or workability or the source of aggregates and/or cement, preliminary tests shall be repeated to determine the revised proportions of the mix to suit the altered conditions. While designing mix proportions, over-wet mixes shall always be avoided.
3. While fixing the value for water/cement ratio for preliminary mixes, assistance may be derived from the graph (Appendix A, IS 456 showing the relationship between the 28-day compressive strengths of concrete mixes with different water/cement ratios and the 7-day compressive strength of cement tested in accordance with IS 269.

b) Preliminary Tests

Test specimens shall be prepared with at least two different water/cement ratios for each class of concrete, consistent with workability required for the nature of the work. The materials and proportions used in making preliminary tests shall be similar in all respects to those to be actually employed in the works as the object of these tests is to determine the properties of cement, aggregates and water necessary to produce concrete of required consistency and to give the specified strength. It will be Contractor's sole responsibility to carry out these tests and he shall therefore furnish to Engineer a statement of proportions proposed to be used for the various concrete mixes. For preliminary tests, the following procedure shall be followed:

- (A) Materials shall be brought to the room temperature and all materials shall be in a dry condition. The quantities of water, cement and aggregates for each batch shall be determined by weight to an accuracy of 1 part in 1000 parts.
- (B) **Mixing Concrete:** It shall be done by hand or in a small batch mixer as per IS 516 in such a manner as to avoid loss of water. The cement and fine aggregate shall first be mixed dry until the mixture is uniform in colour. The coarse aggregate shall then be added, mixed and water added and the whole batch mixed thoroughly for a period of not less than two minutes until the resulting concrete is uniform in appearance. Each batch of concrete shall be of such a size as to leave about 10% excess concrete, after moulding the desired number of test specimens.

- (C) **Consistency:** The consistency of each batch of concrete shall be measured immediately after mixing, by the slump test in accordance with IS.1199. If in the slump test, care is taken to ensure that no water or other material is lost, the material used for the slump test may be remixed with the remainder of the concrete for making the specimen test cubes. The period of re-mixing shall be as short as possible yet sufficient to produce a homogeneous mass.
- (D) **Size of Test Cubes:** Compression tests of concrete cubes shall be made as per IS.516 on 15 cm cubes. Each mould shall be provided with a metal base plate having a plate surface so as to support the mould during filling without leakage. The base plate shall be preferably attached to the mould by springs or screws. The parts of the mould when assembled shall be positively and rigidly held together. Before placing concrete, the mould and base plate shall be cleaned and oiled. The dimensions and internal faces of the mould shall be accurate within the following limits:
- Height and distance between the opposite faces of the mould shall be of specified size ± 0.2 mm. The angle between the adjacent internal faces and between internal faces and top and bottom faces of mould shall be $90^\circ \pm 0.5^\circ$. The interior faces of the mould shall be plane surfaces with a permissible variation of 0.03 mm.
- (E) **Compacting:** Concrete test cubes shall be moulded by placing fresh concrete in the mould and compacted as specified in IS.516.
- (F) **Curing:** Curing shall be as specified in IS.516. The cubes shall be kept in moist air of at least 90% relative humidity at a temperature of $27^\circ\text{C} \pm 2^\circ\text{C}$ for 24 hours ± 2 hours from the time of adding water to the dry ingredients. Thereafter they shall be removed from the moulds and kept immersed in clean, fresh water and kept at $27^\circ\text{C} \pm 2^\circ\text{C}$ temperature until required for test. Curing water shall be renewed every seven days. A record of maximum and minimum temperatures at the place of storage of the cubes shall be maintained during the period they remain in storage.
- (G) **Testing of Specimens:** The strength shall be determined based on not less than five cube test specimens for each age and each water cement ratio. All these laboratory test results shall be tabulated and furnished to the Engineer. The test results shall be accepted by the Engineer if the average compressive strengths of the specimens tested is not less than the compressive strength specified for the age at which specimens are tested subject to the condition that only one out of the five consecutive tests may give a value less than the specified strength for that age. The Engineer may direct the Contractor to repeat the tests if the results are not satisfactory and also make such changes as he considers necessary to meet the requirements specified. All these preliminary tests shall be conducted by the Contractor at his own cost in an approved laboratory.

1.7.2 Proportioning, consistency, batching and mixing of concrete

a) Proportioning

(i) Aggregate:

The proportions which shall be decided by conducting preliminary tests shall be by weight. These proportions of Cement, fine and coarse aggregates shall be maintained during subsequent concrete batching by means of weigh batchers conforming to IS.2722 capable of controlling the weights within one percent of the desired value. Except where it can be shown to the satisfaction of the Engineer that supply of properly graded aggregate of uniform quality can be maintained over the period of work, the grading of aggregate shall be controlled by obtaining the coarse aggregate in different sizes and blending them in the right proportions. The different sizes shall be stocked in separate stock piles. The grading of coarse and fine aggregate shall be checked as frequently as possible, as determined by the Engineer, to ensure maintaining of grading in accordance with the samples used in preliminary mix design. The material shall be stock piled well in advance of use. Cement shall be measured by weight.

(ii) Water:

Only such quantity of water shall be added to the cement and aggregates in the concrete mix as to ensure dense concrete, specified surface finish, satisfactory workability, consistent with the strength stipulated for each class of concrete. The water added to the mix shall be such as not to cause segregation of materials or the collection of excessive free water on the surface of the concrete.

(iii) Definition of Water/Cement Ratio:

The water cement (W/C) ratio is defined as the weight of water in the mix (including the surface moisture of the aggregates) divided by the weight of cement in the mix.

(iv) Water/Cement Ratio:

The actual water cement ratio to be adopted shall be determined in each instance by the Contractor and approved by the Engineer.

(v) Proportioning by Water/Cement Ratio:

The W/C ratio specified for use by the Engineer shall be maintained. The Contractor shall determine the water content of the aggregates as frequently as directed by the Engineer as the work progresses and as specified in IS.2386 (Part III) and the amount of mixing water added at the mixer shall be adjusted as directed by the Engineer so as to maintain the specified W/C Ratio. To allow for the variation in weight of aggregates due to variation in their moisture content, suitable adjustments in the weights of aggregates shall also be made.

b) Consistency and Slump

1. Concrete shall be of a consistency and workability suitable for the conditions of the job. After the amount of water required is determined, the consistency of the mix shall be maintained throughout the progress of the corresponding parts of the work and approved tests e.g. slump tests, compacting factor tests, in accordance with IS.1199, shall be conducted from time to time to ensure the maintenance of such consistency.
2. The range of slumps which shall generally be used for various types of construction unless otherwise instructed by the Engineer will be as specified in Clause 4.4.2 of CPWD Specifications (Volume I).

c) Batching and Mixing of Concrete

1. The materials and proportions of concrete materials as established by the preliminary tests for the mix designs shall be rigidly followed for all concrete on the Works and shall not be changed except when specifically permitted by the Engineer.
2. Concrete shall be produced only by weigh batching the ingredients. The mixer and weigh batchers shall be maintained in clean, serviceable condition. The accuracy of weigh batchers shall be periodically checked. They shall be set up level on a firm base and the hopper shall be loaded evenly. The needle shall be adjusted to zero when the hopper is empty. Fine and coarse aggregates shall be weighed separately.
3. Volume batching will not be permitted. However, the Engineer may permit volume batching by subsequent conversion of the weights of the aggregate into their equivalent volumes knowing their bulk densities, only in the case of small and less important pours involving concrete of not more than 0.25 cubic metres, on days when other pours involving weigh batching are not likely to be taken up.
4. Concrete shall be of strength stipulated in the respective items. All concrete shall be mixed in mechanically operated batch mixers complying with IS:1791 and of the approved make with suitable provision for correctly controlling the water delivered to the drum.
5. The quantity of water actually entering the drum shall be checked with the reading of the size or valve setting, when starting a job. The test should be made while the mixer is running. The volume of the mixed material shall not exceed the manufacturer's rated mixer capacity.
6. The batch shall be charged into the mixer so that some water will enter the drum in advance of cement and aggregates. All water shall be in the drum by the end of the first 15 seconds of the specified mixing time.
7. Each batch shall be mixed until the concrete is uniform in colour, for a minimum period of two minutes after all the materials and water are in the drum. The entire contents of the drum shall be discharged in one operation before the raw materials for the succeeding batches are fed into the drum.
8. Each time the work stops, the mixer shall be cleaned out and when next commencing the mixing, the first batch shall have 10% additional cement to allow for sticking in the drum.

1.7.3 Sampling and testing concrete in the field

All tests as specified in the relevant codes to be carried by the contractor in presence of Engineer in charge/Third party QAQC personnel.

- a) Facilities required for sampling materials and concrete in the field, if the Engineer / QAQC Technical personnel so desires, shall be provided by the Contractor at no extra cost. The equipment's with operator shall be made available at the Engineer's/ QAQC Technical personnel request (all must be in serviceable condition.) and with valid calibration certificates for the devices to be used for testing.)
- b) No concrete of any kind may be placed until the field concrete testing laboratory as specified is provided to the satisfaction of the Engineer/QAQC personnel. The Contractor shall notify the Engineer in advance of all concrete and concrete material testing as provided in the Clause to provide the Engineer / his representative with an opportunity to witness all prescribed tests. TNPCB/ executives shall be permitted to use the contractor's lab to perform all the tests pertaining to the contract.

c) Sampling for Strength of Concrete:

- 1. A random sampling procedure shall be adopted to ensure that each concrete batch shall have a reasonable chance of being tested that is, the sampling should be spread over the entire period of concreting and cover all mixing units.
- 2. Frequency of sampling will be as per (Clause 15.2.2; IS456:2000).
- 3. Three test specimens shall be made for each sample for testing at 28 days. Additional samples may be required for various purposes such as to determine the strength of concrete at 7 days or at the time of striking the formwork, or to determine the duration of curing, or to check the testing error. Additional samples may also be required for testing samples cured by accelerated methods as described in relevant IS code.
- 4. The specimen shall be tested as described in relevant IS code. The laboratory test results shall be tabulated and furnished to the Engineer. The test results of the sample shall be the average of the strength of three specimens. The individual variation should not be more than 15 percent of the average. If more, the test results of the sample are invalid.

- d) **Consistency:** Slump tests shall be carried out as often as requested by the Engineer and invariably from the same batch of concrete from which the test cubes are made. Slump tests shall be done immediately after sampling.

1.7.4 Other Tests

If the work cubes do not give the required strengths, the Engineer reserves the right to ask the Contractor to dismantle such portions of the work which in his opinion are unacceptable and re-do the work to the standard required at the Contractor's cost.

- a) The following tests shall be conducted whenever the required to be strength achieved on cement, sand, coarse aggregate, water in accordance with the relevant Indian Standards.
1. Tests on cement shall include (i) fineness test, (ii) test for normal consistency, (iii) test for setting time, (iv) test for soundness, (v) test for tensile strength, (vi) test for compressive strength, (vii) test for heat of hydration (by experiment and by calculations) in accordance with I.S. 269.
 2. Tests on sand shall include (i) sieve test, (ii) test for organic impurities, (iii) decantation test for determining clay and silt content, (iv) specific gravity test, (v) test for unit weight and bulk age factor, (vi) test for sieve analysis and fineness modulus.
 3. Tests on coarse aggregate shall include (i) sieve analysis, (ii) specific gravity and unit weight of dry, loose and rodded aggregate, (iii) soundness and alkali aggregate reactivity, (iv) petrography examination, (v) deleterious materials and organic impurities, (vi) test for aggregate crushing value.
- b) Load Test on Members or any other Tests
1. In the event of any work being suspected of material or workmanship or both, the Engineer requiring its removal and reconstruction may order, or the Contractor may request that it should be load tested in accordance with the following provisions:
 2. The test load shall be 125 percent of the maximum superimposed load for which the structure was designed. Such test load shall not be applied before 56 days after the effective hardening of concrete. During the test, struts strong enough to take the whole load shall be placed in position leaving a gap under the members, The test load shall be maintained for 24 hours before removal.
 3. If within 24 hours of the removal of the load, the structure does not show a recovery of at least 75 percent of the maximum deflection shown during the 24 hours under load, the test loading shall be repeated after a lapse of at least 72 hours, the structure shall be considered to have failed to pass the test if the recovery after the second test is not at least 75 percent of the maximum deflection shown during the second test. If the structure is certified as failed by the Engineer, the cost of the load test shall be borne by the Contractor.
 4. Any other tests, e.g. taking out in an approved manner concrete cores, examination and tests on such cores removed from such parts of the structure as directed by the Engineer, sonic testing, etc., shall be carried out by the Contractor, if so directed, at no extra cost.
- c) Unsatisfactory Tests: Should the results of any test prove unsatisfactory, or the structure shows signs of weakness, undue deflection or faulty construction, the Contractor shall remove and rebuild the member or members involved or carry out such other remedial measures as may be required by the Engineer, without any additional cost.

1.7.5 Admixtures

- a) **General:** Admixtures may be used in concrete where required, only with the approval of the Engineer based upon evidence that, with the passage of time, neither the compressive strength nor its durability reduced. Calcium chloride shall not be used for

- accelerating set of the cement for any concrete containing reinforcement, or embedded steel parts. When calcium chloride is permitted to be used, such as in mass concrete works, it shall be dissolved in water and added to the mixing water in an amount not to exceed 1.5 percent of the weight of the cement in each batch of concrete. When admixtures are used, the designed concrete mix shall be corrected accordingly. Admixtures shall be used as per manufacturer's instruction and in the manner and with the control specified by the Engineer.
- b) **Air Entraining Agents:** Neutralized vinsol resin or other approved air entraining agent may be used to produce the specified amount of air in the concrete mix and these agents shall conform to the requirements of ASTM standard 6-260; Air Entraining Admixtures for Concrete. The recommended total air content of the concrete is $4\% \pm 1\%$. The method of measuring air content shall be as per I.S. 1199
 - c) **Water Reducing Admixtures:** Water reducing lingo sultanate admixture may be added in quantities approved by the Engineer. The admixtures shall be added in the form of a solution.
 - d) **Retarding Admixtures:** Retarding agents may be added to the concrete mix in quantities approved by the Engineer.
 - e) **Water-proofing Agent:** Water proofing agents shall conform to I.S.2645.
 - f) **Other Admixtures:** The Engineer may at his discretion allow the Contractor to use any other admixture in the concrete.

1.7.6 Preparation prior to concrete placement, final inspection and approval

- a) Before the concrete is actually placed in position, the insides of the formwork shall be inspected to see that they have been cleaned and oiled. Temporary openings shall be provided to facilitate inspection, especially of bottoms of columns and wall forms, to permit removal of sawdust, wood shavings, binding wire, dirt, etc. Openings shall be placed or holes drilled so that these materials and water can be removed easily. Such openings/holes shall be suitably plugged later.
- b) The various trades shall be permitted ample time to install drainage and plumbing lines, floor and trench drains, conduits, hangers, anchors, inserts, sleeves, bolts, frames and other miscellaneous embodiments to be cast in the concrete as specified or required or as is necessary for the proper execution of the work.
- c) All embedded parts, inserts, etc., supplied by the Contractor shall be correctly positioned and securely held in the forms to prevent displacement during depositing and vibrating of concrete.
- d) All anchor bolts shall be positioned and kept in place with the help of properly manufactured templates unless specifically waived in writing by the Engineer.
- e) Slots, openings, holes, pockets, etc., shall be provided in the concrete work in the positions specified or required or as directed by the Engineer.
- f) Reinforcement and other items to be cast in concrete shall have clean surfaces that will not impair bond.
- g) Prior to concrete placement, all work shall be inspected and approved by the Engineer

and if found unsatisfactory, concrete shall not be poured until after all defects have been corrected.

- h) Approval by the Engineer of any and all materials and work as required herein shall not relieve the Contractor from his obligation to produce finished concrete in accordance with the requirements of the Specifications.
- i) Rain or Wash Water: No concrete shall be placed in wet weather or on a water-covered surface. Any concrete that has been washed by heavy rains shall be entirely removed, if there is any sign of cement and sand having been washed away from the concrete mixture. To guard against damage which may be caused by rains, the works shall be covered with tarpaulins immediately after the concrete has been placed and compacted before leaving the work unattended. Any water accumulating on the surface of the newly placed concrete shall be removed by approved means and no further concrete shall be placed thereon until such water is removed. To avoid flow of water over/around freshly placed concrete, suitable drains and sumps shall be provided.
- j) Bonding Mortar: Where the concrete has not fully hardened, all laitance shall be removed by scrubbing the wet surface with wire or bristle brushes, care being taken to avoid dislodgement of particles of aggregate. The surface shall be thoroughly wetted and all free water removed. The surface shall then be coated with neat cement slurry. On this surface, a layer of concrete not exceeding 150mm in thickness shall first be placed and shall well rammed against old work, particular attention being paid to corners and close spots; work thereafter shall proceed in the normal way.

1.7.7 Transportation

- a) **General:** All buckets, containers or conveyors used for transporting concrete shall be mortar-tight. Irrespective of the method of transportation adopted, concrete shall be delivered with the required consistency and plasticity without segregation or loss of slump. However, chutes shall not be used for transport of concrete without the written permission of the Engineer and concrete shall not be re-handled before placing.
- b) **Re-tempered or Contaminated Concrete:** Concrete must be placed in its final position before it becomes too stiff to work. On no account, water shall be added after the initial mixing. Concrete which has become stiff or has been contaminated with foreign materials shall be rejected and disposed off as directed by the Engineer.
- c) **Cleaning of Equipment:** All equipment used for mixing, transporting and placing of concrete shall be maintained in clean conditions. All pans, buckets, hoppers, chutes, pipelines and other equipment shall be thoroughly cleaned after each period of placement.

1.7.8 Procedure for placing of concrete

- a) **Engineer's approval of Equipment & Methods:** Before any concrete is placed, the entire placing programme, consisting of equipment, layout, proposed procedures and methods shall be submitted to the Engineer for approval if so demanded by the Engineer and no concrete shall be placed until Engineer's approval has been received.

Equipment for conveying concrete shall be of such size and design as to ensure a practically continuous flow of concrete during depositing without segregation of materials, considering the size of the job and placement location.

- b) **Time Interval between Mixing and Placing:** Concrete shall be placed in its final position before the cement reaches its initial set and concrete shall normally be compacted in its final position within thirty minutes of leaving the mixer and once compacted, it shall not be disturbed.
- c) **Avoiding Segregation:** Concrete shall, in all cases, be deposited as nearly as practicable directly, in its final position, and shall not be re-handled for caused to flow in a manner which will cause segregation, loss of materials, displacement of reinforcement, shuttering or embedded inserts, or impair its strength. For locations where direct placement is not possible, and in narrow forms, the Contractor shall provide suitable drop and "Elephant Trunks" to confine the movement of concrete. Special care shall be taken when concrete is dropped from a height, especially if reinforcement is in the way, particularly in column and thin walls.
- d) **Placing by Manual Labour:** Except when otherwise approved by the Engineer, concrete shall be placed in the shuttering by shawls or other approved implements, and shall not be dropped from a height more than 1.0 m or handled in a manner which will cause segregation.
- e) **Placing by Mechanical Equipment:** The following specification shall apply when placing or concrete by use of mechanical equipment is warranted considering the nature of work involved. The control of placing shall begin at the mixer discharge. Concrete shall be discharged by a vertical drop into the middle of the bucket or hopper and this principle of a vertical discharge of concrete shall be adhered to throughout all stages of delivery until the concrete comes to rest in its final position.
 - 1. **Types of Buckets:** Central-bottom-dump buckets of a type that provides for positive regulation of the amount and rate of deposition of concrete in all dumping positions, shall be employed.
 - 2. **Operation of Bucket:** In placing concretes in large open areas, the bucket shall be spotted directly over the position designated and then lowered for dumping. The open bucket shall clear the concrete already in place and the height of drop; shall not exceed 1.0 m. The bucket shall be opened slowly to avoid high vertical bounce. Dumping of buckets on the swing or in any manner which results in separation of ingredients of disturbance of previously placed concrete will not be permitted.
- f) **Placement in Restricted Forms:** Concrete placed in restricted forms by barrows, bugles, cars, short chutes or hand shovelling shall be subject to the requirement for vertical delivery of limited height to avoid segregation and shall be deposited as nearly as practicably in its final position.
- g) **Chuting:** Where it is necessary to use transfer chutes, specific approval of Engineer must be obtained to type, length, slopes, baffles, vertical terminals and timing of operations. These shall be so arranged than an almost continuous flow of concrete is obtained at the

discharge and without segregation. To allow for the loss of mortar against the sides of the chutes, the first mixes shall have less coarse aggregate. During cleaning of chutes, the waste water shall be kept clear of the forms. Concrete shall not be permitted to fall from the end of the chutes by more than 1.0 m. Chutes, when approved for use, shall have slopes not flatter than 1 vert : 3 horiz. and not steeper than 1 vert. : 2 horiz. Chutes shall be of metal or metal lined end of rounded cross section. The slopes of all chute sections shall be approximately the same. The slopes of all chute sections shall be approximately the same. The discharge end of the chutes shall be maintained above the surface of the concrete in the forms.

h) Placing by Pumping/Pneumatic Placers:

1. Concrete may be conveyed and placed by mechanically operated equipment e.g., pumps or pneumatic placers only with the written permission of the Engineer. The slump shall be held to the minimum necessary for conveying concrete by this method.
2. When pumping is adopted, before pumping of concrete is started, the pipeline shall be lubricated with one or two batches of mortar composed of one-part cement and two parts sand. Care shall be taken to avoid stoppages in work once pumping has started.
3. When a pneumatic placer is used, the manufacturer's advice on layout of the pipeline shall be followed to avoid blockages and excessive wear. Restraint shall be provided at the discharge box to cater for the reaction at this end.
4. Manufacturer's recommendations shall be followed regarding concrete quality and all other related matters when pumping/pneumatic placing equipment is used.

i) **Concrete in Layers:** Concreting, once started, shall be continuous until the pour is completed. Concrete shall be placed in successive horizontal layers of uniform thickness ranging from 15 cm to 90 cm as directed by Engineer. These shall be placed as rapidly as practicable to prevent the formation of cold joints or planes of weakness between each succeeding layer within the pour. The thickness of each layer shall be such that it can be deposited before the previous layer has stiffened. The bucket loads or other units of deposit, shall be spotted progressively along the face of the layer with such overlap as will facilitate spreading the layer to uniform depth and texture with a minimum shovelling. Any tendency to segregation shall be corrected by shovelling stones into mortar rather than mortar on to stones. Such a condition shall be corrected by redesign of mix or other means, as directed by the Engineer.

j) **Bedding of Layers:** The top surface of each pour and bedding planes shall be approximately horizontal unless otherwise instructed.

k) **Compaction:** Concrete shall be compacted during placing, with approved vibrating equipment until the concrete has been consolidated to the maximum practicable density, is free of pockets of coarse aggregate and fits tightly against all form surfaces, reinforcement and embedded fixtures. Particular care shall be taken to ensure that all concrete placed against the form faces and into corners of forms or against hardened concrete at joints is free from voids or cavities. The use of vibrators shall be consistent with the concrete mix and caution exercised not to over vibrate the concrete to the

point that segregation results.

1. **Type of Vibrators:** Vibrators shall conform to IS specifications. Type of vibrator to be used shall depend on the structures where concrete is to be placed. Shutter vibrators to be effective, shall be firmly secured to the form work which must be sufficiently rigid to transmit the vibration and strong enough not to be damaged by it. Immersion vibrators in sufficient numbers and each of adequate size shall be used to properly consolidate all concrete. Tapping or external vibrating of forms by hand tools or immersion vibrators will not be permitted.
2. **Use of Vibrators:** The exact manner of application and the most suitable machines for the purpose must be carefully considered and operated by experienced men. Immersion vibrators shall be inserted vertically at points not more than 450 mm apart and withdrawn when air bubbles cease to come to the surface. Immersion vibrators shall be withdrawn very slowly. In no case shall immersion vibrators be used to transport concrete inside the forms. Particular attention be paid to vibration at the top of a lift e.g. in a column or wall.
3. **Melding Successive Batches:** When placing concrete in layers, which are advancing horizontally as the work progresses, great care shall be exercised to ensure adequate vibration, blending and melding of the concrete between the succeeding layers.
4. **Penetration of Vibrators:** The immersion vibrator shall penetrate the layer being placed and also penetrate the layer below while the under layer is still plastic to ensure good bond and homogeneity between the two layers and prevent the formation of cold joints.
5. **Vibrating against Reinforcement:** Care shall be taken to prevent contact of immersion vibrators against reinforcement steel. Immersion vibrators shall not be allowed to come in contact with reinforcement steel after start of initial set. They shall also not be allowed to come in contact with forms or finished surfaces.
6. **Use of Form Attached Vibrators:** Form attached vibrators shall be used only with specific authorization of the Engineer.
7. **Use of Surface Vibrators:** The use of surface vibrators will not be permitted under normal conditions. However, for thin slabs and similar construction, surface vibrating by specially designed vibrators may be permitted, upon approval of Engineer.
8. **Stone Pockets and Mortar Pongees:** The formation of stone pockets and mortar pondages in corners and against faces of forms shall not be permitted. Should these occur, they shall be dug out, reformed and refilled to sufficient depth and shape for thorough bonding, as directed by the Engineer.
- l) **Placement Interval:** Except when placing with slip forms, each placement of concrete in multiple lift work shall be allowed to set for at least 24 hours after the final set of concrete and before the start of a subsequent placement.
- m) **Special provision in placing:** When placing concrete in walls with openings, in floors of integral slabs and beam construction and other similar conditions, the placing shall stop when the concrete reaches the top of the opening in walls or bottom horizontal surface

of the slab, as the case may be. Placing shall be resumed before the concrete in place takes initial set, but not until it has had time to settle as determined by the Engineer.

- n) **Placing concrete through reinforcing steel:** When placing concrete through reinforcing steel, care shall be taken to prevent segregation of the coarse aggregate. Where the congestion of steel makes placing difficult, it may be necessary to temporarily move the top steel aside to get proper placement and restore reinforcing steel to design position and the Engineer's approval shall be obtained prior to adopting this method.
- o) Bleeding or free water on top of concrete being deposited into the forms, shall be cause to stop the concrete pour and the conditions causing this defect corrected before any further concreting is resumed.

1.7.9 Construction joints

- a) A construction joint is defined as a joint in the concrete introduced for convenience in construction at which special measures are taken to achieve subsequent continuity without provision for further relative movement.
- b) All the drawings submitted for the Engineer's approval shall indicate position of all construction joints and lifts. No concreting shall be started until the Engineer has approved the method of placing, the positions and form of the construction joints and lifts. The construction joints shall be so located as not to impair the strength of the structure. Rebates, keys or notches shall be formed and water stops inserted as the Engineer may require. The position of construction joints and the size of the formwork panels shall be so coordinated that where possible, the line of any construction joints coincides with the line of a formwork joint and that in any case all construction joint lines and formwork joint lines appear as a regular and uniform series. For all exposed horizontal joints and purposely inclined joints, a uniform joint shall be formed with a batten of approved dimensions to give a straight and neat joint line.
- c) Concrete placed to form the face of a construction joint shall have all laitance removed and the aggregate exposed prior to the placing of fresh concrete. The laitance shall wherever practicable be removed by spraying the concrete surface with water under pressure and be removed while the concrete is still green. The whole of the concrete surface forming part of the joint shall be hacked to expose the aggregate. Where aggregate is damaged during hacking, it shall be removed from the concrete face by further hacking. All loose matter shall be removed and the exposed surface thoroughly cleaned by wire brushing, air blasting or washing, leaving the surface thoroughly cleaned by wire brushing, air blasting or washing, leaving the surface clean and damp. Immediately before fresh concrete is placed, a 12 mm. thick layer of sand/cement mortar mixed in the same proportions as in the concrete shall be spread in the horizontal face of the construction joint. A drier mix shall be used for the top lift of horizontal pours to avoid laitance. The new concrete shall be well worked against the prepared face before the mortar sets. Special care shall be taken to obtain thorough

compaction and to avoid segregation of the concrete along the joint plane.

1.7.10 Movement Joints

- a) Movement joints are defined as all joints intended to accommodate relative movement between adjoining parts of a structure, special provision being made where necessary for maintaining the water tightness of the joint. The Contractor shall comply with the instructions of manufacturers of proprietary jointing materials and shall, if required by the Engineer, demonstrate that the jointing materials can be applied satisfactorily.
- b) The Contractor shall show locations of all movement joints and details thereof on drawings submitted for the Engineer's approval.
- c) The surface of set concrete in a movement joint shall, as shown on the Drawings, be painted with two coats of bituminous paint and new concrete shall be placed against it only when the paint is dry. Expansion joints shall be formed by a separating strip of approved preformed joint filler.
- d) Caulking grooves shall be provided as shown on the specification drawings. At all joints where a caulking groove is formed, immediately prior to caulking, the groove shall be wire brushed and loose material removed and blown out by compressed air. After the groove has dried, it shall be primed and caulked with approved sealing compound applied in accordance with the manufacturer's instructions. At all caulked joints, the face of the strip and a width of concrete on either side shall be painted with two coats of paint having the same base as the sealing compound.

1.7.11 Water stops and joint fillers

- a) At all construction, contraction and expansion joints in the water retaining structures and wherever specified or directed by the Engineer, water stops shall be provided. The water stops shall be P.V.C., type or of any other equivalent material as approved by the Engineer. P.V.C., water stops shall have a tensile strength of not less than 14 MN/m² and elongation at break of not less than 300%. Water stops shall not be exposed to direct sunlight for long periods. Before being concreted in, water stops shall be cleaned of all foreign materials. Wherever provided, water stops shall be placed in such a manner that they are embedded in the adjacent sections of the panels for equal width.
- b) As far as possible, jointing on site shall be confined to the making of butt joints in straight runs of water stops. Where it is agreed with the Engineer that it is necessary to make an intersection or change of direction of any joint, other than a butt joint in a straight run on site, a preliminary joint, intersection or change of direction piece shall be made and submitted to such tests as the Engineer may require.
- c) Flexible water stops shall be fully supported in the formwork, free of nails and clear of reinforcement and other fixtures. Damaged water stops shall be replaced and during concreting care shall be taken to place the concrete so that water stops do not bend or distort.

- d) The different types of water stop to be used in liquid retaining structures will be as follows:

Type of joint	Type of water stops
Partial/complete contraction joint in walls and slabs	150mm wide, ribbed with hollow center bulb & 5mm minimum thickness
Expansion joints in walls and slabs	225mm wide, ribbed with hollow center bulb & 9mm minimum thickness
Construction joint in raft	225mm wide, ribbed with hollow center bulb & 9mm minimum thickness
Construction joint in wall	150 mm wide, ribbed with hollow center bulb & 5mm minimum thickness
Expansion joint raft	225mm wide, ribbed with hollow center bulb & 9mm minimum thickness

- e. Jointing Fillers: Joint fillers shall be of durable, compressible, and non-extruding material.

1.7.12 Sealing compounds

Horizontal joints shall, where used in water retaining structures be sealed with a cold pouring polysulphide rubber sealing compound of quality equal to, or better than serviced "Paraseal". Horizontal joints in roofs, floors and other non-water retaining structures shall be sealed with an approved sealant with properties equal to or better than serviced "Paraplastic 41". Vertical joints and joints in the soffits of slabs in both water retaining as well as non-water retaining structures shall be sealed with a trowel or gun applied polysulphide rubber sealing compound such as serviced "Vertiseal" or equivalent. Sealing compounds shall be fully cured before water is permitted to come in contact. At 40° Centigrade, the curing time would be approximately 7 weeks for polysulphide compound.

1.7.13 Curing, protecting, repairing and finishing

a) Curing

1. All concrete shall be cured by keeping it continuously damp for the period of time required for complete hydration and hardening to take place. Preference shall be given to the use of continuous sprays, or ponded water, continuously saturated coverings of sacking, canvas, hessian or other absorbent materials, or approved effective curing compounds applied with spraying equipment capable of producing a smooth, even - textured coat. Extra precautions shall be exercised in curing concrete during cold and hot weather as outlined hereinafter. The quality of curing water shall be the same as that used for mixing concrete.
2. Certain types of finish or preparation for over flying concrete must be done at certain stages of the curing process and special treatment may be required for specific concrete surface finish.
3. Curing of concretes made of high alumina cement and super-sulphated cement shall be carried out as directed by the Engineer.

(A) Curing with Water: Fresh concrete shall be kept continuously wet for a minimum period of 10 days from the date of placing of concrete, following a lapse of 12 to 14 hours after laying concrete. The curing of horizontal surfaces exposed to the drying winds shall however begin as soon as the concrete has hardened. Water shall be applied to formed surfaces immediately upon removal of forms. Quantity of water applied shall be controlled so as to prevent erosion of freshly placed concrete.

(B) Continuous Spraying: Curing shall be assured by use of an ample water supply under pressure in pipes, with all necessary appliance of hose, sprinklers and spraying devices. Continuous fine mist spraying or sprinkling shall be used, unless otherwise specified or approved by the Engineer.

(C) Alternate curing methods:

Whenever, in the judgment of the Engineer, it is necessary to omit the continuous spray method, a covering of clean sand or other approved means such as wet gunny bags, which will prevent loss of moisture from the concrete, may be used. No type of covering will be approved which would stain or damage the concrete during or after the curing period. Covering shall be kept continuously wet during the curing period.

1. For curing of concretes in pavements, side-walks, floors, flat roofs of other level surfaces, the ponding method of curing is preferred. The method of containing the ponded water shall be approved by the Engineer. Special attention shall be given to edges and corners of the slabs to ensure proper protection to these areas. The ponded areas shall be kept continuously filled with water during the curing period.
2. Curing compound: Surface coating type curing compounds shall be used only by special permission of Engineer. Curing compounds shall be liquid type white pigmented, conforming to U.S. Bureau of Reclamation specification. No curing compound shall be used on surfaces where future blending with concrete, water of acid proof membrane, or painting is specified.
3. Curing equipment: All equipment and materials required for curing shall be on hand and ready for use before concrete is placed.

b) Protecting Fresh Concrete

Fresh concrete shall be protected from defacements and damage due to construction operations by leaving forms in place for an ample period as specified later in this specification. Newly placed concrete shall be protected by approved means such as tarpaulins from rain, sun and winds. Steps as approved by the Engineer shall also be taken to protect immature concrete from damage by debris, excessive lading, vibration, abrasion or contact with other materials, etc., that may impair the strength and/or durability of the concrete. Workmen shall be warned against and prevented from disturbing green concrete during its setting period. If it is necessary that the workmen enter the area of freshly placed concrete, the Engineer may require that bridges be placed over the area.

c) Repair and Replacement of Unsatisfactory Concrete

1. Immediately after the shuttering is removed, the surface of concrete shall be very carefully gone over and all defective areas called to the attention of the Engineer who may permit patching of the defective areas or also reject the concrete unit either partially or in its entirety. Rejected concrete shall be removed and replaced by the contractor. Holes mortar composed of one part of cement to one and half parts of sand passing 2.36 mm I.S. sieve after removing any loose stones adhering to the concrete. Concrete surfaces shall be finished as described in specifications or as directed by the Engineer.
2. Superficial honey combed surfaces and rough patches shall be similarly made good immediately after removal of shuttering, in the presence of the Engineer and superficial water and air holes shall be filled in. The mortar shall be well worked into the surface with a wooden float. Excess water shall be avoided. Unless instructed otherwise by the Engineer, the surface of the exposed concrete placed against shuttering shall be rubbed down immediately on removal of shuttering to remove fine or other irregularities, care being taken to avoid damaging the surface. Surface irregularities shall be removed by grinding.
3. If reinforcement is exposed or the honeycombing occurs at vulnerable positions e.g., ends of beams or columns, it may be necessary to cut out the member completely or in part and reconstruct. The decision of the Engineer shall be final in this regard. If only patching is necessary, the edges being cut perpendicular to the affected surface or with a small undercut if possible. Anchors, tees or dovetail slots shall be provided whenever necessary to attach the new concrete securely in place. An area extending several centimetres beyond the edges and the surfaces of the prepared voids shall be saturated with water for 24 hours immediately before the patching material is placed.
4. Use of Epoxy: The use of epoxy for bonding fresh concrete used for repairs will be permitted upon written approval of the Engineer. Epoxies shall be applied in strict accordance with the instructions of the manufacturer.
5. Method of Repair: Small size holes having surface dimensions about equal to the depth of the hole, holes left after removal of form bolts, grout insert holes and slots cut for repair of cracks shall be repaired as follows. The hole to be patched shall be roughened and thoroughly soaked with clean water until absorption stops. A 5 mm thick layer of grout of equal parts of cement and sand shall be well brushed into the surface to be patched, followed immediately by the patching concrete which shall be well consolidated with a wooden float and left slightly proud of the surrounding surface. The concrete patch shall be built up in 10 mm thick layers. After an hour or more, depending upon weather conditions, it shall be worked off flush with a wooden float and a smooth finish obtained by wiping with hessian. A steel trowel shall be used for this purpose. The mix for patching shall be of the same materials and in the same proportion as that used in the concrete being repaired, although some reduction in the maximum size of the coarse aggregates may be necessary and the mix shall be kept as dry as possible. Mortar filling by air pressure (guniting) shall be used for repair of areas too large and / or too shallow for patching with mortar. Patched surfaces shall be given a final treatment to match the colour and texture of the surrounding concrete. White

cement shall be substituted for ordinary cement, if so directed by the Engineer, to match the shade of the patch with the original concrete.

6. Curing of Patched Work: The patched area shall be covered immediately with an approved non-staining, water-saturated material such as gunny bags which shall be kept continuously wet and protected against sun and wind for a period of 24 hours. Thereafter, the patched area shall be kept wet continuously by a fine spray, or sprinkling for not less than 10 days. All fillings shall be tightly bonded to the concrete and shall be sound, free from shrinkage cracks after the fillings have been cured and dried.
7. Approval by the Engineer: All materials, procedures and operations used in the repair work shall be subject to the approval of the Engineer.

d) Finishing

1. The type of finish for formed concrete surfaces shall be as follows, unless varied by the Engineer.
2. When the structure is in service all the surfaces shall receive no special finish, except repair of damaged or defective concrete, removal of fine and abrupt irregularities, filling defective concrete, removal of fins and abrupt irregularities, filling of holes left by form ties and rods and clean-up of loose or adhering debris.
3. Surfaces which will be exposed to the weather and which would normally be level, shall be sloped for drainage. Unless a horizontal surface or the slope required is specified, the tops of narrow surfaces such as stair treads, walls, curbs and parapets shall be sloped across the width approximately 1 in 30. Broader surfaces such as walkways, roads, parking areas and platforms shall be sloped about 1 in 50. Surfaces that will be covered by backfill or concrete, sub floors to be covered with concrete topping, terrazzo or quarry tiles, and similar surfaces shall be smooth screeded and levelled to produce even surfaces. Surface irregularities shall not exceed 6 mm. Surfaces which will not be covered by backfill, concrete or tile toppings such as outside decks, floors of galleries and sumps, parapets, gutters, side-walks, floors and slabs, shall be consolidated, screeded and floated. Excess water and laitance shall be removed before final finishing. Floating may be done with hand or power tools and started as soon as the screeded surface has attained a stiffness to permit finishing operations and these shall be the minimum required to produce a surface uniform in texture and free from screed marks or other imperfections. Joints and edges shall be tooled as specified or as directed by the Engineer.
4. Standard finish for exposed concrete: Exposed concrete shall mean any concrete, other than floors or slabs, exposed to view upon completion of the works. Unless otherwise specified, the standard finish for exposed concrete shall be a smooth finish. A smooth finish shall be obtained with the use of lined or plywood forms having smooth and even surface and edges. Panels of forms shall be of uniform size and be as large as practicable and installed with closed joints.

TENDER DOCUMENT
(National Competitive Bidding)

For
Tender no. TNPCB/TCCL/2025

**Request for Proposal (RFP) for Execution of Interim
Remediation at the Chrome Contaminated site at M/s TCCL,
Ranipet, Tamil Nadu.**

Volume-II, Section 2
Technical specifications



TAMIL NADU POLLUTION CONTROL BOARD
76, Mount Salai, Guindy, Chennai – 600 032

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1. Executive Summary:

M/s Tamil Nadu Chromates and Chemicals Limited (TCCL), located in the SIPCOT Industrial Estate, Ranipet, Vellore, Tamil Nadu, operated from 1976 to 2001, generated and disposed large quantities of hexavalent chromium-bearing chromite ore processing residue in open land. The improper storage and disposal of this hazardous waste led to the leaching of hexavalent chromium, resulting in subsurface percolation and widespread contamination of soil and groundwater in the surrounding areas.

The TCCL site covers approximately 18.27 acres (73,916 m²), while the TCCL-SWDA spans around 12.63 acres (51,108 m²). The study area, encompassing a micro-watershed of 5.93 km², is located between the coordinates 12°55'2.13" N, 79°16'45" E in the north and 12°59'5" N, 79°18'22" E in the south, within Vellore, Tamil Nadu. This region falls under the Survey of India Topo sheet number D44T5.

TCCL operated between 1975 and 1995, manufacturing chemicals such as sodium dichromate, basic chromium sulphate, and sodium sulphate. The primary raw materials used included chromite ore, limestone, and soda ash. An estimated 2.27 lakh tons of hazardous waste were generated and stored on-site, with hexavalent chromium identified as the key contaminant, as per the Environmental Resources Management (ERM) assessment.

The primary goal of the interim remedial measures is to prevent rainfall from infiltrating the stockpiled hazardous waste and further contaminating groundwater. Additionally, the design aims to control and prevent surface water runoff during rainfall events from flowing beyond the TCCL premises into the neighbouring areas.

The interim remedial measures include the following steps:

1.0 Phase 1 - Capping the Existing Hazardous Waste Stockpile

As shown on attached design drawings in Volume –IV, a significant portion of the hazardous waste is stockpiled towards the northern end of the property (referred to as Area 1). The stockpiled area needs to be covered with a 2mm HDPE, double-side textured geo-membrane and a soil cover with a vegetative cover. Contractor to move waste stockpile away from the existing boundary wall approximately 6 meters inward to facilitate construction of a storm water drainage system which will also be used to anchor the HDPE membrane.

2.0 Phase 2 - Capping New Stockpile Area

Contractor needs to create a new area south of Area 1 to stockpile sluffed hazardous waste and construction debris (referred to as Area2). This area will be used to stockpile all the demolished construction waste and sluffed hazardous

waste. Contractor must demolish all the concrete and iron structures and then segregate waste for stockpiling. Anything that can be recycled shall be recycled. The stockpiled area needs to be covered with a 2mm HDPE double-side textured geo-membrane and a soil cover with a vegetative cover. As shown on attached design drawings, contractor has to move stockpile away from the existing boundary wall approximately 6 meters inward to facilitate construction of a storm water drainage system which will also be used to anchor the HDPE membrane.

3.0 Phase 3 - Construction of Drainage Network

A storm water drainage is required to ensure that the rainwater run-off from the capped area does not enter the waste. A storm water drain per specifications provided, with a slope towards the pond needs to be constructed. The goal is not to allow rainwater to flow outside the TCCL property. All storm water will flow towards and will be stored in a pond created onsite.

4.0 Phase 4 - Construction of A Collection Pond for Rainwater Collection

The area identified as Area 3 will be graded to native soil and graded to allow all flow towards the pond. A collection pond (100 x 50 x 3) meter to be constructed in the south west end of the TCCL property. The excavation should have a slope of 3H/1V. A 30-cm thick soil layer will be placed at the bottom of the excavation and compacted. A 2mm HDPE geo-membrane layer will then to be placed on top of the soil subsurface per design drawings. All constructed drainage networks should be connected with this layered pond. An overflow outlet shall be constructed in the south of the pond.

5.0 Phase 5 - Construction of New Boundary Wall, Access Pathway and Cut-off Trench

A new 8 feet high boundary wall needs to be constructed per design details around the perimeter of the site. Contractor to provide a 15 feet-wide gate and 6 feet wide gate (direction) for access in the wall. (as shown in drawings) In addition, contractor to provide metal spikes on top of the wall to prevent climbers. A 3-meter depth cut-off trench needs to be constructed in the Northern boundary of the site to reduce the groundwater movement into the site from the upstream. Cut-off trench will be routed with the storm water drainage inside the site. In addition, two piezo metric wells shall be installed in both up gradient (two wells) and down gradient (two wells) to monitor the groundwater.

The scope of work involves the construction of a 2-meter wide paver block pathway starting from the south side entrance gate, leading to Area 1. The pathway will encircle the boundary of Area 1, ensuring future pedestrian access,

and will connect back to the entrance. Given the available space around Area 1, the width of the pathway around this area will be 1 meter, while the remainder of the pathway will have a width of 2 meters. The work includes clearing and preparing the ground, laying a solid foundation, installing high-quality paver blocks, and ensuring proper drainage. The pathway will be finished with edge restraints, sand filling between pavers to ensure stability.

2. Technical Specification:

The technical specification provided herein along with the design drawings shall be used by the contractor to develop costs and execute the interim remediation. Design and drawing for the implementation of the interim remediation are attached in Annexure A.

Submittals

- The Contractor shall submit the following during installation:
 1. Quality control documentation prior to deployment of the FML.
 2. Soil Analytical data 5 days prior to construction start date.
 3. Implementation schedule
 4. Plan for transport and storage of membrane rolls
 5. Plan for transport and storage of cover layer soil.
 6. Welding equipment and method to be used.
- Shall submit after completion of installation:
1. Certification that the FML was installed in accordance with the manufacturer and project specifications with a listing of changes and Engineer's written approval of such, where applicable.
 2. Manufacturer's materials warranty and the contractor's installation warranty.
 3. As-built drawings depicting the locations of panels, seams, patches, and destructive sample locations, including numbering associated with said locations.

2.1. Allowable Slopes

Any permanent slopes required for the project shall not exceed 3H:1V (horizontal to vertical) unless otherwise specified on the plans.

2.2. Erosion Control

The faces of cut and fill slopes shall be prepared and maintained to control against erosion. No sediment shall be allowed to leave the site. A warranty of two year of erosion free construction has to be provided by the contractor.

2.3. Earthwork

General soil fill will be used for capping soil layer over the HDPE-geo-membrane liner. Fill soils used in cover construction shall be entirely upon Engineer approval. In addition, the suitability of all fill materials intended for use shall be subject to approval by the Engineer based on the laboratory analytical data. General soil fill shall be free of bush, roots, sod, or other deleterious or unsuitable materials. The Contractor shall provide all necessary labour and equipment to remove such materials from borrow and fill areas.

Contractor should procure the soil that is not contaminated. The contractor shall provide the source of soil to be procured. Imported soil will be stored on visqueen or tarp on site to ensure that the fresh soil will not come into contact with waste. All analytical reports need to be maintained onsite. Contractor must arrange grass on top of the soil cover.

The low permeability (10⁻⁷ cm/sec) vegetative cover soil layer to be placed on top of the HDPE- geo-membrane liner, shall be a minimum of 60 cm thick. A geocell (30cm height) network to be placed on the geo-membrane liner prior to slop protection. The vegetative soil layer shall be nominally compacted to approximately 85 percent of maximum dry density. The objective of the compaction is to ensure that landslide or sluffing of soil layer does not occur. Soil should not erode during normal rain fall events. Fill shall be placed and compacted over the HDPE- geo-membrane liner using low-ground-pressure equipment and in such a manner as to protect the HDPE-geo-membrane liner from potential damage. The Contractor shall present a plan for cover soil placement and compaction to the Engineer for approval prior to implementation.

2.4. Geo-membrane Liner

The flexible membrane liner (FML) shall consist of 2 mm thickness high density polyethylene (HDPE) double side textured. The geo-membrane shall be manufactured, installed and inspected in accordance with the manufacturer's recommendations and these specifications.

Alternate HDPE products can be used but all products should be only Made in India products. The total quantity of FML required is estimated to be 50,000 Square meters.

Contractor to procure, transport and store the membrane rolls as per specifications included in Annexure B.

2.5. General

1. The Contractor shall furnish and install the FML and all miscellaneous materials incidental to the liner system and install in accordance with specifications provided in Annexure B, and Drawings.
2. The alignment and areas for FML placement are shown in the Drawings. Exact locations and lengths may vary to suit field conditions and approved by the Engineer.
3. The Contractor shall furnish sufficient material to provide the finished FML's shown in the Drawings; including material for all seams, laps and anchors. The Contractor shall balance the actual project FML requirements, as determined from the Contractor's quantity take-offs, against those shown in the Drawings.
4. Installation and testing shall be in general conformance with Standard Specification for "Test Methods, Test Properties and Testing Frequency for High Density Polyethylene (HDPE) Textured Geo-membranes – included in Annexure B."

2.6. Delivery, Storage, and Handling

Transportation:

The FML shall be packaged and shipped in such a manner that the material is not damaged or exposed to damaging substances. Transportation shall be the responsibility of the Contractor of shipment of FML to the project site.

Off-Loading:

Off-loading of the FML is the responsibility of the Contractor. No off-loading of FML shall be performed unless the TNPCB or Engineer representative is present. Any damage to the rolls during off-loading shall be documented by the TNPCB/Engineer representative and the Contractor. All damaged rolls must be stored separate from the undamaged rolls. The rolls shall be unrolled to determine the extent of the damage. The use of the roll or portions of the damaged roll shall be only at the approval of the Engineer. The cost of evaluating, replacing or repairing rolls damaged during off-loading shall be the sole responsibility of the Contractor.

Storage:

The FML shall be stored such that it is protected from UV rays, puncture, dirt, grease, gasoline, diesel fuel, water, moisture, mud, mechanical abrasion, excessive heat, and other causes of damage to the FML material.

The rolls shall be stored on a prepared surface. Storage on wooden pallets or other

surfaces that may damage the FML material shall not be permitted. The rolls shall not be stacked more than two high and shall be stacked in such a manner that the roll number of each roll is easily visible.

Rolls without the proper documentation (specific roll number) shall be stored separately until all the required documentation is received and approval given for deployment by the Engineer.

2.7. Warranty

All workmanship furnished by the Contractor under this Specification shall be guaranteed by the Contractor against failure due to improper installation for a period of not less than five (5) years. All permanent HDPE materials furnished by the Contractor under this Specification shall be guaranteed by the Contractor and the FML manufacturer for a period of not less than ten (10) years.

Upon written notice that the material fails to meet scope of work identified herein, or of failure of guaranteed materials or workmanship during the guarantee period the Contractor shall promptly furnish and install new materials and/or furnish the workmanship necessary to correct the failure at the expense of the Contractor.

2.8. Deployment

Prior to the deployment of a FML roll the following documentation must be reviewed and approved by the Engineer:

- 1.0 The visible roll or panel number on the roll or panel prior to unrolling or unfolding.
- 2.0 Manufacturer's certification sheet.
- 3.0 Production Log which includes that particular roll or panel number.
- 4.0 The resin certificate for the resin that was used to manufacture that particular roll of FML.

No roll of liner will be deployed without the review and acceptance by the Engineer of the required documentation. The contractor shall be solely responsible for expenses incurred for replacing rolls without the required documentation, replacing rolls that are rejected upon review of the required documentation, delays in deployment while waiting for the required documentation, and expenses incurred due to excessive amounts of time spent by the TNPCB/Engineer representative reviewing incomplete documentation.

2.9. Handling of FML During Deployment:

1. The contractor shall use equipment that will not damage the FML during moving and deployment.

2. The Contractor shall not use equipment or methods that damage the subgrade (during pond construction) during deployment.
3. The Engineer will be the final authority regarding damage to the FML or to the subgrade. Items that will be considered as damage to the FML will include, but not be limited to, abrasions, tears, punctures, gas, oil or fuel spills, and creases
4. Dragging the FML across the ground will not be permitted.
5. Methods that are found to consistently damage the liner or the subgrade, as determined by the Owner and Engineer, will be terminated immediately upon written notice to the Installation Contractor.
6. Damage to the FML or to the subgrade, during handling and deployment of the FML shall be repaired at the sole expense of the Installation Contractor.

2.10. Deployment Pattern:

1. The deployment pattern used shall be according to the deployment plan submitted as required by the Engineer. Any deviation from the Engineer approved deployment pattern must be approved, in writing, by the Engineer prior to implementation.
2. The individual panels shall be numbered in a logical sequence and in accordance with the Engineer approved deployment plan.
3. The deployment pattern shall be such that seams are oriented parallel to any slopes. No horizontal seams shall be placed along a slope. Overlap seams shall be placed in a downhill direction.
4. No base or T-seams shall be placed within 5 feet from the toe of the slope.
5. To the extent possible, the number of field seams shall be minimized.

2.11. Tension and Excess Material:

1. The FML shall be installed in a relaxed condition and shall be free of tension or stress on completion of the installation.
2. Stretching of the liner to fit will not be allowed.
3. The installation shall contain slack material sufficient to allow for shrink and wrinkles as per the FML manufacturer's recommendations. Individual wrinkles shall take the form of undulations of the FML material and shall not be enough to allow the material to fold back over itself. Corrective measures will depend upon the cause and severity and will be determined by the Engineer.

2.12. Daily Deployment:

1. No more FML shall be deployed than can be seamed during that day's shift.
2. FML that is deployed and not welded during that day will be inspected prior to seaming for damage and moisture. If it is determined that the FML has been damaged, it shall be repaired or removed. If it is determined that the FML has become wet or moist it shall be dried prior to seaming to the satisfaction of the Engineer.

2.13. Weather Conditions:

1. No FML shall be deployed during periods of precipitation, high winds, in the presence of excessive moisture, or in areas of excessive moisture in the subgrade.
2. No geo-membrane material shall be seamed when the sheet temperature is > 60 °C as measured by an infrared thermometer or surface thermocouple unless otherwise approved by the Engineer/TNPCB representative.

2.14. Ballast Against Wind Uplift:

1. Adequate loading shall be placed on the FML to prevent uplift due to wind. Loading may be temporarily achieved through the use of sand bags filled with clean sand and securely tied. The bags shall not be tied with any type of material that will potentially damage the FML. The removal of spilled sand bag contents will be the responsibility of the Installation Contractor.
2. Loose piles of material will not be allowed for use as ballast.
3. Sand bags shall not be thrown on top of or allowed to slide down slopes on top of the FML.

2.15. General Workmanship

1. The Contractor shall perform all installation Work to conform to the best recognized practice to achieve a neat and functional installation.
2. The Contractor shall exercise proper precaution to verify all measurements before laying out the work. He will be held responsible for any errors that otherwise might have been avoided. He shall carry on the Work systematically and so manage it at all times as to secure substantial progress.
3. No equipment will be allowed upon the FML that has not been previously approved by the Engineer. Additional equipment shall only be placed upon the FML with the prior written approval of the Engineer.

4. Splash pads shall be placed beneath any type of equipment that may leak or splash gasoline, diesel fuel, oil or grease upon the FML. The splash pad shall be large enough to contain any splash or leak. Prior to placing the splash pad the FML shall be swept clean of any material that may puncture or abrade the FML.
5. All hoses or cords that are used upon the liner shall be handled such that they will not damage the FML. Hoses and cords shall not be dragged across material that may abrade the FML.
6. If tired equipment is approved by the Engineer for use upon the FML, the tired equipment shall be moved with extreme care. The tired equipment shall be lifted over berms and wrinkles in the FML.
7. There shall be no smoking on the FML.
8. There shall be no hard objects allowed to come in contact with the FML without the use of a rub sheet.

2.16. Welding

Please refer to manufacturers specification for welding of FML included in Section 3. Contractor shall comply with manufacturers specification for welding of FML. Additional details of welding and related equipment is included herein.

Only welding equipment that is on the equipment list and has been approved by the Engineer shall be used to seam the FML. As a minimum the welding and related equipment shall meet the following requirements:

1. The welding equipment shall be equipped with a means of measuring and displaying the temperature within the apparatus and at the nozzle for extrusion welders and hot air welders, and at the wedge(s) for fusion welders.
2. The power source shall be capable of providing a constant voltage under varying line loads.
3. Contractor should arrange of necessary electric power. No power is available onsite.

2.17. Welding Procedures for HDPE:

Welding shall only be performed when the FML is clean and free of dust, dirt, grease, oil, other foreign substances, and dry. Solvents or adhesives shall not be used unless the specific product is approved, in writing, by the Engineer.

1. Fusion Welding:
 - a. The geo-membrane shall have an overlap between 100 to 150 millimetres.

- b. The area shall be prepared by wiping the area with a clean dry cloth to remove any foreign matter.
- c. The welder shall be inserted at one end of the seam, then the pressure rollers are to be clamped down and the wedge engaged and drive motor turned on. If the welder is interrupted during the seaming process, the area affected shall be marked and repaired.

The welding machine is aligned and set to the required temperature depending on the material thickness and air-temperature, and the machine travel speed is to set to the required setting for the applicable material thickness and surface temperature.

2. Extrusion Welding:

- a. Surface of FML to be extrusion welded shall be ground.
- b. Grind surface no more than thirty (30) minutes prior to seaming.
- c. Grind such that surface oxidation is removed, no more than 5 percent of the liner thickness is removed, and the ground area shall extend no more than 1/4 inch beyond the edge of the extruded bead.
- d. Extruded bead shall be placed over excess grinder marks.
- e. Grinder marks shall be perpendicular to the seam.
- f. The edge of the overlying panel shall be bevelled.
- g. The two pieces to be seamed together shall be tack bonded together such that they will be held in place during seaming, there is no damage to the FML, seaming operations can be effectively performed, and construction quality control CQC testing can be effectively performed.
- h. Extrudate shall be extruded from the welder to expose fresh Extrudate prior to welding.

3. Restarting Procedures:

- a. Overlap existing weld a minimum of 2 inches.
- b. Grind existing weld prior to initiating welding.
- c. Resume welding where the grinding starts.
- d. Resuming seaming operations on any weld that is over five (5) minutes old shall be considered a restart.

4. Reseaming Procedures:

- a. Grind existing seam prior to re-welding.
- b. Bevelling shall be performed during grinding.

- c. No seam shall be re-seamed without grinding prior to initiating Reseaming operations.

T-Welds:

No T- welds will be allowed and all intersections of liner that form a T or + shall be capped with no exceptions.

Fish-Mouths and Wrinkles:

Fish-mouths and excessive wrinkles shall be cut and overlapped a minimum of 5 inches and seamed. Where the 5-inch overlap cannot be achieved, a cap shall be seamed in-place extending a minimum of 6 inches beyond the cut.

2.18. Defects and Repairs

The FML sheet and the welding shall be observed by the Engineer and Installation Contractor for holes, creases, underlying projections, blisters, undispersed raw materials, contamination by foreign material, and other damage.

If installed FML becomes covered or too dirty for effective inspection, the Installation Contractor shall clean the FML such that effective inspection can be made.

Repairs made due to damage resulting from the Installation Contractor's activities shall be at the sole expense of the Installation Contractor.

Areas of the FML installation that cannot be effectively repaired, shall be capped or the damaged FML removed and replaced, as determined by the Engineer.

Areas of LHDPE FML requiring repair shall be ground prior to welding as required in these Specifications.

All FML surfaces shall be clean and dry prior to repairing.

All HDPE patches and caps shall be ground, bevelled, extend at least 6 inches beyond the cut or defect, and have rounded corners.

Repair methods are listed below for specific types of damage. Types of damage that are not covered in these Specifications shall be repaired as determined by the Owner.

Small Holes: A hole or defect smaller than 1/4-inch in diameter in its largest dimension shall be considered a small hole. Small holes shall be repaired by grinding and extruding a bead for HDPE. The repair shall be non-destructively tested.

Large Holes: A hole or defect larger than 1/4-inch in its smallest dimension shall be considered a large hole. Large holes shall be repaired by patching. The repair shall be non- destructively tested. For large patches, the repair may be destructively tested.

Projections Under the FML: Projections under the FML shall be removed by making as small a cut as possible. The cut shall be patched. The repair shall be non-destructively tested.

Abrasions and Nicks: Abrasions and cuts requiring repair shall be determined by the

Engineer, and will include grinding and extruding a bead for HDPE.

Creases: Creases requiring repair shall be determined by the Engineer, and will include grinding and extruding a bead for HDPE.

Bridging of the FML:

Excessive bridging (trampolining) of the FML will not be allowed. The degree of excessive bridging will be determined during cool morning weather. Excessive bridging of the FML liner will be determined by the weight of a 180-pound man standing on the bridged FML. If the imposed weight does not force the liner to the underlying subgrade during the coolest site conditions encountered during construction, bridging must be removed. The installation Contractor will take steps to remove excessive bridging without cutting and patching the FML, unless approved by the Engineer. Such steps will include temporarily sandbagging the toe of the slopes over a period of a day or more, and/or temporarily removing material from anchor trenches to allow adjustment.

2.19. Vegetative Cover

The final cover of the area shall be planted with a mixture of native grasses to establish a vegetative cover. Native plant material selected to be shallow rooted, low nutrient requiring and drought resistant. The grass seed mix shall be applied at a rate of 15 kg per acre. Prior to planting, the soil surface shall be prepared for a seed bed using a disc or rototiller device. The vegetative mixture shall be distributed along with the appropriate nutrients (nitrogen, phosphorus, and potassium) by hydro seeding onto the cover. Straw mulch shall be placed on the surface to facilitate germination and control soil erosion.

2.20. Boundary wall

A new 8 feet high boundary brick wall shall be constructed around the perimeter of the site. Details of wall dimensions and other material details are presented in the design drawings.

2.21. Cut-off Trench

A cut-off trench shall be constructed on the northern boundary of the site, with details of the trench, including dimensions and other specifications, presented in the design drawings. In addition, four piezometric wells shall be installed to monitor groundwater, with two wells placed in the up-gradient area and two in the down-gradient area. The wells shall be installed with two having a depth of 6 metres and the other two having a depth of 10 metres, all with a standard piezometric well

diameter. The total depth and well screening will be finalised based on site conditions at the time of drilling.

2.22. Access Pathway

The scope of work involves the construction of a 2-meter wide paver block pathway starting from the south side entrance gate, leading to Area 1. The pathway will encircle the boundary of Area 1, ensuring future pedestrian access, and will connect back to the entrance. Given the available space around Area 1, the width of the pathway around this area will be 1 meter, while the remainder of the pathway will have a width of 2 meters. The work includes clearing and preparing the ground, laying a solid foundation, installing high-quality paver blocks, and ensuring proper drainage. The pathway will be finished with edge restraints, sand filling between pavers to ensure stability.

3. Specification for Geo-membrane installation

3.1. Standard test method

This specification refers to the following Standards, Specifications.

ASTM D 6392: Standard Test Methods for Determining the Integrity of Non-Reinforced

Geo-membrane Seams Produced Using thermos Fusion Methods

ASTM D 5641: Standard Practice for Geo-Membrane Seam Evaluation by Vacuum Chamber

ASTM D 5820: Standard Practice for Pressurized Air Channel Evaluation of Dual Seamed Geo-membranes

ASTM D 7240: Standard Practice for Leak Location using Geo-membranes with an Insulating Layer in Intimate Contact with a Conductive Layer via Electrical Capacitance Technique (Conductive Geo-membrane Spark test).

ASTM D 6497: Standard Guide for Mechanical Attachment of Geo-membrane to penetrations or Structures.

GRI Test Method GM 9: Standard Practice for Cold Weather Seaming of Geo-membranes

GRI Test Method GM13: Test Properties, Testing Frequency and Recommended Warranty for High Density Polyethylene (HDPE) Smooth and Textured Geo-membranes.

GRI Test Method GM 17: Test Properties, Testing Frequency and Recommended Warranty for Linear Low Density Polyethylene (LLDPE) Smooth and Textured Geo-membranes.

GRI Test Method GM14: Selecting Variable Intervals for taking Geo-Membrane Destructive Seam Samples Using the Method of Attributes

GRI Test Method GM19: Standard Specification for Seam Strength and related Properties of thermally Bonded Polyolefin Geo-membranes

3.2. Geo-membrane Installation Quality Assurance

Subgrade Preparation

The surface must be smooth; with no rapid changes of grade such as steps or settlement next to concrete structures. All slopes and surfaces must be compacted to ensure the integrity of the membrane. While the geomembrane should be designed to withstand some differential settlement, an analysis of these areas must be made by the engineer to ensure that the stresses on the membrane are acceptable.

The surface of the subgrade must be free of sharp rocks, penetrating debris or other appurtenances that could damage the membrane. Typically, finished subgrade is achieved using a smooth steel drum roller or other method as approved by the Project Engineer. Under certain conditions a cushion geotextile can be evaluated and approved for use by the Project Engineer, to provide protection for the geomembrane.

The installer site engineer shall approve the surface (daily basis) on which the geomembrane will be installed and acceptance of the finished subgrade surface shall be recorded.

Geo-membrane Panel Deployment

Prior to the beginning of a project, it is recommended that the geo-membrane installer produces a proposed panel layout showing the location of all geo-membrane panels. This panel layout is submitted for information purposes based on the project plans and specifications.

During the geo-membrane liner installation, the panel deployments might differ from the proposed layout as construction progresses. The actual field panel deployments to be shown on the as-built drawing of the project.



Picture 1: Geo-membrane Panel Deployment

Please follow below International Regulations for the Geo-membrane panel deployment

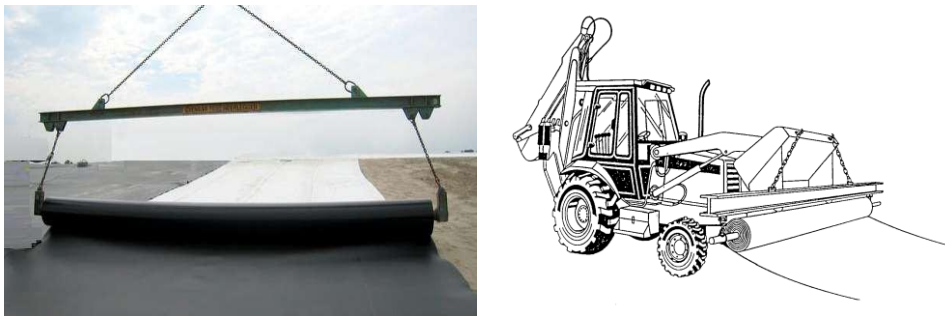
- No geo-membrane material shall be unrolled and deployed if the material temperatures are lower than 0° C unless otherwise approved by the Owner's Representative. The specified minimum temperature for material deployment may be adjusted by the Owner's Representative based on recommendations from the manufacturer. Typically, only the quantity of geo-membrane that will be anchored and seamed together in one day should be deployed.
- Nobody shall be allowed to walk on the geo-membrane when wearing shoes that could puncture, scratch or otherwise damage the geo-membrane.
- No vehicular traffic shall travel on the geo-membrane other than an approved low ground pressure all- terrain vehicle or equivalent.
- Sand bags or equivalent ballast shall be used as necessary temporarily to hold the geo-membrane material in position under the foreseeable and reasonable-expected wind conditions. Sand bag material shall be sufficiently close-woven to prevent soil fines from escaping from the bags and discharging onto the geo-membrane.

Picture2: hold the geo-membrane with sand bag in position



- Geo-membrane placement shall not be done if moisture prevents proper sub grade preparation, panel placement, or panel seaming.
- Geo-membrane placement shall not be done if moisture prevents proper sub grade preparation, panel placement, or panel seaming.
- Damaged panels which have been rejected shall be marked and their removal from the work area recorded.
- The geo-membrane shall not be allowed to "bridge over" (also called trampolining) voids or low areas in the sub grade. In these areas, the geo-membrane shall rest in intimate contact with the sub grade.
- Wrinkles caused by panel placement or thermal expansion should be minimized.

Picture 6: Deployment of 8.00 m wide geo-membrane Panel process



- In general: seams shall be oriented parallel to the line of the maximum slope. In corners and odd shaped geometric locations, the total length of field seams shall be minimized. Seams shall not be located at low points in the subgrade unless geometric requires seaming at such locations and if approved by the Owner's Representative.
- The panels shall be overlapped prior to seaming. The special white lines parallel to the edges shows the correct overlapping dimension to avoid mistakes.
- The geo-membrane panels shall not be deployed and left unseamed overnight. Whenever weather conditions are uncertain or not adequate for field seaming, no geo-membrane deployment shall take place. Extreme temperatures, high humidity, rain, high winds etc. are all conditions not favourable to field seaming. The site foreman and QC-Inspector should determine whether seaming can be conducted properly so as to achieve quality seams.
- The geo-membrane material must be homogeneous, free from pores and, apart from a surface texture that may be required, it must be smooth and watertight. The Geo-membranes shall be produced in the factory with protective plastic film strips along the bonded edges which can be removed without a residue.

Site seaming procedure

All seams must be oriented down the slope, not across. Tie - in seams (perpendicular to the slope) should be located within 1.5 m (5 feet) from the toe of the slope in the flat area. The field documentation of the seaming procedures shall include the seam number, date and time of seaming, technician welder, seam length and reference to the corresponding test seam (calibration). Seaming shall be a continuous process with a minimum of interruptions along any given seam. Prior to seaming, the geo-membrane shall be overlapped a minimum of 100 millimetres for extrusion welding and 100 to 150 millimetres for fusion welding. Any geo-membrane area showing injury due to excessive scuffing, puncture, or distress from any cause shall, at the discretion of the installer's onsite supervisor, be repaired or replaced with an additional piece of geo-membrane.

No geo-membrane material shall be seamed when the sheet temperature is > 60 °C as measured by an infrared thermometer or surface thermocouple unless otherwise approved by the Owner's representative.

Seaming shall primarily be performed using automatic fusion welding equipment and techniques. Extrusion welding shall be used where fusion welding is not possible such as pipe penetrations, patches, repairs and short (less than a roll width) runs of seams.



Picture 3: welding of geo-membrane

Excessive wrinkles at the seam overlaps shall be minimized and when necessary cut along the ridge of the wrinkles into the panel so as to effect a flat overlap. The cut shall be terminated with a keyhole cut (nominal 10 mm (½ in) diameter hole) so as to minimize crack/tear propagation. The overlay shall subsequently be seamed. The key hole cut shall be patched with an oval or round patch of the same base geo-membrane material extending a minimum of 150 mm (6 in.) beyond the cut in all directions.

Welding Procedures

Environmental Conditions

Liner deployment is permitted at ambient temperatures between 5°C and 45°C. Above 45°C, seaming is allowed only after verification. Deployment is restricted during excessive moisture (fog, rain, dew) or strong winds. Seaming humidity limits: <83% for extrusion welding, <90% for hot wedge welding. Higher humidity requires verification before proceeding.

Fusion Welding

Geo-membrane overlap: 100–150 mm. The area must be cleaned before welding. The welder is positioned at the seam's start, pressure rollers engaged, and the drive motor activated. Any interruption requires marking and repair. Machine settings vary based on material thickness and temperature:

Wedge temperature: 310°C–420°C

Mating force: 30–40 N/mm

Speed: 1–2 m/min

Trial welds are conducted per shift, ensuring seam integrity through peel and shear strength tests.

Extrusion Welding

Weld areas must be sanded (≤ 0.02 mm depth), overlapped (≥ 100 mm), and kept clean and dry. Panels are tack bonded without damage. Grinding should not exceed 10% of liner thickness. The welding rod, made from the same resin, is melted inside the extrusion welder, with preheating ensured for effective bonding. Operators monitor speed and alignment throughout.

Cross-Seams

Excess overlap is removed, and cross seams are ground smooth before welding. Fusion-welded cross seams require an additional extrusion bead extending 100 mm in all directions, forming a "T" shape. Seams run parallel to slopes.

Cold Weather Seaming

Welding in colder temperatures (-10°C to -20°C) requires equipment adjustments and trial welds in the same conditions. Preheating may be necessary in cloudy or windy weather. No welding is allowed during precipitation. Snow and ice must be removed with rubber-tired blowers or plastic shovels to prevent slippage.

4. Quality Control

4.1 Site Test Equipment

The installer shall maintain functional testing equipment, including:

- **Field Tensiometer** – Certified motor-driven unit for tensile testing (certified within 12 months).
- **Vacuum Box** – Rigid housing with a transparent window, neoprene gasket, vacuum gauge, and soap solution for seam inspection.
- **Air Pressure Test Equipment** – Used for split hot wedge seams, capable of generating 1.5–2.0 bars with a manometer for pressure monitoring.

4.2 Non-Destructive Testing

- **Test Seams** – Conducted at the start of each shift and after equipment breakdowns. Seams are tested for peel and shear strength.
- **Vacuum Testing** – Used for extrusion and "T" seams. A vacuum of 0.2 bars is maintained to detect leaks using a soap solution. Defective seams are marked and repaired.
- **Air Pressure Testing** – Double wedge seams are pressurized to 2.0 bars for 60 seconds. Pressure loss exceeding 0.2 bars results in seam rejection and required repairs.
- **Spark Testing** – Conducted using ASTM D 7240 if a conductive geo-membrane is used.
- **Full-Length Testing** – All seams undergo non-destructive testing and are documented. Alternative methods require prior approval.

4.1. Destructive Testing

- **Sampling Frequency** – One test per 150 meters of seam, determined by the inspector.
- **Sample Testing** – 300 mm x 1.0 m samples undergo shear and peel tests using a field tensiometer. Strength must meet project specifications (Tables 2 & 3).
- **Lab Testing** – Additional testing may be performed at the owner's discretion.
- **Repair & Retesting** – Sampled areas are repaired and tested using a vacuum box.

4.2. Inspection & Acceptance

- The inspector documents all repairs and ensures compliance with quality standards.
- Seams must pass all tests before acceptance; failed seams require reconstruction.
- The geo-membrane surface is inspected for defects and must be clean for evaluation.
- No overburden is placed until the liner system is inspected and approved.

5. Installation Defects and Repairs

The geo-membrane shall be inspected for defects, including holes, blisters, undispersed raw materials, and contamination. It must remain clean during inspection, and all defects must be repaired and tested before covering.

5.1. Assessment

Suspect areas in seams and non-seam sections undergo non-destructive testing. Failed sections are marked and repaired.

5.2. Repair Procedure

Defective seams: Cap-stripped or replaced.

Small holes: Extrusion welded; patched if >5 mm.

Tears: Patched; sharp-ended tears on slopes must be rounded.

Blisters, large cuts, or raw material defects: Repaired with patches.

Patches: Round/oval, made from the same geo-membrane, extending at least 150 mm beyond defects. Extrusion welding is used, ensuring proper cleaning and grinding.

5.3. Repair Testing

All repairs undergo non-destructive testing. Failed repairs must be redone and retested until passing.

6. Completion of Work

The geo-membrane installation is considered complete when all deployment, seaming, repairs, testing, and site clean-up (including sandbags) are finished. The installer must submit all required certifications, and the Owner or their Representative must confirm compliance with the specified requirements.

TENDER DOCUMENT
(National Competitive Bidding)
For
Tender no. TNPCB/TCCL/2025

**Request for Proposal (RFP) for Execution of Interim Remediation
at the Chrome Contaminated site at M/s TCCL, Ranipet,
Tamil Nadu.**

Volume-III
DRAWINGS



TAMIL NADU POLLUTION CONTROL BOARD
76, Mount Salai, Guindy, Chennai – 600 032

Attachments:

ANNEXURE A: - DESIGN DRAWINGS

Attachment - Design Drawing 1 - Detailed Site Plan

Attachment - Design Drawing 2 - Preliminary surface, Drainage, Cutoff trench and Pond creation

Attachment - Design Drawing 3 - Existing Sluffed Waste Area and Proposed Building Debris and Sluffed Waste Stockpile Area

Attachment - Design Drawing 4 – Destruction of Existing On-Site Buildings

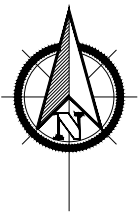
Attachment - Design Drawing 5 – Construction Details for Pond

Attachment - Design Drawing 6 - Existing Sluffed Waste Area and Liner Details – Area 1

Attachment - Design Drawing 7 - Proposed Stockpile Area of Debris and Sluffed Waste and Liner Details – Area 2

Attachment - Design Drawing 8 - Construction Details of New Boundary Wall and Cutoff Trench Details

ATTACHMENT
ANNEXURE - A
DESIGN DRAWINGS



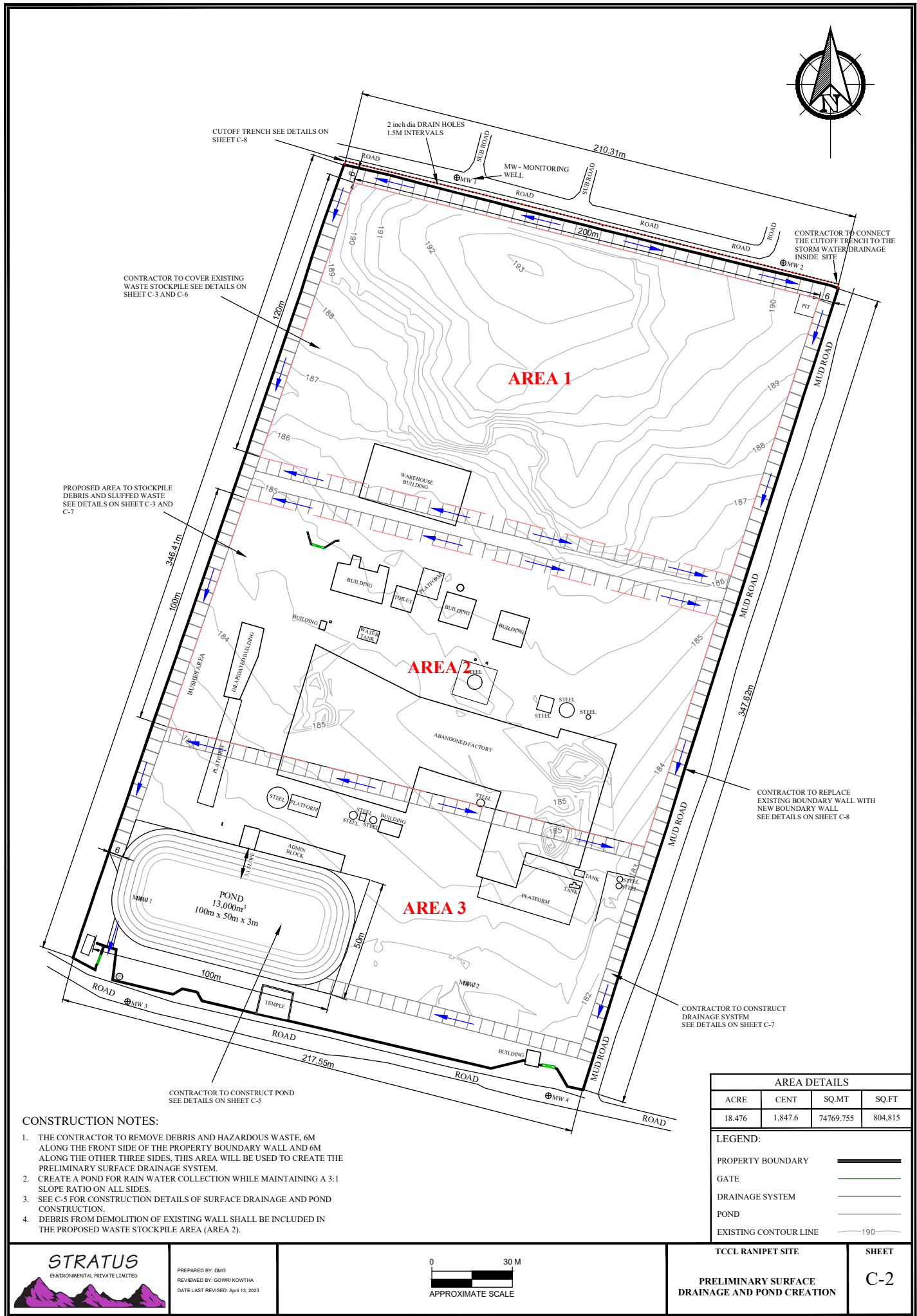
LEGEND:	
PROPERTY BOUNDARY	=====
GATE	=====



PREPARED BY: DMG
 REVIEWED BY: GOWRI KOWTHA
 DATE LAST REVISED: March 13, 2023

TCCL RANIPET SITE
 SITE LOCATION MAP

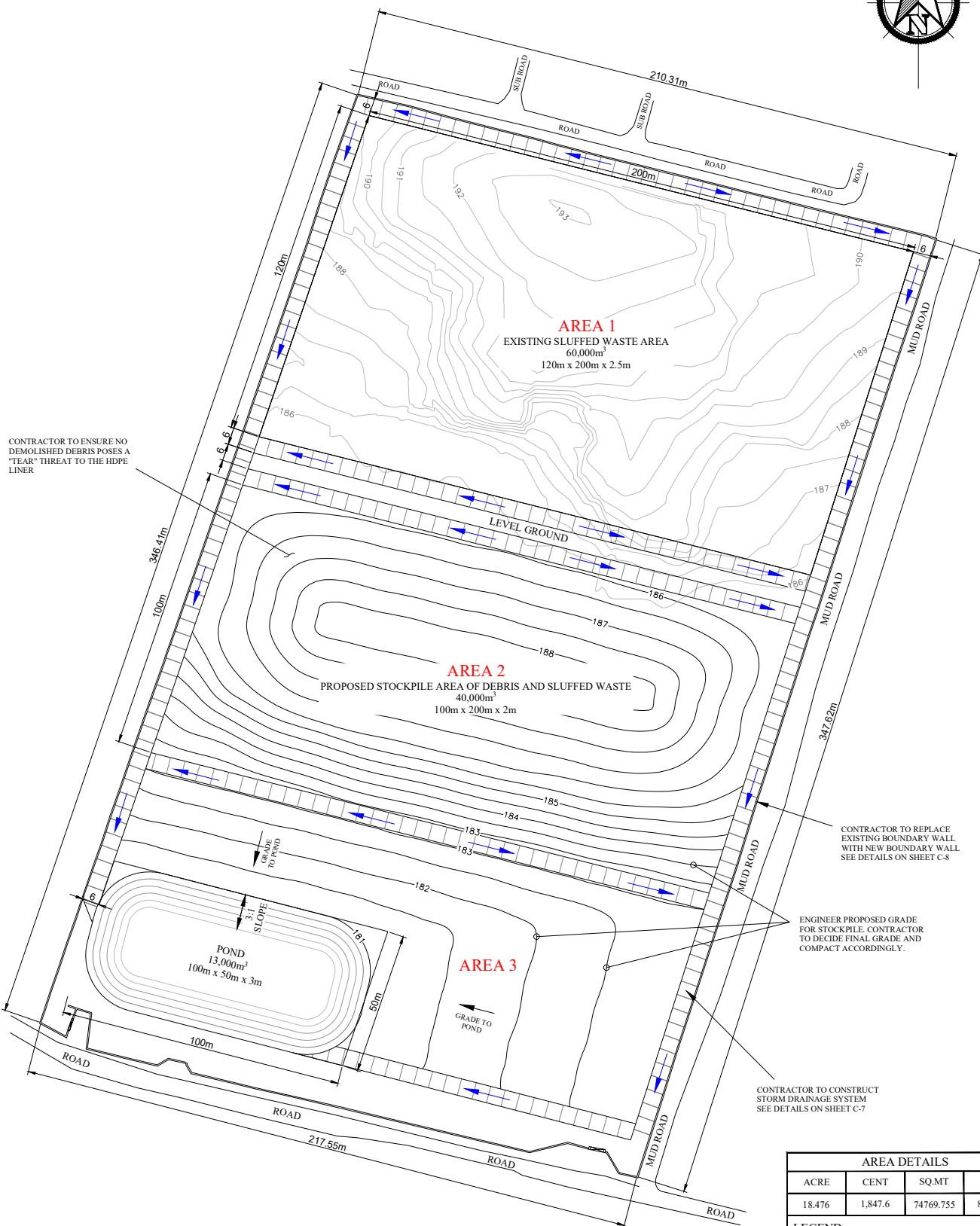
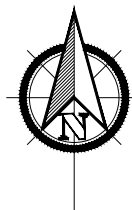
SHEET
 C-1



1. THE CONTRACTOR TO REMOVE DEBRIS AND HAZARDOUS WASTE, 6M ALONG THE FRONT SIDE OF THE PROPERTY BOUNDARY WALL AND 6M ALONG THE OTHER THREE SIDES, THIS AREA WILL BE USED TO CREATE THE PRELIMINARY SURFACE DRAINAGE SYSTEM.
2. CREATE A POND FOR RAIN WATER COLLECTION WHILE MAINTAINING A 3:1 SLOPE RATIO ON ALL SIDES.
3. SEE C-5 FOR CONSTRUCTION DETAILS OF SURFACE DRAINAGE AND POND CONSTRUCTION.
4. DEBRIS FROM DEMOLITION OF EXISTING WALL SHALL BE INCLUDED IN THE PROPOSED WASTE STOCKPILE AREA (AREA 2).

AREA DETAILS			
ACRE	CENT	SQ.MT	SQ.FT
18.476	1,847.6	74769.755	804,815

LEGEND:	
PROPERTY BOUNDARY	
GATE	
DRAINAGE SYSTEM	
POND	
EXISTING CONTOUR LINE	

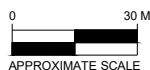


CONSTRUCTION NOTES:

1. THE CONTRACTOR TO COMBINE AND COMPACT DEMOLISHED DEBRIS (BUILDINGS, ETC.) AND SLUFFED WASTE TO FORM A NEW HILL OF WASTE IN THE PROPOSED STOCKPILE AREA (AREA 2).
2. CONTRACTOR TO USE IMPORTED SOIL AS NECESSARY FOR CREATION OF SECOND STOCKPILE AND TO ATTAIN ADEQUATE COMPACTION.



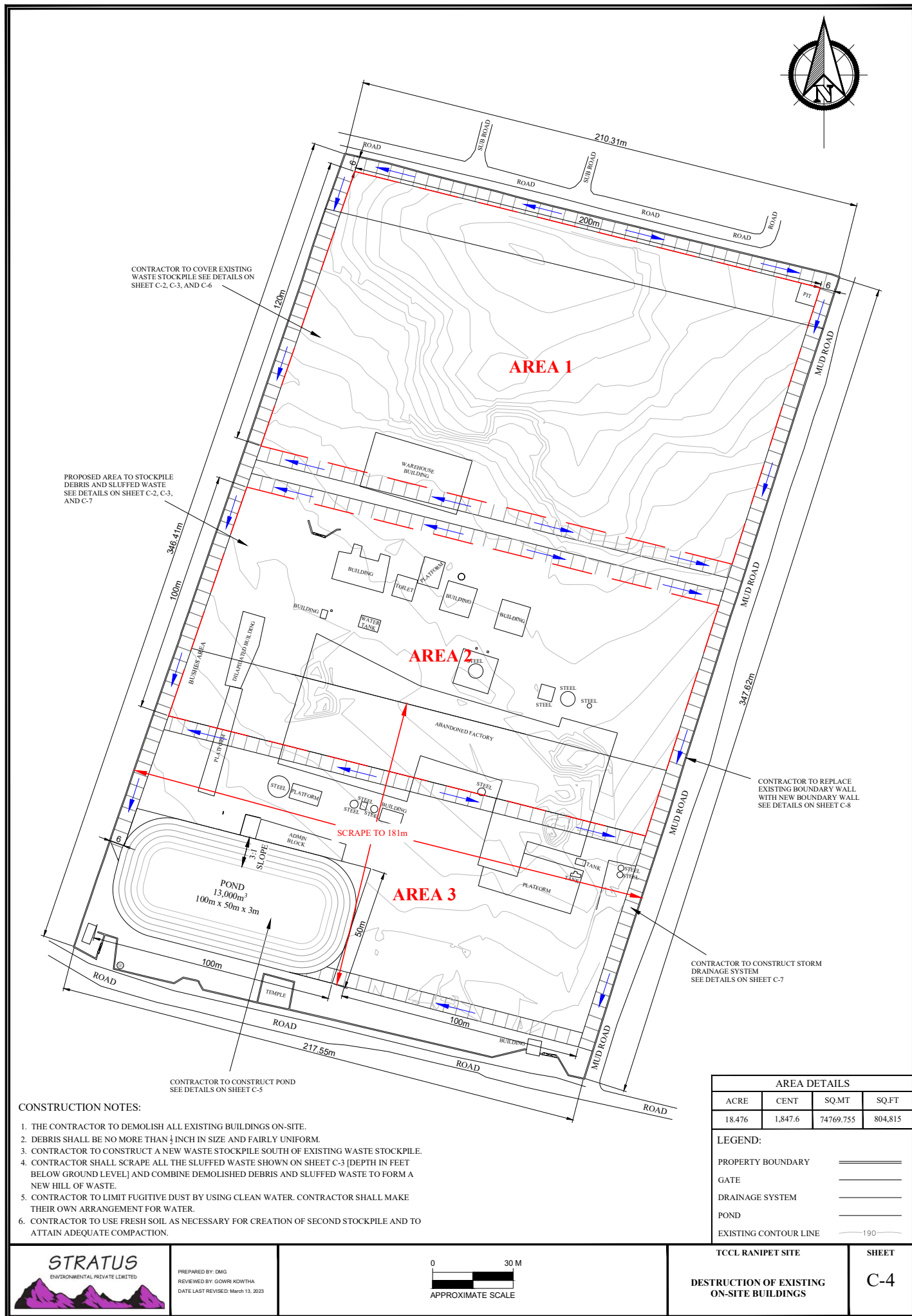
PREPARED BY: DMG
REVIEWED BY: GOWRI KOWTHA
DATE LAST REVISED: March 13, 2023



TCCL RANIPET SITE
EXISTING SLUFFED WASTE AREA
AND PROPOSED BUILDING
DEBRIS AND SLUFFED WASTE
STOCKPILE AREA

SHEET
C-3

AREA DETAILS			
ACRE	CENT	SQ.MT	SQ.FT
18.476	1,847.6	74769.755	804,815
LEGEND:			
PROPERTY BOUNDARY		=====	
GATE		=====	
DRAINAGE SYSTEM		=====	
POND		=====	
EXISTING CONTOUR LINE		—190—	
PROPOSED CONTOUR LINE		—182—	



1. THE CONTRACTOR TO DEMOLISH ALL EXISTING BUILDINGS ON-SITE.
2. DEBRIS SHALL BE NO MORE THAN $\frac{1}{4}$ INCH IN SIZE AND FAIRLY UNIFORM.
3. CONTRACTOR TO CONSTRUCT A NEW WASTE STOCKPILE SOUTH OF EXISTING WASTE STOCKPILE.
4. CONTRACTOR SHALL SCRAPE ALL THE SLUFFED WASTE SHOWN ON SHEET C-3 [DEPTH IN FEET BELOW GROUND LEVEL] AND COMBINE DEMOLISHED DEBRIS AND SLUFFED WASTE TO FORM A NEW HILL OF WASTE.
5. CONTRACTOR TO LIMIT FUGITIVE DUST BY USING CLEAN WATER. CONTRACTOR SHALL MAKE THEIR OWN ARRANGEMENT FOR WATER.
6. CONTRACTOR TO USE FRESH SOIL AS NECESSARY FOR CREATION OF SECOND STOCKPILE AND TO ATTAIN ADEQUATE COMPACTION.

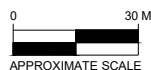
AREA DETAILS			
ACRE	CENT	SQ.MT	SQ.FT
18.476	1,847.6	74769.755	804,815

LEGEND:

PROPERTY BOUNDARY	=====
GATE	=====
DRAINAGE SYSTEM	=====
POND	=====
EXISTING CONTOUR LINE	————— 190 —————



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REVIEWED BY: GOWRI KOWTHA
DATE LAST REVISED: March 13, 2023

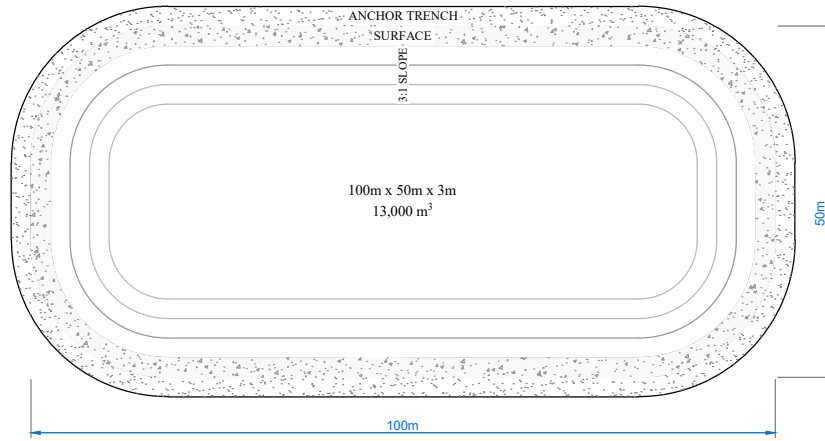
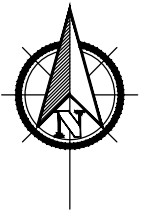


TCCL RANIPET SITE

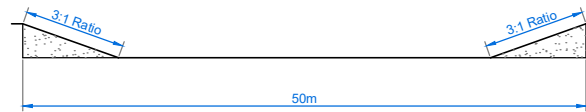
DESTRUCTION OF EXISTING ON-SITE BUILDINGS

SHEET

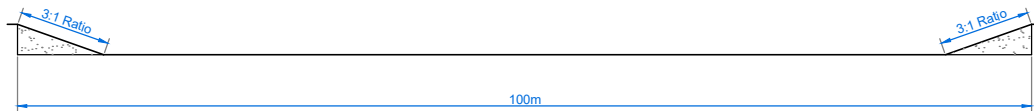
C-4



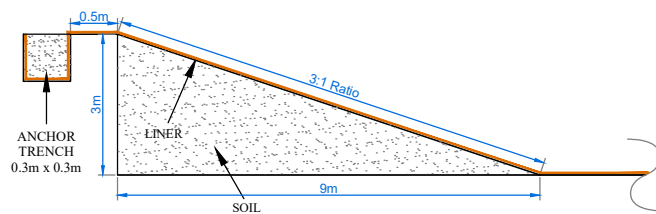
POND DEPTH ELEVATION PLAN VIEW



POND CROSS SECTION OF N-S



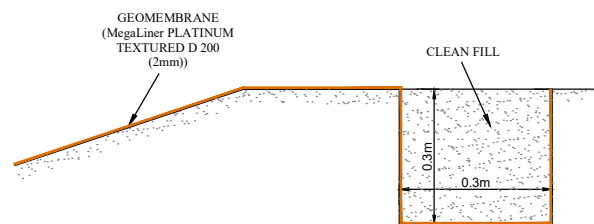
POND CROSS SECTION OF E-W



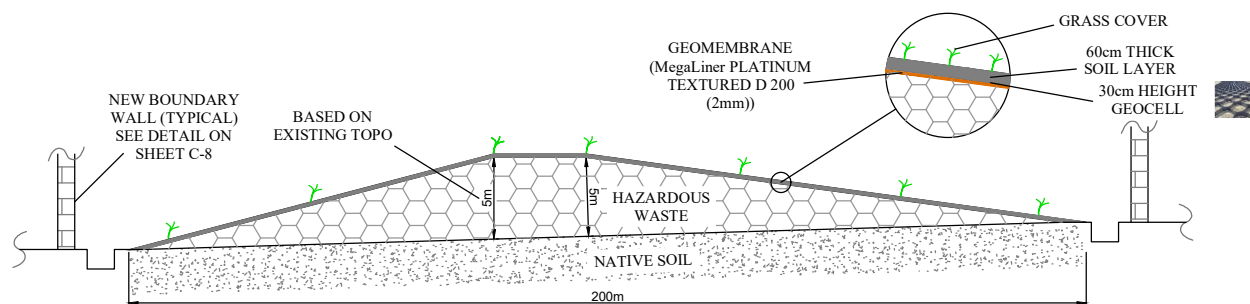
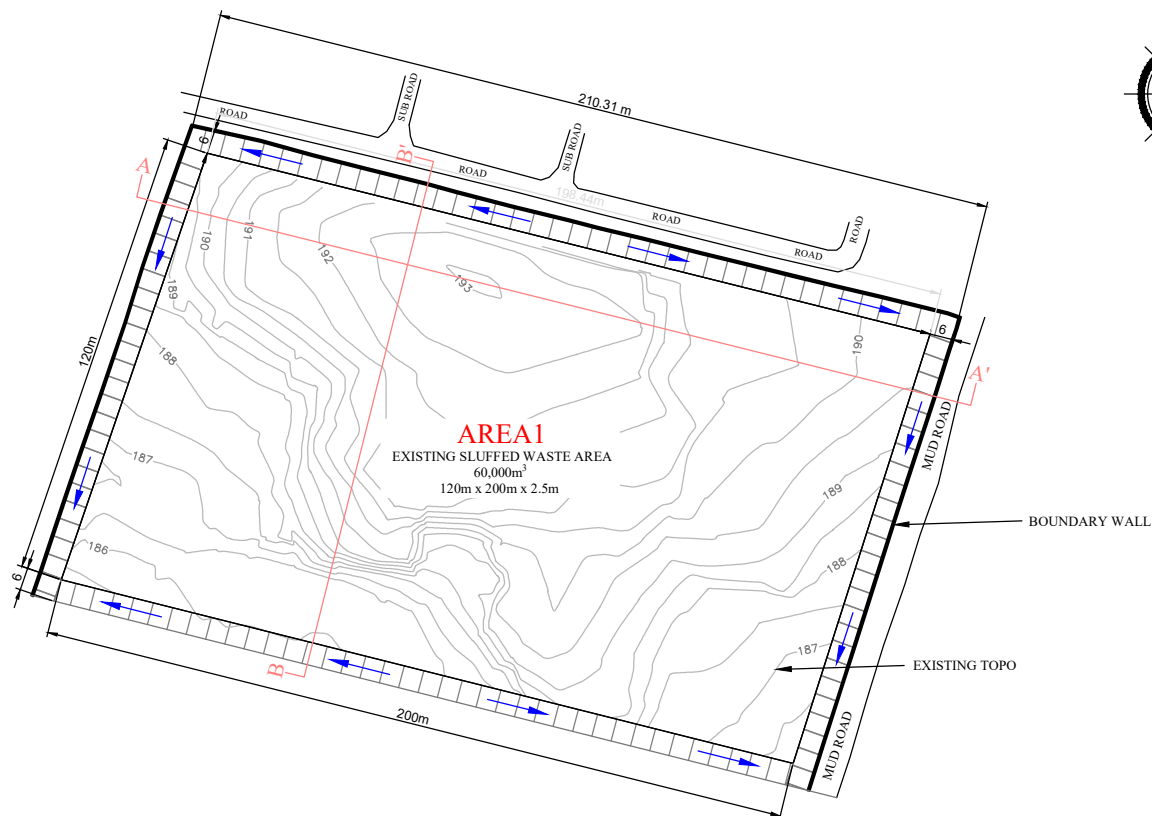
POND SLOPE AND LINER ANCHOR DETAILS

CONSTRUCTION NOTES:

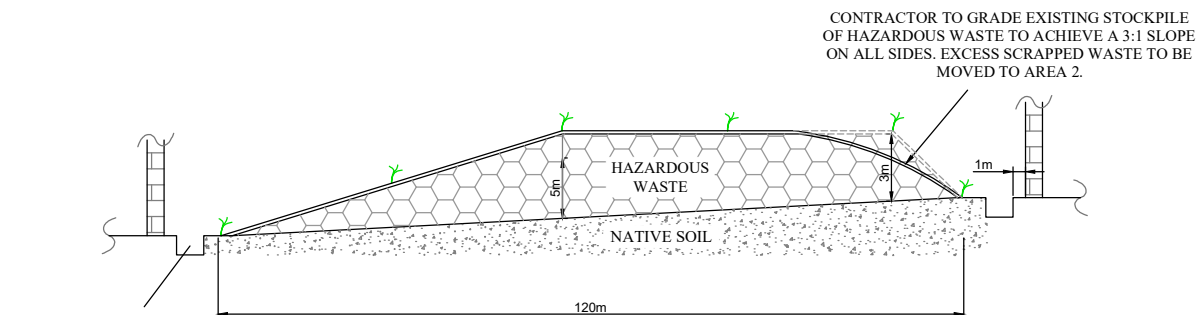
1. THE CONTRACTOR TO CONSTRUCT A TRENCH AROUND THE POND FOR LINER ANCHORING.
2. POND SLOPE RATIO 3:1 SHALL BE MAINTAINED ON ALL SIDES.
3. HDPE LINER WILL BE EXPOSED
4. AN OVER FLOW OUTLET TO PROVIDE SAME AS ANCHOR TRENCH SPECIFICATIONS



POND ANCHOR TRENCH DETAIL
NOT TO SCALE

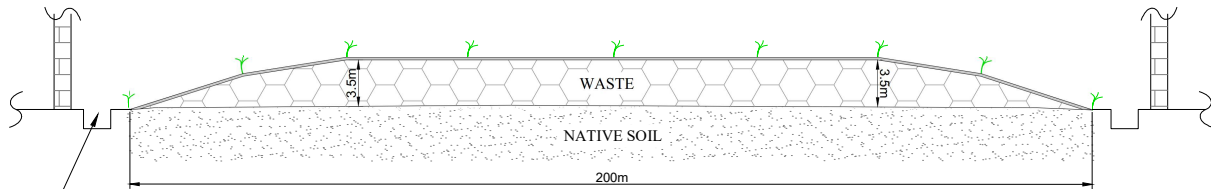
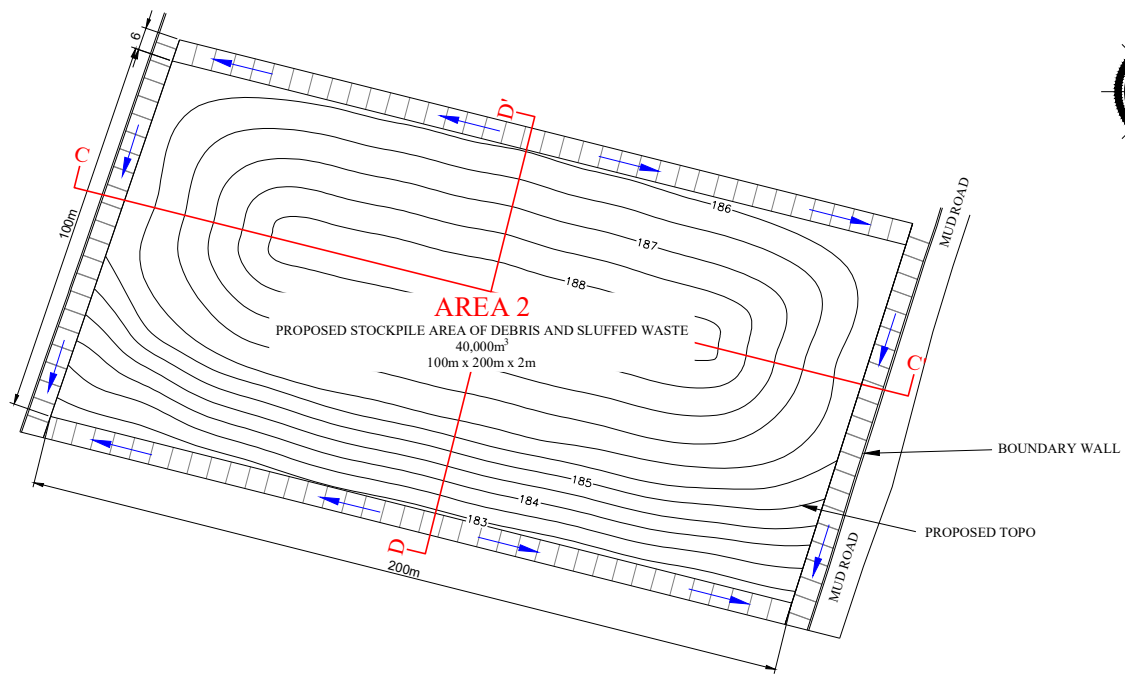
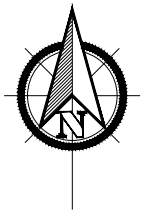


CROSS SECTION OF A-A'
NOT TO SCALE

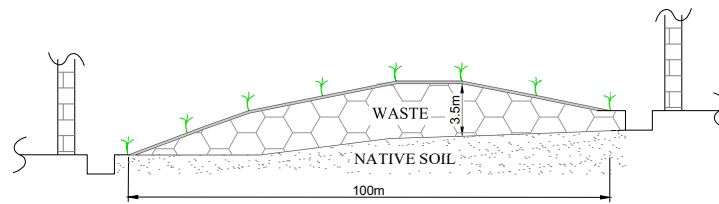


CROSS SECTION OF B-B'
NOT TO SCALE

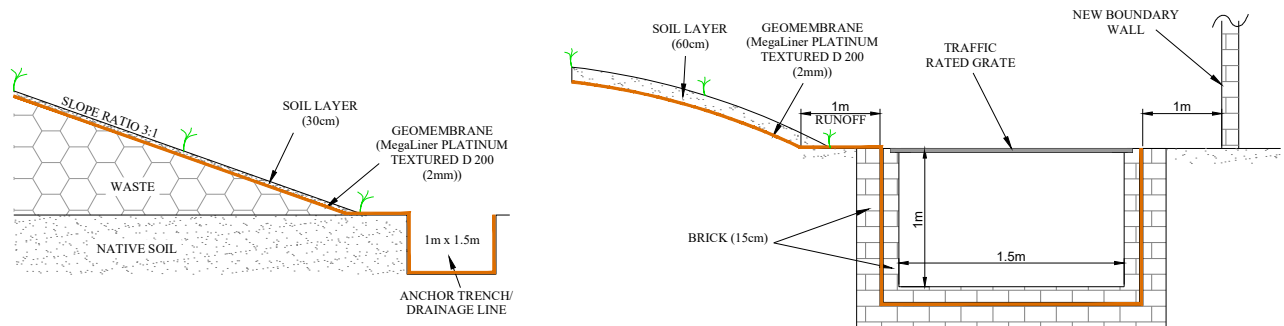
SEE ANCHOR
TRENCHING DETAIL
(TYPICAL) ON SHEET C-7



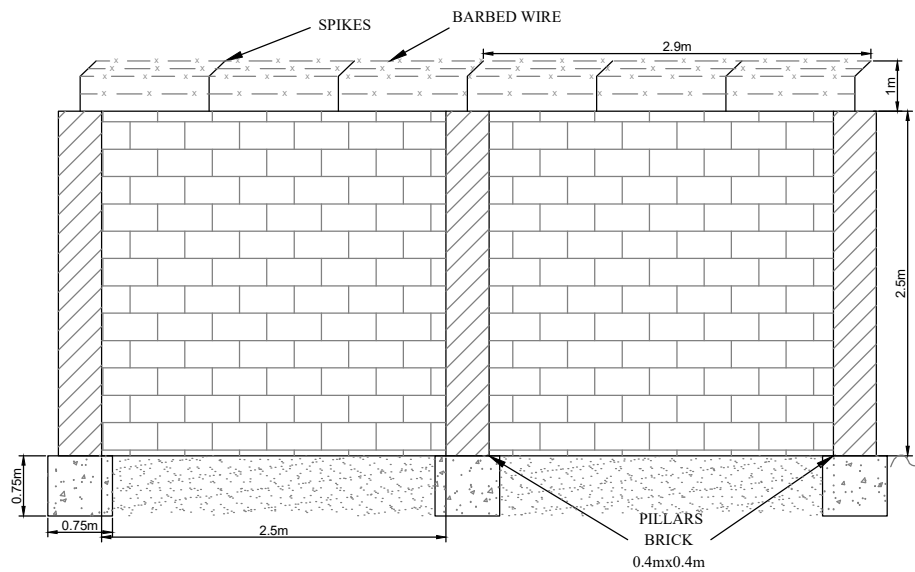
PROPOSED CROSS SECTION OF C-C'
NOT TO SCALE



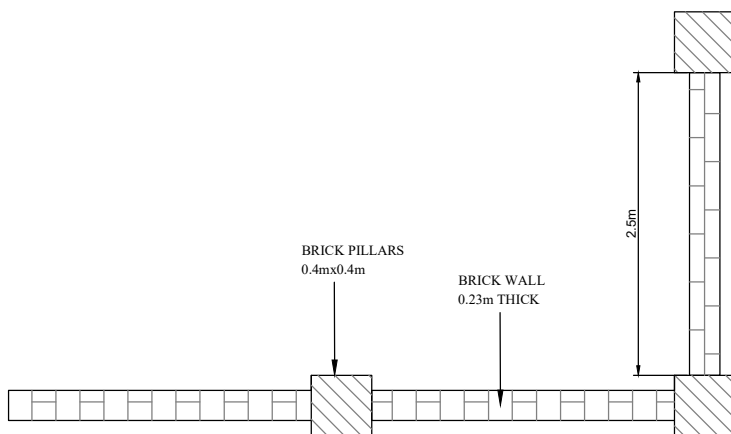
PROPOSED CROSS SECTION OF D-D'
NOT TO SCALE



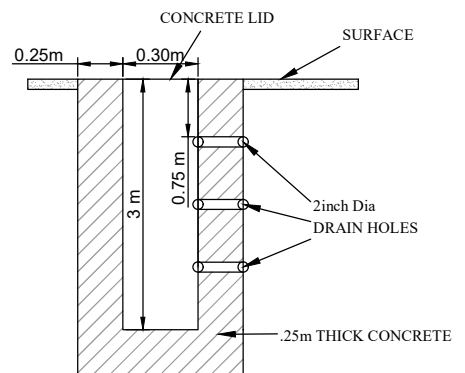
CROSS SECTION OF ANCHOR TRENCH/DRAINAGE LINE



NEW BOUNDARY WALL: CROSS SECTION VIEW
NOT TO SCALE



NEW BOUNDARY WALL: PLAN VIEW
NOT TO SCALE

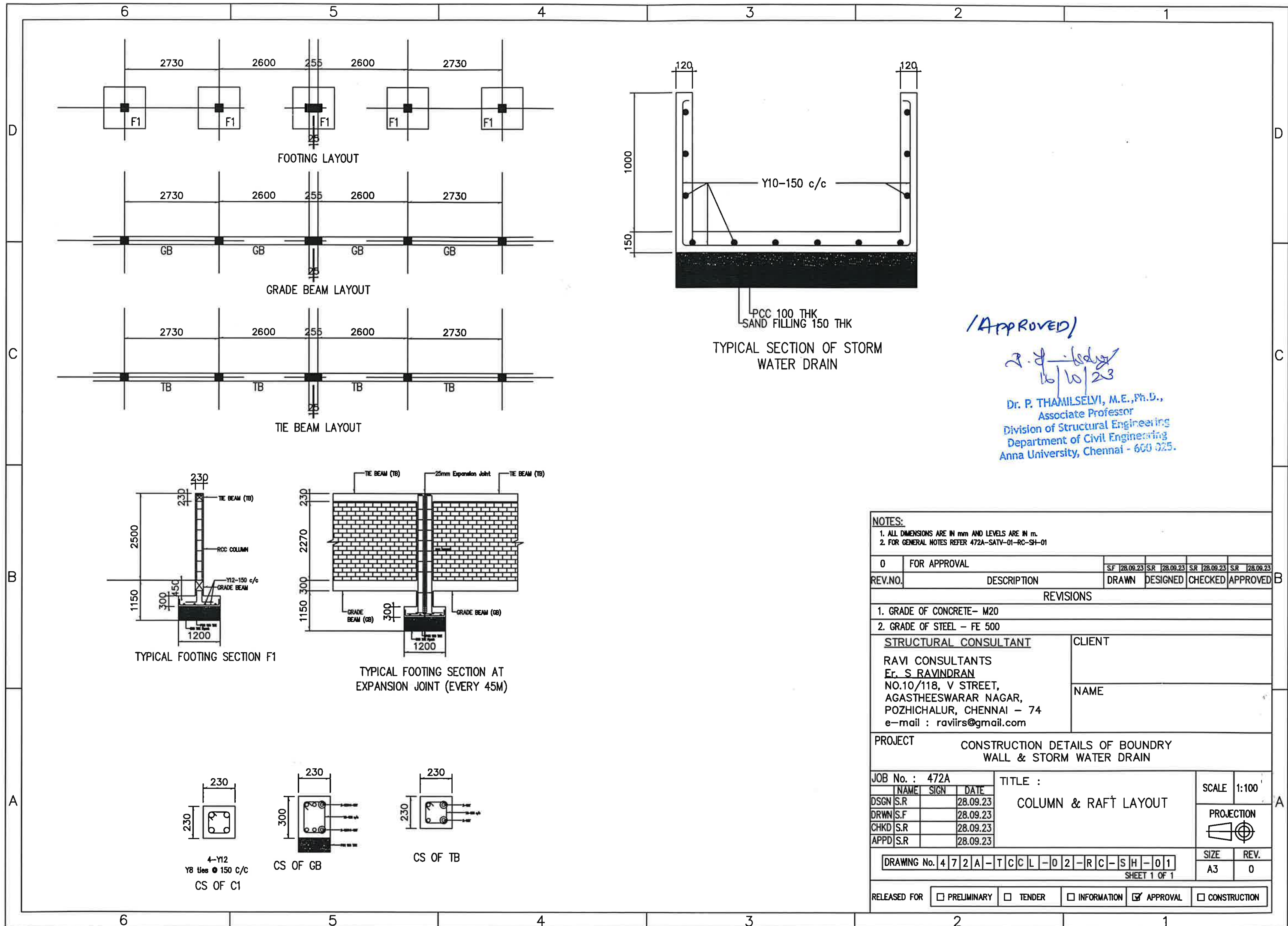


CUTOFF TRENCH: CROSS SECTION VIEW
NOT TO SCALE

CONSTRUCTION NOTES:

1. CONTRACTOR TO MAINTAIN NEW BOUNDARY WALL HEIGHT AT 2.5m (8 FEET) ABOVE GROUND SURFACE, CONTRACTOR TO USE A STEP DOWN TECHNIQUE AS NECESSARY TO MAIN WALL HEIGHT.
2. CONTRACTOR TO USE THE CONCRETE LID TO COVER THE TRENCH. DRAIN HOLES SHALL BE SPACED 1 M * 1 M. DISTANCE

- * The following changes to be made as per the drawing in the next pages:
1. Use of concrete as a replacement for brick in storm water drainage.
 2. Boundary wall with basement to be placed at a depth of 1.45 meters.



1.0 GENERAL NOTES

- 1.1 ALL STRUCTURAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER ENGINEERING DRAWINGS, SPECIFICATION, WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. DISCREPANCIES SHALL BE REFERRED TO THE CONSULTANT FOR DECISION BEFORE PROCEEDING WITH THE WORK.
- 1.2 ALL DIMENSIONS ARE IN mm. & LEVELS ARE IN METERS UNLESS OTHERWISE SPECIFIED.
- 1.3 ALL DIMENSIONS SHALL BE VERIFIED BY THE CONTRACTOR BEFORE CONSTRUCTION AND FABRICATION IS COMMENCED.

2.0 SOIL AND FOUNDATIONS

- 2.1 ALL FOUNDATIONS SHALL BE PLACED ON UNDISTURBED SOIL WHICH SHOULD HAVE ALLOWABLE BEARING CAPACITY NOT LESS THAN DESIGN VALUE UNLESS SPECIFICALLY APPROVED BY THE CONSULTANT / GEOTECHNICAL ENGINEER. (REFER SBC VALUE IN FOOTING DRAWING.
- 2.2 ALL FOUNDATIONS SHOULD REST ON HARD STRATA WITH MIN. BEARING CAPACITY AND DEPTH OF FOUNDATION AS SPECIFIED IN RELEVANT STRUCTURAL DRAWINGS. IF THE HARD STRATA IS AVAILABLE AT MORE DEPTHS, THE REMAINING EXCAVATED SOIL SHALL BE REPLACED WITH PLUM CONCRETE OR PLAIN CONCRETE OF GRADE C-10.
- 2.3 BACKFILLING SHALL BE DONE UNIFORMLY PLACED IN LAYER NOT EXCEEDING 150mm AND COMPACTED BY VIBRATION TO THE SATISFACTION OF THE ENGINEER. BACKFILLING MATERIAL TO BE APPROVED BY THE ENGINEER.
- 2.4 ALL WATER MUST BE REMOVED FROM THE FOUNDING PIT AND SOFTENED MATERIAL IF ANY MUST BE REMOVED AND REPLACED WITH LEAN-MIX CONCRETE OR BLINDING OR COMPACTED APPROVED GRANULAR FILL TO ENGINEER'S DIRECTION.
- 2.5 NO CHANGE IN SPECIFIED DEPTH TO FOUNDATION LEVELS SHALL BE MADE WITHOUT THE ENGINEER'S AUTHORITY.
- 2.6 ANTI-TERMITE TREATMENT:
THE CONTRACTOR SHALL PROVIDE ANTI-TERMITE TREATMENT BELOW FOUNDATION, BELOW GRADE SLAB, AROUND THE BUILDING AND OTHER LOCATIONS AS PER SPECIFICATIONS.

3.0 CONCRETE WORKS

- 3.1 UOS MINIMUM CLEAR COVER TO REINFT. SHALL BE AS FOLLOWS :

ITEM	COVER
FOR COLUMNS	40 mm
FOR BEAMS	30 mm
FOR SLABS	20 mm
FOR WALLS	20 mm
FOR WATER TANK INNER SURFACES IN CONTACT WITH WATER	50 mm
FOR WATER TANK OUTER SURFACES	40 mm
IN CONTACT WITH EARTH FACE	40 mm

- 3.2 FOR SPECIAL CASES WHERE GROUND WATER CONTAINS SULPHATES/CHLORIDES MORE THAN THE ALLOWABLE LIMITS THEN COVER SHOULD BE SUITABLY INCREASED AS PER THE INSTRUCTION OF ENGINEER IN CHARGE.
- 3.3 WATER STOP SHALL BE 150mm WIDE WITH MIN. 6 THK. RIBBED/SERRATED PVC WITH A CENTRAL BULB UOS.
- 3.4 FOR P.C.C M10, R.C.C CONCRETE GRADE M20 ALL RCC ELEMENTS CONFIRMING TO IS: 456-2000, STEEL GRADE Fe-500 CONFIRMING TO IS: 1786-2008
- 3.5 CONCRETING FOR WATER RETAINING AND BASEMENT STRUCTURES SHALL BE MADE WATER-TIGHT BY ADDITION OF WATER PROOFING ADMIXTURES IN THE PROPORTION AS RECOMMENDED BY MANUFACTURER.
- 3.6 FORMWORK SHALL REMAIN IN POSITION FOR NOT LESS THAN THE MINIMUM PERIODS.
AS PER IS 14687-1999
- 3.7 ALL SAMPLING AND TESTING OF SITE MIXED CONCRETE SHALL BE CARRIED OUT BY THE CONTRACTOR UNDER THE ENGINEER'S SUPERVISION IN ACCORDANCE WITH RELEVANT IS CODES.
- 3.8 THE TEST CUBES SHALL BE SEPARATELY CAST AND CURED IN ACCORDANCE WITH RELEVANT IS CODES METHOD OF TESTING CONCRETE.
- 3.9 THE MINIMUM RATE OF SAMPLING AND TESTING OF CUBES SHALL BE AS INDICATED IN THE SPECIFICATION.

4.0 REINFORCEMENT

- 4.1 'T' DENOTES HYSD BARS (Fe 500) CONFIRMING TO IS : 1786-2008.
- 4.2 LAPS FOR HYSD BARS SHALL BE AS FOLLOWS

DIA. OF BAR	LAP LENGTH (50D)
8	400
10	500
12	600
16	800
20	1000
25	1250
32	1600

FOR PLAIN BARS THE ABOVE VALUES SHOULD BE INCREASED BY 60%.

- 4.3 LAPS SHALL BE STAGGERED SUCH THAT NOT MORE THAN 50% OF STEEL IS LAPPED AT ANY LOCATION.
- 4.4 TOP BARS IN BEAMS & SLABS SHALL BE LAPPED AT MID SPAN AND NOT AT SUPPORT ZONE.
- 4.5 BOTTOM BARS IN BEAMS & SLABS SHALL BE LAPPED AT END ONE THIRD SPAN.
- 4.6 ALL HOOKS, BENDS AND LAPS SHALL BE AS PER IS : 2502 UNLESS OTHERWISE INDICATED.
- 4.7 REINFT. SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS, OPENINGS, CUTOUTS, ETC. AS APPROVED BY THE ENGINEER AT SITE UNLESS SHOWN OTHERWISE IN THE DRAWINGS.

/Approved/

P. P. Thamilselvi
16/09/23

Dr. P. THAMILSELVI, M.E., Ph.D.,
Associate Professor
Division of Structural Engineering
Department of Civil Engineering
Anna University, Chennai - 600 025.

NOTES:

1. ALL DIMENSIONS ARE IN mm AND LEVELS ARE IN m.

1	FOR APPROVAL	S.F. 28.09.23	S.R. 28.09.23	S.R. 28.09.23	S.R. 28.09.23
REV.NO.	DESCRIPTION	DRAWN	DESIGNED	CHECKED	APPROVED

REVISIONS

1. GRADE OF CONCRETE- M20
2. GRADE OF STEEL - FE 500

STRUCTURAL CONSULTANT	CLIENT
RAVI CONSULTANTS Er. S RAVINDRAN NO.10/118, V STREET, AGASTHEESWARAR NAGAR, POZHICALUR, CHENNAI - 74 e-mail : raviirs@gmail.com	NAME

PROJECT CONSTRUCTION DETAILS OF BOUNDARY
WALL & STORM WATER DRAIN

JOB No. : 472A	TITLE : GENERAL NOTES	SCALE : 1:100
DSGN S.R.	NAME	PROJECTION
DRWN S.F.	SIGN	DATE
CHKD S.R.	DATE	DATE
APPD S.R.	DATE	DATE

DRAWING No. 472A-TCL-01-RCSH-01
SHEET 1 OF 1

RELEASED FOR	☐ PRELIMINARY	☐ TENDER	☐ INFORMATION	☒ APPROVAL	☐ CONSTRUCTION
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